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Complex Analysis

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COMPLEX ANALYSIS

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CHAPTER 1

FUNDAMENTAL CONCEPTS OF COMPLEX ANALYSIS

1.1. Introduction

In the field of real numbers, the equation $x^2+1=0$ has no solution . To permit the solution of this and similar equations, the real number system was extended to the set of complex numbers. Euler introduced the symbol i with property that $i^2=-1$. He also called i as the imaginary unit. A number of the form $a+ib$, where a,b are real numbers, was called complex number.

If we write $z=x+iy$, then z is called a complex variable. Also x and y are respectively called real and imaginary parts of z. Sometimes we express z as $z=(x,y)$.

we also write

$\text{Re}(z)=x, \text{Im}(z)=y.$

If $x=0$, i.e., $z=iy$, the z is called purely imaginary number.

If $y=0$ i.e., $z=x$ then z is called purely real number.

The complex conjugate, or briefly conjugate, of

$z=x+iy$ is $\overline{z}=x-iy$.

For example conjugate of $-3-5i$ is $-3+5i$.

It is easy to verify that

$$\text{Re}(z)=x=\frac{z+\overline{z}}{2}, \text{Im}(z)=y=\frac{z-\overline{z}}{2i}.$$

1.2. Geometrical Representation of Complex Numbers

Consider the complex number $z=x+iy$,

A complex number can be regarded as an ordered pair of reals i.e. $z=(x,y)$

This form of z suggests that z can be represented by a point P whose co-ordinates are x and y relative to rectangular axes X and Y.

To each complex number there corresponds one and only one point in the xy-plane and conversely to each point in the plane there exists one and only one complex number. Due to this fact, the complex number z is referred to the point z in this plane. This plane is called complex plane or Gaussian plane or Argand plane. The representations of complex numbers is called Argand diagram .The complex number $x+iy$ is called complex co-ordinate; and x and y axes are respectively called real and imaginary axes. The distance between the points $z_1=x_1+iy_1$ and $z_2=x_2+iy_2$, is

$$|z_1-z_2|=\sqrt{\left[(x_1-x_2)^2+(y_1-y_2)^2\right]}$$

The complex number z is called affix of the point (x,y) which represent it.

1.3. Polar Forms of a Complex Numbers $P(x, y)$

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Consider a point P in the complex plane corresponding to a complex number $z = x + iy$. From the adjoining figure.

$x = r\cos\theta$, $y = r\sin\theta$

Then $r = \sqrt{(x^2+y^2)} = |x+iy| = |z|$

$\theta = \tan^{-1}(y/x)$.

If follows that $z = x + iy$

$= r(\cos\theta + i\sin\theta) = re^{i\theta}$ ...(1)

It is called polar form of the complex number z.

r and θ are called polar co-ordinates of z. r is called modulus or absolute value of z.

(Modulus of any complex number z is equal to distance of that point from the origin). The angle θ which the line OP makes with the positive x-axis is called argument or amplitude of z. It is also written as

$\theta = amp(z)$ or $\theta = \arg(z)$.

The argument of z is not unique, since the equation (1) does not alter, if we replace θ by $2\pi + \theta$. So θ can have infinite number of values which differ from each other by 2π. In order to specify a unique value of $\arg(z)$, we may restrict its value to some interval of length 2π. For this we introduce the concept of "principal value" of $\arg(z)$ as follows:

For an arbitrary $z \neq 0$, the principal value of Arg z is defined to be the unique value of z that satisfies $-\pi < \arg z \leq \pi - \theta \leq \pi$

and it will be denoted by arg z. Thus the relation between Arg z and arg z is given by $\arg z = Arg\, z + 2k\pi; k = 0, \pm1, \pm2, \ldots\ldots$

For convenience the set of all the values of Arg z is denoted by* $\arg z$.

In fact while inverting second equation in we should note the following

$$Arg\ z = \begin{cases} \text{Arctan}\left(\dfrac{y}{x}\right) & if\ x > 0,\ y \geq 0 \\ \pi + \text{Arctan}\left(\dfrac{y}{x}\right) & if\ x > 0,\ y > 0 \\ & and\ x < 0, y \geq 0 \\ -\pi + \text{Arctan}\left(\dfrac{y}{x}\right) & if\ x < 0,\ y < 0, \\ \dfrac{\pi}{2} & if\ x = 0,\ y > 0 \\ -\dfrac{\pi}{2} & if\ x = 0,\ y < 0 \end{cases}$$

Where $\text{Arctan}\, X$ is the principal value of the arctangent of a real number X, satisfying the inequality. $-\dfrac{\pi}{2} < \arctan X \leq \dfrac{\pi}{2}$.

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Results:

(i) The modulus of the product of two complex numbers is the product of their moduli.

$|z_1.z_2| = |z_1|.|z_2|$

(ii) The modulus of the sum of two complex numbers is less than or equal to sum of their moduli.

$|z_1 + z_2| \le |z_1| + |z_2|$..

(iii)The modulus of the difference of two complex numbers is greater than or equal to the difference of their moduli.

$|z_1 - z_2| \ge |z_1| - |z_2|$.

(iv) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left[|z_1|^2 + |z_2|^2\right]$

(v) Equation of a circle the equation of a circle in the Argand plane can be put in the form.

$z\bar{z} + b\bar{z} + \bar{b}z + c = 0$

where c is real and b is complex constant. If $b = \alpha + i\beta$ then centre of the circle is given by $(-\alpha, -\beta)$ & radius $= \sqrt{\alpha^2 + \beta^2 - c}$

1.4. Inverse Points w. r. t. A Circle

Two points $A(z = a)$ and $B(z = b)$ are said to be inverse points of a circle with centre $O(z = z_0)$ and radius r if O, A, B are collinear and $OA.OB = r^2$.

So that $|a - z_0|.|b - z_0| = r^2$

Then $\arg(a - z_0) = \arg(b - z_0) = -\arg\left(\overline{b - z_0}\right)$

This gives are $(a - z_0) + \arg\left(\overline{b - z_0}\right) = 0$

This $\Rightarrow (a - z_0)\left(\overline{b - z_0}\right)$ is real and equal to r^2.

Hence $\Rightarrow (a - z_0)\left(\overline{b - z_0}\right) = r^2$ is the required equation.

Deduction:

(i) If $z_0 = 0$, then a and b are inverse point if $a\bar{b} = r^2$.

Hence z_1, z_2 are inverse points of the circle $|z| = r$ if $z_1\overline{z_2} = r^2$.

Finally $z_1, \frac{1}{z_1}$ are inverse point of each other w. r. t. $|z| = r$.

(ii) $z = 0, z = \infty$ are inverse point of each other w. r. t. $|z| = r$.

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Example: Show that inverse point of a point $z = a$ w. r. t. circle $|z-c| = r$ is the point $c + \frac{r^2}{\bar{a}-\bar{c}}$.

Solution: $|z-c| = r$ ...(1)

Let B (b) is the inverse of A (a) w. r. t. circle (1) whose centre is $C(c)$ and radius r.

Then following conditions hold

(i) $CB.\,CA = r^2$

(ii) Lines CA and CB are collinear

From above

(i) $\Rightarrow |b-c||a-c| = r^2$

(ii) $\Rightarrow \arg(b-c) = \arg(a-c) = -\arg(\bar{a}-\bar{c})$. {For $\arg z = -\arg \bar{z}$}

Or, $\arg\left[(b-c)(\bar{a}-\bar{b})\right] = 0$

For $\arg(z_1 z_2) = \arg z_1 + \arg(z_2)$

This $\Rightarrow (b-c)(\bar{a}-\bar{c})$ is real and positive

$\Rightarrow |(b-c)(\bar{a}-\bar{c})| = \Rightarrow (b-c)(\bar{a}-\bar{c}) > 0$ ...(2)

Now (1) $\Rightarrow$ $|b-c|.|\bar{a}-\bar{c}| = r^2$

$\Rightarrow$ $|(b-c)(\bar{a}-\bar{c})| = r^2$

Using (2) we get

$|(b-c)(\bar{a}-\bar{c})| = r^2$

Or $b = c + \frac{r^2}{\bar{a}-\bar{c}}$.

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CHAPTER 2

STEREOGRAPHIC PROJECTION AND POINT SET TOPOLOGY

We shall extend the complex plane $\mathbb{C}$ by adjoining one extra point at infinity which we shall denote by $z=\infty$, i.e., we consider the extended complex plane $\mathbb{C}\cup\{\infty\}$ as a closed surface having a single point at infinity. We shall then introduce a new metric to describe the behavior of a complex function at infinity and to map the points in $\mathbb{C}$ into the surface of a sphere. This process will be referred to as stereographic projection.

In fact such an extra point '∞' is defined so as to satisfy the following computational properties.

Whatever be $z\in\mathbb{C}$

$$\frac{z}{\infty}=0;\ z+\infty=\infty(z\neq\infty);\ \frac{z}{0}=\infty(z\neq 0);$$

$$z.\infty=\infty(z\neq 0);\frac{z}{\infty}=0(z\neq\infty).$$

We do not define $\frac{0}{0}$, $\infty+\infty$, $\infty-\infty$ and $\frac{\infty}{\infty}$

We use the following construction due to Riemann. There are two commonly used methods, according as the complex plane C is the tangent to the sphere or passes through its centre . We shall use the first one. Now we set up a correspondence between the points of C and those of a sphere of radius $\frac{1}{2}$ with centre at $\left(0,0,\frac{1}{2}\right)$ tangent to this plane. (There is another method of correspondence in which the sphere of radius 1 has centre at $(0,0,0)$ and the plane passes through $(0,0,0)$.

Let C be the complex plane. Through the origin construct a line perpendicular to C. Let this be ξ-axis of a 3-dimensional Euclidean space in which a point has coordinates (ξ,η,ς). Consider the sphere S of radius $\frac{1}{2}$ and centre at $\left(0,0,\frac{1}{2}\right)$. That is

$$S=\left\{(\xi,\eta,\varsigma)\in R^3:\xi^2+\eta^2+\left(\varsigma-\frac{1}{2}\right)^2=\frac{1}{4}\right\}$$

It is a common practice to call the points N and O with coordinates $(0,0,1)$ and $(0,0,0)$ the North Pole and South Pole of the sphere S respectively .The great circle in the plane $\varsigma=\frac{1}{2}$ is called the equator. The plane $\varsigma=0$ coincides with the complex plane C and that ξ and η axes are

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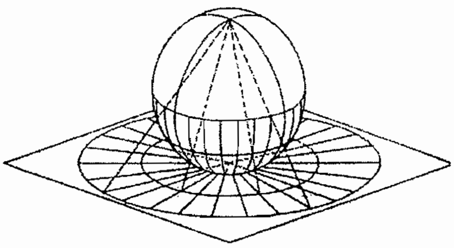
the x and y axes respectively. Let $Q(x,y,0)$ be any point in the plane C. Through the points N and Q we draw a straight line NQ intersecting the sphere S at a point say $P(\xi,\eta,\varsigma)$. Then (ξ,η,ς) is called the stereographic projection, or image of $(x,y,0)$ on the sphere and is considered as the spherical representation of $z=x+iy$. This procedure assigns a unique point on S to every given complex number z. Conversely to each point (ξ,η,ς) on the sphere other than N we can associate the complex number $z=(x,y,0)$ where the line from $(0,0,1)$ through (ξ,η,ς) intersects C. Now we immediately see that there is a one to one correspondence between C and the points of S with one exception, namely the north pole $(0,0,1)$ itself. By assigning to the north pole N of the sphere to correspond to the point at infinity, we obtain then a one to one correspondence between the points of the sphere S on one hand and the point of the extended comples plane $C\cup\{\infty\}$ on the obtain explicit equations expressing ξ,η and ς in terms of x and y. The line in R^3 passing through $(0,0,1)$ and $(x,y,0)$ is given by

$$\{t(0,0,1)+(1-t)(x,y,0):t\in R\};$$

that is $\{((1-t)x,(1-t)y,t):t\in R\}$

Since this line intersects the sphere S, we must have



$$C\to\left(0,0,\frac{1}{2}\right)$$
$$O\to(0,0,0)$$
$$Q\to(x,y,0)$$

$$(1-t)^2x^2+(1-t)^2y^2+\left(t-\frac{1}{2}\right)^2=\frac{1}{4}$$

So that $(1-t)^2|z|^2=t(1-t)$.

If $(\xi,\eta,\varsigma)\neq(0,0,1)$ we arrive at

$$t=\frac{|z|^2}{1+|z|^2},\ \text{i.e.,}\quad 1-t=\frac{1}{1+|z|^2}$$

Using the fact that the points $(0,0,1)$, $N\to(0,0,1)$

(ξ,η,ς) and $(x,y,0)$ are collinear now yields $P\to(\xi,\eta,\varsigma)$

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$$\left.\begin{aligned}\xi &= \frac{x}{1+x^2+y^2} = \frac{z+\bar{z}}{2\left(1+|z|^2\right)};\\ \eta &= \frac{y}{1+x^2+y^2} = \frac{-i(z-\bar{z})}{2\left(1+|z|^2\right)};\\ \varsigma &= \frac{x^2+y^2}{1+x^2+y^2} = \frac{|z|^2}{1+|z|^2};\end{aligned}\right\}$$

that is $z = x+iy \in C$ corresponds to $\equiv(\xi,\eta,0)$

$$\left(\frac{x}{1+x^2+y^2},\frac{y}{1+x^2+y^2},\frac{x^2+y^2}{1+x^2+y^2}\right)\in C$$

Or $\left(\dfrac{z+\bar{z}}{2\left(1+|z|^2\right)},\dfrac{-i(z-\bar{z})}{2\left(1+|z|^2\right)},\dfrac{|z|^2}{1+|z|^2}\right)\in S$

For instance, the images of $1, i, \dfrac{(1-i)}{\sqrt{2}}$ on the sphere are respectively given by Z_1, Z_2, Z_3

Where $Z_1 = \left(\frac{1}{2},0,\frac{1}{2}\right)$

$Z_2 = \left(0,\frac{1}{2},\frac{1}{2}\right)$

$Z_3 = \left(\frac{\sqrt{2}}{4},-\frac{\sqrt{2}}{4},\frac{1}{2}\right)$

Using the three equations in it is easy to see that

$$1-\varsigma = \frac{1}{1+|z|^2};\ \left(i.e., |z|^2 = \frac{\varsigma}{1-\varsigma}\right)$$

So that $x = \dfrac{\xi}{1-\varsigma},\quad y = \dfrac{\eta}{1-\varsigma};$

Or $z = \dfrac{\xi + i\eta}{1-\varsigma}$

That is (ξ,η,ς) corresponds to

$$\left(\frac{\xi}{1-\varsigma}\right)+i\left(\frac{\eta}{1-\varsigma}\right)\in C$$

The map $z \leftrightarrow (\xi,\eta,\varsigma)$ is called stereographic projection of C on $S\setminus\{(0,0,1)\}$ or vice versa.

Example: Prove that the points z and z' in the complex plane will represent symmetric points with respect to the equatorial plane (viz. the plane $\varsigma = \frac{1}{2}$) if and only if $zz' = 1$

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Solution: Note that z and z' correspond to symmetric points with respect to equatorial plane if and only if z corresponds to (ξ,η,ς) and z' corresponds to $(\xi,\eta,1-\varsigma)$ .This holds if the only if

$$z=\frac{\xi+i\eta}{1-\varsigma} \text{ and } z'=\frac{\xi+i\eta}{1-(1-\varsigma)}=\frac{\xi+i\eta}{\varsigma}$$

i.e., if and only if $zz'=\dfrac{\xi^2+\eta^2}{(1-\varsigma)\varsigma}=1$

2.1. Chordal Distance

Let z_1 and z_2 be the points in the argand plane corresponds sterio-graphically to z_1 and z_2 respectively.

Distance between z_1 and z_2 i.e. length of chord joining z_1 and z_2 is referred as chordal distance between z_1 and z_2

Example: Find the chordal distance $z_1=1+i,\ z_2=\infty$

$\left(\frac{1}{3},\frac{1}{3},\frac{2}{3}\right)(0,0,1)$

$$\text{Distance }=\sqrt{\left(\frac{1}{3}-0\right)^2+\left(\frac{1}{3}-0\right)^2+\left(\frac{2}{3}-1\right)^2}$$

$$=\sqrt{\frac{1}{9}+\frac{1}{9}+\frac{1}{9}}$$

$$=\frac{1}{\sqrt{3}}$$

Chordal Distance between z and ∞ is denoted by $\gamma(z,\infty)$ and is equal to

$$\chi(z,\infty)=\frac{1}{\sqrt{1+|z|^2}}$$

Chordal Distance between z_1 and z_2 is calculated by the following formula

$$\chi(z_1,z_2)=\frac{|z_1-z_2|}{\sqrt{1+|z_1|^2}\times\sqrt{1+|z_2|^2}}$$

If $\chi(z_1,z_2)=1$ then

$$\frac{|z_1-z_2|}{\sqrt{1+|z_1|^2}\times\sqrt{1+|z_2|^2}}=1$$

$$|z_1-z_2|=\sqrt{1+|z_1|^2}\sqrt{1+|z_2|^2}$$

Squaring both side

$$(z_1-z_2)(\overline{z}_1-\overline{z}_2)=\left(1+|z_1|^2\right)\left(1+|z_2|^2\right)$$

$$z_1\overline{z}_1-z_1\overline{z}_2-z_2\overline{z}_1+z_2\overline{z}_2=1+z_2\overline{z}_2+z_1\overline{z}_1+z_1z_2\overline{z}_1\overline{z}_2$$

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$$z_1\bar{z}_2 + z_2\bar{z}_1 + z_1 z_2 \bar{z}_1 \bar{z}_2 + 1 = 0$$

$$z_1\bar{z}_2(z_2\bar{z}_1 + 1) + (z_2\bar{z}_1 + 1) = 0$$

$$(z_1\bar{z}_2 + 1)(z_2\bar{z}_1 + 1) = 0$$

$$z_1\bar{z}_2 = -1$$

$$z_2\bar{z}_1 = -1$$

If on the sphere two point diametrically opposite then $z_1\bar{z}_2 = -1$

2.2. Sets of Points in the Complex Plane

Let S be a non-empty set of complex numbers and $z_0 \in S$ be any complex number and δ be a positive real number. Then, we define the following:

1. **Circle:** The set of points which satisfies the equation $|z - z_0| = \delta$ or $(x - x_0)^2 + (y - y_0)^2 = \delta^2$

   Defines a circle C of radius δ, with centre at $z_0 = (x_0, y_0)$. This set consists of all points which lie on the boundary of the circle C. Any point z on this circle has the polar form $z = z_0 + \delta e^{i\theta}$. As θ varies from 0 to $2\pi, z$ traverses once over this circle in the counter clockwise direction. If $z_0 = 0$, then the equation $|z| = \delta$ defines a circle of radius δ about the origin.

2. **Open Disk:** An open disc is defined as $\Delta(z_0;\in) = \{z \in \mathbb{C} : |z - z_0| < \in\}$ with center at the point z_0 and radius $\in$ i.e. it is the collection of all those point inside the circle $|z - z_0| = \in$.

3. **Closed Disk:** An open disc is defined as $\Delta(z_0;\in) = \{z \in \mathbb{C} : |z - z_0| \le \in\}$ with center at the point z_0 and radius $\in$ i.e. it is the collection of all those point inside and on the circle $|z - z_0| = \in$.

4. **Annulus:** The set of points which lie between two concentric circles $C_1 : |z - z_0| = r_1$ and $C_2 : |z - z_0| = r_2$ defines an open annulus or an open circular ring, that is the set of points which satisfies the inequality $r_1 < |z - z_0| < r_2$.

   The set of points which satisfies the inequality $r_1 \le |z - z_0| \le r_2$, defines a closed annulus.

5. **Neighborhood of A Point:** A δ-neighborhood of a point z_0 in the complex plane is the set of all points z which lie in the open disk $|z - z_0| < \delta$. Usually, the δ-neighborhood about the point z_0 is denoted by $N(z_0, \delta)$ or $N_\delta(z_0)$. If we exclude the point z_0 from the open disk $|z - z_0| < \delta$, then it is called the deleted neighborhood of the point z_0 and is written as $0 < |z - z_0| < \delta$.

6. **Interior Point:** A point z is an interior point of S, if for some $\delta > 0, \mathrm{N}(z, \delta) \subseteq S$.

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7. **Exterior Point:** A point z is an exterior point of S, if it is the interior of the set S^{c}.

8. **Frontier Point:** $a \in \mathbb{C}$ is said to be frontier point of $S \subseteq \mathbb{C}$, if it is neither an interior point of S nor an exterior point of S. Set of all frontier point of S is denoted by Fr S

9. **Boundary Point:** Frontier point of S which are member of S are referred as boundary point. Set of all boundary point of S is denoted by ∂S.

   Example: For the set of points defined by $\text{Im}(z) \geq 1$, the points on the line $y = 1$ are the boundary points. The points on the circle $|z - z_0| = r$ are the boundary points for the disk $|z - z_0| \leq r$. The collection of all the boundary points define the boundary of S.

10. **Open Set:** A set S is open, if every point of S is an interior point.

    Example: The sets

    $S = \{z : |z - z_0| < r\}$; $S = \{z : \text{Re}(z) < 0\}$; $S = \{z : 1 < |z| < 2\}$ are open sets.

11. **Closed Set:** A set S is closed, if every boundary point of S belongs to S.

    Example: The sets $S = \{z : |z - z_0| \leq r\}$; $S = \{z : r_1 \leq |z - z_0| \leq r_2\}$ are closed sets.

12. **Bounded Set:** An open set S in bounded, if there exists a positive real number M, such that $|z| \leq M$ for all $z \in S$. Otherwise, the set S is said to be unbounded.

    Example: The set $S = \{z : |z - z_0| < r\}$ is a bounded and $S = \{z : |z - z_0| > 0\}$ is an unbounded.

13. **Connected Set:** A subset S of $\mathbb{C}$ is said to be connected if the only subsets of S which are both open and closed are ϕ and S.

14. **Domain:** An open connected set is called a domain. Usually, a domain is denoted by D.

15. **Region:** A region is a domain together with all, some or none of its boundary points. Thus, a domain is always a region but a region may or may not be a domain.

    Example: An open disk is both domain and region but closed disk is region but not domain. Usually, a region is denoted by R.

16. **Extended Complex Plane:** The complex plane to which the point at $z = \infty$ has been added is called the extended complex plane. The complex plane without the point at $z = \infty$ is called the finite complex plane.

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CHAPTER 3

LIMIT CONTINUITY & DIFFERENTIABILITY

3.1. Limit of A Function

Let $w = f(z)$ be a complex valued function f defined on $D \subseteq \mathbb{C}$ let $z_0 \in D'$. Then f is said to have a limit l as $z \to z_0$ and we write

$\lim_{z \to z_0} f(z) = l$ or $f(z) \to l$ as $z \to z_0$

if and only if for any given $\varepsilon > 0$, $\exists\, a\ \delta > 0$ such that

$|f(z) - l| < \varepsilon$ whenever $z \in D$ & $0 < |z - z_0| < \delta$;

i.e., if and only if for each $\varepsilon > 0$, $\exists\ \delta > 0$ such that

$f(z) \in B(l; \varepsilon)$ whenever $z \in B(z_0, \delta) \setminus \{z_0\} \cap D$.

$z \to z_0$ in complex plane. It is straight forward to state

$\lim_{z \to z_0} f(z) = l \Leftrightarrow \lim_{z \to z_0} |f(z) - l| = 0$

Note:

(i) The function need not to be defined at z_0 in order to have a limit at z_0.

(ii) It is the punctured disk $B(z_0, \delta) \setminus \{z_0\}$ which is involved in D, i.e., z_0 need not to be in D.

(iii) If the condition that $z_0 \in D$ holds, we may have $f(z_0) \neq l$.

3.2. Alter Definition of Limit of A Function

Let $f(z) = u(z) + iv(z)$, where $u(z) = u(x, y)$ & $v(z) = v(x, y)$ are real valued functions, be defined on D except possibly at z_0. Then for l_1 & $l_2 \in R$.

$\lim_{z \to z_0} f(z) = l_1 + il_2$

If and only if $\lim_{(x,y) \to (x_0, y_0)} u(x, y) = l_1$ & $\lim_{(x,y) \to (x_0, y_0)} v(x, y) = l_2$.

Theorem: If $\lim_{z \to z_0} f(z)$ exists, then it is unique.

Examples:

1. Let $z \neq 0$ consider $f(z) = \dfrac{\bar{z}}{z}$, now let us examin the limits of $f(z)$ as $z \to 0$ in many ways. Let m be any real number & allow $z \to 0$ along the line $y = mx$ then

$$\lim_{(x+imx) \to 0} \frac{x - imx}{x + imx} = \frac{1 - im}{1 + im}$$

$\Rightarrow \quad \lim_{z \to 0} f(z)$ depends upon the path as we approach towards z_0 hence limit does not exist at $z = 0$.

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2. Let $f(z)=x\sin\frac{1}{x}+iy\sin\frac{1}{y}$ then its domain of definition is the collection of all those points = which does not lie on x axis & y axis.

Consider $\lim\limits_{z\to iy_0} f(z)=\lim\limits_{x\to 0} x\sin\frac{1}{x}+i\lim\limits_{y\to y_0} y\sin\frac{1}{y}$

$=0+iy_0\sin\frac{1}{y_0}$

hence limit exist.

3. $\lim\limits_{z\to 0} z^m \sin\frac{1}{z}$ does not exist $\forall\, m\in\mathbb{N}$.

4. $\lim\limits_{z\to 0} e^{-\frac{1}{z^m}}$ such that $m\in\mathbb{N}$ does not exist.

Theorem: If $f(z)$ has a finite limit at z_0, then $f(z)$ is a bounded function in some neighborhood of z_0.

Theorem: Let $f(z)$ & $g(z)$ be two functions such that

$\lim\limits_{z\to z_0} f(z)=l_1$ & $\lim\limits_{z\to z_0} g(z)=l_2$

Then

(a) $\lim\limits_{z\to z_0}\left[\alpha f(z)\right]=\alpha l_1$ where α is a real or complex constant.

(b) $\lim\limits_{z\to z_0}\left[f(z)\pm g(z)\right]=l_1\pm l_2$

(c) $\lim\limits_{z\to z_0}\left[f(z)\cdot g(z)\right]=l_1\cdot l_2$

(d) $\lim\limits_{z\to z_0}\left[\frac{1}{g(z)}\right]=\frac{1}{l_2},\ l_2\neq 0$

(e) $\lim\limits_{z\to z_0}\left[f(z)/g(z)\right]=\frac{l_1}{l_2},\ l_2\neq 0$

3.3. Limit of a Function at $z=\infty$

The function $f(z)$ has a limit L as $z\to\infty$, if for any arbitrary small real number $\varepsilon>0$, $\exists$ a real number $\delta>0$ such that

$|f(z)-l|<\varepsilon$ whenever $|z|>\frac{1}{\delta}$

Alternately, we substitute $z=\frac{1}{\xi}$, since $\xi\to 0$ as $z\to\infty$ we obtain

$\lim\limits_{z\to\infty} f(z)=\lim\limits_{\xi\to 0} f\left(\frac{1}{\xi}\right)$

3.4. Continuous Function

A function $f:D\to\mathbb{C}$ is continuous at $z_0\in D$ if and only if $\lim\limits_{z\to z_0} f(z)$ exists & equals to the functional value $f(z_0)$. We say that f is continuous on D or $f:D\to\mathbb{C}$ is continuous when f is continuous at all points of D i.e., for a given $\varepsilon>0$, there $\exists$ a $\delta>0$ such that

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$|f(z)-f(z_0)|<\varepsilon$ whenever $z\in D$ & $|z-z_0|<\delta$ or equivalently,

$f(z)\in B(f(z_0);\ \varepsilon)$ whenever $z\in B(z_0;s)\cap D$.

Results on Continuity

1. A function is continuous by default at all isolated points of the domain.
2. Continuous function maps connected sets to connected set.
3. Continuous function maps compact set to compact set.
4. If a continuous function is defined on a compact set then it is uniformly continuous.
5. If the function $f(x)=u(x,y)+iv(x,y)$ is continuous at $z_0=x_0+iy_0$, then the real functions $u(x,y)$ & $v(x,y)$ are also continuous at the point (x_0,y_0).
6. If $f(z)$ & $g(z)$ are continuous at a point z_0, then the function $f(z)\pm g(z),\ f(z).g(z),|f|,\dfrac{f(z)}{g(z)}$ where $g(z_0)\neq 0$ are also continuous at z_0.
7. If $f(z)$ is continuous in a closed region S, then it is bounded i.e., $|f(z)|\leq M\ \ \forall z\in S$.
8. The function $f(z)$ is continuous at $z=0$ if the function $f\left(\dfrac{1}{\xi}\right)$ is continuous at $\xi=0$.

   Composite Function: Let a function $g(z)$ be defined in the neighborhood of a point z_0 & let the image of $g(z)$ in this neighborhood be contained in a region in which $f(z)$ is defined. Then the composite function $f(g(z))$ is defined for all z in the neighborhood of the point z_0.

9. The composition of two continuous function is continuous i.e., if $f:D_1\rightarrow D_2$ is continuous at $z_0\in D_1$ & if $g=D_2\rightarrow\mathbb{C}$ is continuous at $w_0=f(z_0)$, then gof defined by $gof(z)=g(f(z))$ is continuous at z_0.
10. A function f is continuous at a point $z_0\in D$ if and only if $f(z_0)=\lim\limits_{n\to\infty}f(z_n)$ for every sequence $\langle z_n\rangle$ such that $z_n\in D$ for $n=1,2,....$ & $z_n\rightarrow z_0$ as $n\rightarrow\infty$.
11. Let $f:D\rightarrow\mathbb{C}$ be a function. Then f is continuous on D if and only if for every open set $O\subseteq\mathbb{C}$ $f^{-1}(O)=\{z\in D:f(z)\in O\}$ is open in D.

Notes

Examples:

(i) The functions $e^z, \sin z, \cos z$ are continuous function for all z.

Consider $e^z = e^{x+iy}$

$= e^x \cos y + i e^x \sin y = u + iv$

$\sin z = \sin x \cosh y + i \cos x \sinh y = u + iv$

$\cos z = \cos x \cosh y - i \sin x \sinh y = u + iv$

since the real valued functions u & v of two real variables x & y are continuous $\forall\, x \,\&\, y$ in each case, the given function of a complex variable z are continuous for all z.

(ii) Show that the function $f(z)$ is not continuous at $z = 0$, where

$$f(z) = \begin{cases} \dfrac{\operatorname{Im} z}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

Solution: $\lim\limits_{z \to 0} \dfrac{\operatorname{Im}(z)}{|z|} = \lim\limits_{\substack{x \to 0 \\ y \to 0}} \dfrac{y}{\sqrt{x^2 + y^2}}$

Consider the path $y = mx$. Then

$$\lim_{\substack{x \to 0 \\ y \to 0}} \frac{mx}{\sqrt{x^2 + m^2 x^2}} = \frac{m}{\sqrt{1 + m^2}}$$

Since the limit of $f(z)$ is path dependent so function is not continuous.

3.5. Uniform Continuity

A function $f(z)$ is said to be uniformly continuous in a region S if for a given real number $\varepsilon > 0$ depending only on ε such that

$|f(z_1) - f(z_2)| < \varepsilon$ whenever $0 < |z_1 - z_2| < \delta$

where z_1 & z_2 are any two points in the region S.

Note: It is meaningless to talk about uniform continuity at a point.

(i) If $f(z)$ is uniformly continuous in a region S, then $f(z)$ is continuous in S.

Example: Show that the function $f(z) = z^2$ is uniformly continuous in the region $|z| < 1$.

Solution: We need to show that for any given $\varepsilon > 0$, $\exists$ a $\delta > 0$ such that $|z_1^2 - z_2^2| < \varepsilon$ whenever $0 < |z_1 - z_2| < \delta$ where z_1 & z_2 are any two points in the region $|z| < 1$, we have

$$|z_2^2 - z_1^2| = |z_2 - z_1||z_2 + z_1| \le (|z_2| + |z_1|)(|z_2 - z_1|) \le 2|z_2 - z_1|$$

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Thus when $\left|z_1^2 - z_2^2\right| < \varepsilon$ it follows that $\left|z_2 - z_1\right| < \frac{\varepsilon}{2}$. Thus, $\delta < \frac{\varepsilon}{2}$ depends only on ε & not on the choice of the points z_1, z_2 in the region. Hence $f(z) = z^2$ is uniformly continuous in the region $|z| < 1$.

3.6. Differentiability

A complex function f defined on a non-empty set D is differentiable at $z_0 \in D$ if the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} \quad \ldots(i)$$

exists & the value of the limit denoted by $f'(z_0)$ is called the derivative of f at z_0. The function f is differentiable on D if it is differentiable at every point of D or in terms of $\varepsilon - \delta$ notation, the limit in equation (i) exists if and only if given $\varepsilon > 0$, $\exists$ a $\delta = \delta(\varepsilon, z_0) > 0$ such that

$$\left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| < \varepsilon \text{ whenever } 0 < |z - z_0| < \delta$$

Examples:

(i) $p(z) = \sum_{i=1}^{n} a_i z^i = f_1(z)$

(ii) $\sin z = f_2(z)$

(iii) $\cos z = f_3(z)$

(iv) $e^z = f_4(z)$

(v) $\alpha f_i(z) + \beta f_j(z) = f_5(z) i, j = 1$ to 4

(vi) $\frac{f_i(z)}{f_4(z)}$, $i = 1$ to 5.

The functions defined above are all differentiable on the complex plane

Note: Differentiability $\Rightarrow$ Continuity

3.7. Cauchy–Riemann Equations

Let $f : G \to \mathbb{C}$ be differentiable & let $u(x,y) = \operatorname{Re} f(z), v(x,y) = \operatorname{Im} f(z)$ for $z \in G$. Let us evaluate the limit $f'(z) = \lim_{h \to 0} \frac{f(z+h) - f(z)}{h}$ in two different ways. First let $h \to 0$ through real values of h for $h \neq 0$ & h real we get

$$\frac{f(z+h) - f(z)}{h} = \frac{f(x+h+iy) - f(x+iy)}{h}$$

$$= \frac{u(x+h,y) - u(x,y)}{h} + i\frac{v(x+h,y) - v(x,y)}{h}$$

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Letting $h \to 0$ gives

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \qquad ...(i)$$

Now let $h \to 0$ through purely imaginary values; i.e., for $h \neq 0$ and h real,

$$\frac{f(z+ih)-f(z)}{ih} = -\frac{i\left(u(x,y+h)-u(x,y)\right)}{h} + \frac{v(x,y+h)-v(x,y)}{h}$$

Thus $f'(z) = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$ ...(ii)

equating the real & imaginary parts of (i) & (ii) we get the $C-R$ equations.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \;\;\&\;\; \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

3.8. Polar Form of $C-R$ Equation

Let $f(z) = u + iv$ is a differentiable function & $z = re^{i\theta}$ then $C-R$ equations are given by

$$u_r = \frac{v_\theta}{r} \;\;\&\;\; v_r = \frac{u_\theta}{r}$$

3.9. Complex Form of $C-R$ Equation

Let $f(z) = u + iv$ be a differential function.

Let $z = x + iy = z(x,y)$

$\& \; \bar{z} = x - iy = \bar{z}(x,y)$

$$x = \frac{z+\bar{z}}{2}, \;\; y = \frac{z-\bar{z}}{2i}$$

i.e. $x = x(z,\bar{z}), \; y = y(z,\bar{z})$

$$\Rightarrow \frac{\partial f}{\partial \bar{z}} = \frac{\partial u}{\partial \bar{z}} + i\frac{\partial v}{\partial \bar{z}}$$

$$= \frac{\partial u}{\partial x}.\frac{\partial x}{\partial \bar{z}} + \frac{\partial u}{\partial y}.\frac{\partial y}{\partial \bar{z}} + i\frac{\partial v}{\partial x}.\frac{\partial x}{\partial \bar{z}} + i\frac{\partial v}{\partial y}.\frac{\partial y}{\partial \bar{z}}$$

$$= \frac{1}{2}u_x + \frac{1}{2}u_y + i\frac{1}{2}v_x + i\frac{1}{2}v_y$$

$$= \frac{1}{2}\left[\left(u_x - v_y\right) + i\left(u_y + v_x\right)\right]$$

Since f is differentiable, so it must satisfy the $C-R$ equation i.e., $u_x = v_y$ & $u_y = v_x$

$\Rightarrow \frac{\partial f}{\partial \bar{z}} = 0$ is called $C-R$ equation in complex form.

3.10. Necessary Condition for Differentiability

Let $f(z)=u+iv$ be a function then necessary condition f_0 it be differentiable is that u_x, u_y, v_x & v_y exist & satisfy

$u_x = v_y$ & $u_y = -v_x$ i.e., $C-R$ equation

i.e. $\frac{\partial f}{\partial \bar{z}} = 0$

3.11. Sufficient Condition for Differentiability

Let $f: S \to \mathbb{C}$ where $S \subset \mathbb{C}$ be such that $f(z)=u+iv$ & $(x_0+iy_0)=z_0 \in S \cap S'$ then f is differentiable if

(i) u_x, u_y, v_x, v_y exist

(ii) All the partial derivative are continuous.

(iii) Also satisfy the $C-R$ equation.

Theorem: A real valued function of a complex variable either has derivative zero or the derivative does not exist.

Proof: Suppose that $f(z)$ is a real-valued function of complex variable whose derivative exists at a point a. Then

$$f'(a) = \lim_{h \to 0} \frac{f(a+h)-f(a)}{h}.$$

If we take the limit $h \to 0$ along the real axis, then $f'(a)$ is real. If we take the limit $h \to 0$ along the imaginary axis, then $f'(a)$ becomes a purely imaginary number. We must have $f'(a)=0$.

Theorem: If f and g are differentiable at z_0, then their sum $f+g$, difference $f-g$, product fg, quotient f/g $(\text{where } g(z_0) \neq 0)$ and the scalar multiplication cf, are also differentiable at z_0 and

$$(f \pm g)' = f' \pm g', \ (fg)' = f'g + fg',$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \ (cf)' = cf'$$

Where c is a complex constant.

More generally, a finite linear combinations (of the form $\alpha_1 f_1 + \alpha_2 f_2 + \ldots + \alpha_n f_n, \alpha_j \in \mathbb{C}, j = 1,2,\ldots n$) and finite products of functions differentiable at z_0 are also differentiable at z_0.

Example: Let $f(z) = \operatorname{Re} z$ and z_0 be an arbitrary fixed point in $\mathbb{C}$. The $h = h_1 + ih_2 (\neq 0)$,

$$\frac{f(z_0+h)-f(z_0)}{h} = \frac{\operatorname{Re}(h)}{h} = \begin{cases} 1 \text{ for } h = h_1 + i.0 \in \mathbb{R} \setminus \{0\} \\ 0 \text{ for } h = 0 + ih_2 \in i(\mathbb{R} \setminus \{0\}) \end{cases}$$

So that $\lim\limits_{h \to 0} \frac{f(z_0+h)-f(z_0)}{h}$

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does not exist. Therefore, $f(z)=\operatorname{Re} z$ is nowhere differentiable even though it is continuous in $\mathbb{C}$.

Similarly, we see that $\operatorname{Im} z, |z|, \bar{z}$ and $\operatorname{Arg} z$ are all nowhere differentiable in $\mathbb{C}$. Is each of the functions listed here continuous in $\mathbb{C}$? Is each of these functions infinitely often real differentiable in $\mathbb{R}^2$? Is a product of two nowhere differentiable functions always nowhere differentiable? How about $g(z)=(\bar{z})^2$?

Example: Each of the functions

$f_1(z)=|z|,\ f_2(z)=\operatorname{Re} z$ and $f_3(z)=\operatorname{Im} z$,

is a non-constant real-valued function defined in $\mathbb{C}$. Each of them is nowhere analytic. If we rewrite these function as

$f_1(x,y)=\sqrt{x^2+y^2},\ f_2(x,y)=x,\ f_3(x,y)=y,$

then, except f_1, each of these functions are real differentiable in $\mathbb{R}^2$.

Example: It is easy to see that the function f defined by

$$f(z)=|\operatorname{Re} z \operatorname{Im} z|^{1/2}$$

satisfies the *C-R* equations at the origin, but is not differentiable at this point. To see this, we may rewrite the given function as

$$f(z)=\frac{\left|z^2-\bar{z}^2\right|^{1/2}}{2} \text{ or } f=u+iv \text{ with } u(x,y)=|xy|^{1/2} \text{ and } v(x,y)=0.$$

Note that f is identically zero on the real and imaginary axes. Therefore, it is trivial to see that $u_x(0,0)=u_y(0,0)=v_x(0,0)=v_y(0,0)=0$. For example,

$$u_x(0,0)=\lim_{s\to 0}\frac{u(s,0)-u(0,0)}{s}=0.$$

Thus, the *C-R* equations hold at $z=0$. However, taking $h=re^{i\theta}\neq 0$ with $r\to 0$, we find that

$$\lim_{h\to 0}\frac{f(h)-f(0)}{h}=\lim_{r\to 0}\frac{\left|r^2\cos\theta\sin\theta\right|^{1/2}}{r(\cos\theta+i\sin\theta)}=\frac{e^{-i\theta}|\sin 2\theta|^{1/2}}{\sqrt{2}}$$

Which is clearly depending upon θ (e.g. take $\theta=0$ and $\theta=\pi/4$). We conclude that f is not differentiable at $z=0$ even though f satisfies the *C-R* equations at the origin. Here, since $v(x,y)=0$, v is a C^∞- function in $\mathbb{R}^2$. Are the partial derivatives u_x and u_y continuous at the origin? How about the functions

$$f(z)=|\operatorname{Re} z \operatorname{Im} z|^{1/3} \text{ and } f(z)=|\operatorname{Re} z \operatorname{Im} z|^{1/4}?$$

Example: The function $f(z)=|z|$ is now here differentiable

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Since $f(z)=|z|=\sqrt{x^2+y^2}=\sqrt{z.\bar{z}} \Rightarrow \dfrac{\partial f}{\partial \bar{z}}=\sqrt{z}\times\dfrac{1}{2\sqrt{\bar{z}}}\neq 0$

$\Rightarrow f(z)=|z|$ does not satisfy the $C-R$ equation hence it is nowhere differentiable.

Example: The function $f(z)=|z|^2$ is differentiable only at origin.

Since $f(z)=|z|^2=x^2+y^2=z.\bar{z}$

$\Rightarrow \dfrac{\partial f}{\partial \bar{z}}=z$, so if $f(z)$ wants to be differentiable then it must satisfy the $C-R$ equation i.e., $\dfrac{\partial f}{\partial \bar{z}}=0 \Rightarrow z=0$

Consider $\lim\limits_{z\to 0}\dfrac{f(z)-f(0)}{z-0}=\lim\limits_{z\to 0}\dfrac{z.\bar{z}}{z}=\lim\limits_{z\to 0}\bar{z}=0$

Hence $f(z)$ is differentiable only at origin.

Example: Function $f(z)=\dfrac{1}{z}$ is differentiable at all points except at $z=0$.

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CHAPTER-4

SINGULARITES OF ANALYTIC FUNCTIONS

4.1. Regular Point

Let f is defined in D and z_0 is an interior point in D. We say f is regular on z_0, or equivalently, z_0 is a regular point of f if $\exists\, \delta > 0$ such that $f'(z)$ exists $\forall\, z \in |z - z_0| < \delta$

Statement: If f is regular at z_0, then $\exists$ a open disc around z_0 such that every point of the disc is regular point.

$f'(z)$ Exists $\forall\, z \in |z - z_0| < \delta$

Let $z_1 \in |z - z_0| < \delta$ and $|z_0 - z_1| = \delta_1, \delta_2 = \delta - \delta_1$

$\delta' = \min\{\delta_1, \delta_2\}$

$|z - z_1| < \delta' \subset |z - z_0| < \delta$

$\Rightarrow$ z_1 is regular point of f

4.2. Analytic Function

f is said to be analytic in D if it is regular at every point of D.

Notes:

(i) If f is analytic in $D \Rightarrow D$ is open

(ii) If it is said that f is analytic at z_0 it means f is analytic in an open disc around z_0. i.e. z_0 is a regular point.

(iii) If f is defined on D where D is open and connected then f is analytic on D iff f is differentiable at every point of D. .

(iv) If f is analytic on $D-\{a\}$ where $a \in D$ then we say f is not analytic on a, if $\lim\limits_{z\to a} f(z)$ either doesn't exist if exist then not equal to $f(a)$.

4.3. Singularity

Let f is defined on D. Then a limit point of D i.e. the limit point of the set of regular point is defined as singular point if it is not regular itself.

If $D =$ Domain of definition

$R =$ The set of regular points

$R' =$ The set of limit point of R

Then if $\alpha \in R' - R$

$\Rightarrow$ α is a limit point of R but $\alpha \notin R$ then α is a singular point.

$S = R' - R =$ the set of all the singular point

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Example: Let $f(z)=\log z$

then $D=\{z\in\mathbb{C}-\{x+iy:x\le 0, y=0\}\}$

$D=\mathbb{C}-\{x+iy:x\le 0, y=0\}$

$R=D$

$R'=\mathbb{C}$

$S=\{x+iy:x\le 0, y=0\}$

Result: Set of all singular points is a closed set i.e. $S'=S$.

Proof: Define $S'=$ The set of limit point of S

If $\alpha\in S' \Rightarrow \alpha\notin R$ (i)

$\Rightarrow |z-\alpha|<\delta\cap S$ is infinite for some $\delta>0$

$\Rightarrow |z-\alpha|<\delta\cap R'$ is infinite

$\Rightarrow |z-\alpha|<\delta\cap R$ is infinite

$\Rightarrow \alpha\notin R'$

$\Rightarrow \alpha\in S \Rightarrow S'\subseteq S$.

4.4. Classification of Singular Points

Positional Classification

I. Isolated Singularity: Let $z_0\in S$ i.e. a singular point is called an isolated singularity if $\exists\,\delta>0$ such that f is regular in $0<|z-z_0|<\delta$

i.e., $I=S=S'=$ The set of isolated points in S.

Examples: Let $\tan\frac{1}{z}=\frac{\sin\frac{1}{z}}{\cos\frac{1}{z}}$, then $f(z)$ have singularities at those points where $\cos\frac{1}{z}=0$ and at $z=0$.

$$i.e.\ S=\left\{\frac{1}{(2n+1)\pi/2},\ n\in\mathbb{Z}\right\}\cup\{0\}$$

$S'=\{0\}$

$\Rightarrow$ 0 is not an isolated singularity.

II. Non-isolated Singularity: The limit point of singularity is defined as non-isolated singularity.

So, α is non-isolated singularity iff $\alpha\in S'$.

i.e., α is non-isolated singularity iff it is not isolated singularity

Examples:

(i) $f(z)=\log z$

Every singularity of $\log z$ is non-isolated singularity

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(ii) $f(z)=\dfrac{1}{\sin(\pi/z)}, S=\left\{\dfrac{1}{n}, n\in\mathbb{Z}-\{0\}\right\}\cup\{0\}$

$S'=\{0\}$

$\Rightarrow \{0\}$ is non-isolated singularity, rest are isolated singularity.

(iii) $f(z)=\dfrac{1}{\sin \pi z}$

$S=\mathbb{Z},\ S'=\phi$

Notes:

(i) $|z|>R$ is a neighborhood of ∞ for any $R>0$.

(ii) In extended $\mathbb{C}, \infty$ is limit point of every unbounded set.

Examples: In extended $\mathbb{C}$, $\mathbb{Z}'=\{\infty\}$

$N'=\{\infty\}$

(iii) The behavior of f at ∞ is analyzed with help of that $g(z)$ at $z=0$ where $g(z)=f\left(\dfrac{1}{z}\right)$

Example: Let $f(z)=\sin z$

Then $f\left(\dfrac{1}{z}\right)=\sin\dfrac{1}{z}$ has isolated singularity at $z=0$

$\Rightarrow f(z)$ has isolated singularity at ∞.

(iv) If g is regular at 0 then f is regular at ∞ where $f\left(\dfrac{1}{z}\right)=g(z)$

(v) If set of singular point of f is unbounded then ∞ is non-isolated singularity of f.

Example: $f(z)=\dfrac{1}{\sin z-\cos z}$

It has infinite singularity which are at the points $n\pi+\dfrac{\pi}{4}$, $n\in\mathbb{Z}$ and ∞ is non isolated singularity.

4.5. Character Based Classification of Singularity

1. Removable Singularity: $z_0\in S$ is said to be removable singularity.

If $\lim\limits_{z\to z_0} f(z)$ exists finitely, where f is defined in $0<|z-z_0|<\delta$ for some $\delta>0$

Examples:

(i) $f(z)=\sin z \quad z\neq 0$

(ii) $f(z)=\dfrac{\sin z}{z}$

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(iii) $f(z)=\begin{cases}\dfrac{\sin z}{z} & , z\neq 0\\ 0 & , z=0\end{cases}$

$f(z)=z,\ |z|<1$

$D=|z|<1$

$R=D,\ R'=|z|\leq 1$

$S=|z|=1$

Note: Limit of Branch point is not defined.

Pole: Let $z_0\in S$ and $\lim\limits_{z\to z_0} f(z)$ does not exist, then z_0 is called a pole of order m. if $\lim\limits_{z\to z_0}(z-z_0)^m f(z)$ exists non-zero.

Example: $f(z)=\dfrac{1}{z}$ – order 1 pole

$f(z)=\dfrac{\sin z}{z^3}$ – order 2 pole

$f(z)=\dfrac{\cos z}{z^3}$ – pole order 3

Order one pole is called simple pole.

2. **Essential Singularity:** Which is neither the above two is called essential singularity.

Example: $\log z$ has essential singularity at the points on the negative x axis.

Example: $f(z)=\begin{cases}1 & |z|<1\\ 2 & 1\leq|z|<2\end{cases}$

$D=|z|=2$

So, $|z|=1$ is essential singularity and non-isolated.

$\left|(z-z_0)^m\cdot f(z)\right|\leq 2|z-z_0|^m<\varepsilon$

As $D=|z|<2$

$R=D-\{|z|=1\}$

$R'=|z|\leq 2$

Result: If $f(z)$ is defined in a neighborhood of z_0 except possibly at z_0.

Case (i): When z_0 is regular point. Then

(a) $f(z_0), f'(z_0)$ exist.

(b) $f'(z)$ exist $\forall\ z\in|z-z_0|<\delta$ for some $\delta<0$

(c) f is bounded in a neighborhood of z_0

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Case (ii): If z_0 is removable singularity

(a) By definition it is an isolated singularity.

(b) As limit exist in a neighborhood of z_0 function is bounded.

Case (iii): If z_0 is pole

(a) By definition it is an isolated singularity.

(b) $\lim_{z \to z_0} f(z) = \infty$ i.e., in a neighborhood of pole functions are unbounded

(c) If $z = z_0$ is a pole of order n of $f(z)$ then $(z-z_0)^n f(z)$ has a removable singularity at $z = z_0$ i.e. $\lim_{z \to z_0} (z-z_0)^m f(z)$ exists and non zero.

Result: If f & g are two analytic function defined on some domain D such that f has a zero of order n & g has a zero of order m at $z = z_0$ then

(i) $f(z) = (z-z_0)^n f_1(z)$ & $g(z) = (z-z_0)^m g_1(z)$ where

$f_1(z_0) \neq 0$ & $g_1(z_0) \neq 0$

(ii) Define $h(z) = \dfrac{f(z)}{g(z)}$ then

(a) $z = z_0$ is a removable singularity if $n \geq m$.

(b) If $n < m$ then at $z = z_0$ $h(z)$ has a pole of order m-n.

Consider $\dfrac{\pi}{z_n} = 2n\pi + \dfrac{\pi}{6}$

$$\Rightarrow z_n = \frac{1}{2n + \frac{1}{6}} \to 0$$

$f(z_n) \to \dfrac{1}{2}$ (Essential)

Case (iv): When z_0 is isolated essential singularity.

f is unbounded in a neighborhood of z_0. In this case

$A = \{\lim f(z) \text{ can vary} : z_n \to z_0\} = \mathbb{C}$

4.6. Entire Function

f is said to be entire if it is analytic on $\mathbb{C}$ i.e. regular on $\mathbb{C}$ i.e. $R = \mathbb{C}$

i.e. ∞ is the only possible singularity.

If f is entire then $e^{f(z)}$ is entire. In fact singularities of f and e^f are same positionally not character wise.

If z_0 is R.S. of $f \Rightarrow z_0$ is R.S. of e^f

Put Your Own Notes

If z_0 is pole or essential of $f \Rightarrow z_0$ is essential of e^f.

Example: Let $f(z)=\frac{1}{z}$ then $\frac{1}{z}$ has a pole of order 1.

$\Rightarrow$ $e^{1/z}$ has essential singularity at $z=0$

4.7. Results on Analyticity

Let f be an analytic function defined on a domain D (open and connected) then

1. $f'(z)=0 \ \forall z \in D$ then f is constant.
2. $f(z)=u+iv$, If any of u and v is constant then f is constant.

   Let $u(x,y)=k$

   $u_x=0 \;\&\; u_y=0$

   Since $u_x=v_y=0 \;\&\; u_y=-v_x=0$

   $\Rightarrow f'(z)=0 \Rightarrow f(z)=k, k\in\mathbb{C}$
3. If $|f|$ is constant then f is constant.

   $|f|=c \Rightarrow u^2+v^2=c$

   If $c=0$ then done

   If $c\neq 0 \Rightarrow 2uu_x+2vv_x=0$

   & $\quad uu_y+vv_y=0$

   i.e. $vv_x+uv_y=0$

   $-uv_x+vv_y=0$

   $$\Rightarrow \begin{bmatrix} v & u \\ -u & v \end{bmatrix}\begin{bmatrix} v_x \\ v_y \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

   $Ab=0$ where $A=\begin{bmatrix} v & u \\ -u & v \end{bmatrix} \& b=\begin{bmatrix} v_x \\ v_y \end{bmatrix}$

   Since $|A|=u^2+v^2\neq 0 \Rightarrow$ system of equation has unique solution

   $\Rightarrow v_x=v_y=0$ i.e. $v(x,y)$ is a constant function and hence f is constant.

Case (iv): If $\arg f=$constant then f is constant.

Let $\arg f=\tan^{-1}\frac{v}{u}=k \quad \Rightarrow \frac{v}{u}=\tan k=\lambda$

$\lambda u-v=0$

$\Rightarrow \lambda u_x - v_x=0 \quad \Rightarrow \lambda u_x + u_y=0$

& $\lambda u_y - v_y=0 \quad \Rightarrow -u_x+\lambda u_y=0$

$\Rightarrow \begin{bmatrix} \lambda & 1 \\ -1 & \lambda \end{bmatrix}\begin{bmatrix} u_x \\ u_y \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ Which is system of equation and have trivial solution i.e. $u(x,y)$ constant and hence function $f(z)$ is constant.

Put Your Own Notes

4. If $au+bv=$ constant then function is constant.

   Let $f(z)=u+iv$

   & $au+bv=c$

   $$g(z)=(a+ib)f(z)=(a-ib)(u+iv)$$

   $=(au+bv+i(av-bu))$

   $=U+iV$, since $U=au+bv$ is constant

   $\Rightarrow$ $g(z)$ is constant and hence $f(z)$ is constant.

5. If u and v lie on circle then function is constant.
6. If u and v are harmonic conjugate of each other then f is constant i.e. $f=u+iv$ is constant.
7. If $f=u+iv$ is analytic then u and v are harmonic functions of x and y

   i.e., u and v satisfies Laplacian equation

   i.e., $\Delta^2 u=0$ i.e. $u_{xx}+u_{yy}=0$

   $\Delta^2 v=0$ i.e. $v_{xx}+v_{yy}=0$

   Example: Let $f(z)=x^2+y^2+2xyi=u+iv$

   Since u is not harmonic $\Rightarrow$ Not analytic

   $f(z)=2xy+(x^2-y^2)i=u+iv$

   Then $u_{xx}+u_{yy}=0$ but f is not analytic

8. If $f(z)=u+iv$ is analytic then $u(x,y)=\alpha$, $v(x,y)=\beta$ represents orthogonal family of curves with α and β aress parameter.

4.8. Construction of Analytic Function

Method 1: Milne's Thomson's method.

We have $z=x+iy$ so that $x=\dfrac{z+\overline{z}}{2}$, $y=\dfrac{z-\overline{z}}{2i}$.

$w=f(z)=u+iv=u(x,y)+iv(x,y)$

or $f(z)=u\left(\dfrac{z+\overline{z}}{2},\dfrac{z-\overline{z}}{2i}\right)+iv\left(\dfrac{z+\overline{z}}{2},\dfrac{z-\overline{z}}{2i}\right)$

In fact, this relation is formal identity in two independent variables z and $\overline{z}$.

By setting $x=z$, $y=0$ so that $z=\overline{z}$, we obtain

$$f(z)=u(z,0)+iv(z,0) \quad \ldots(1)$$

We know that

$f'(z)=\dfrac{\partial w}{\partial z}=\dfrac{\partial u}{\partial x}i\dfrac{\partial u}{\partial x}=ux-iuy$

$=\dfrac{\partial w}{\partial x}$, by Cauchy Riemann equations.

Put Your Own Notes

Taking $\frac{\partial u}{\partial x}=\phi_1(x,y)=\phi_1(z,0)$

$\frac{\partial u}{\partial y}=\phi_2(x,y)=\phi_2(z,0)$

We get $f(z)=\phi_1(z,0)-i\phi_2(z,0)$

Integration yields the result.

$f(z)=\int[\phi_1(z,0)-i\phi_2(z,0)]dz+c,$

Where c is a constant. We can calculate $f(z)$ directly if u is known. Similarly if $v(x,y)$ is given, then it can be proved that

$f(z)=\int[\phi_1(z,0)-i\phi_2(z,0)]dz+c,$

Where $\Psi_1=\frac{\partial v}{\partial x},\psi_2=\frac{\partial v}{\partial x}$

Formula for obtaining $f(z)$ when real part of $f(z)$ is given

Let $f(z)=u(x,y)+iv(x,y)$ ...(*i*)

$\overline{f(z)}=u(x,y)-iv(x,y)$

$f(\overline{z})=u(x,y)-iv(x,y)$ ...(*ii*)

Putting $x=0,\ y=0$ in above equation

We get $\overline{f}(0)=u(0,0)-ic$ ...(*iii*)

$2u(x,y)=f(z)+f(\overline{z})$ after addition

$2u(x,y)=f(x+iy)+\overline{f}(x-iy)$ ...(*iv*)

Replacing $x=\frac{z}{2},\ y=\frac{z}{2i}$ then the equation (*iv*) becomes

$x=\left(\frac{z}{2},\frac{z}{2i}\right)=f(z)+\overline{f}(0)$

Substituting the value of $\overline{f}(0)$ from equation (*iii*), we get

$f(z)=2u\left(\frac{z}{2},\frac{z}{2i}\right)-\overline{f}(0)$

$f(z)=2u\left(\frac{z}{2},\frac{z}{2i}\right)-u(0,0)+ic$

Note: Function should be defined on $(0,0)$

Formula for finding $f(z)$ when v is given

Subtracting equation number (*ii*) from equation number (*i*)

We get $f(z)-f(\overline{z})=2iv(x,y)$

$$2iv(x,y) = f(x+iy) - \bar{f}(x,iy) \qquad \ldots(v)$$

Replacing $x = \frac{z}{2},\ y = \frac{z}{2i}$, in equation number (v), we get

$$2iv\left(\frac{z}{2},\frac{z}{2i}\right) = f\left(\frac{z}{2}+\frac{z}{2}\right) - \bar{f}(0)$$

$$2iv\left(\frac{z}{2},\frac{z}{2i}\right) = f(z) - \bar{f}(0)$$

$$f(z) = 2iv\left(\frac{z}{2},\frac{z}{2i}\right) + \bar{f}(0)$$

We know from

$$f(\bar{z}) = u(x,y) - iv(x,y)$$

$$\bar{f}(0) = c - iv(0,0)$$

$$\therefore f(z) = 2iv\left(\frac{z}{2},\frac{z}{2i}\right) + c - iv(0,0)$$

$$f(z) = 2iv\left(\frac{z}{2},\frac{z}{2i}\right) - iv(0,0) + c$$

Example 1: Find the analytic function $f(z) = u + iv$ of which the real part $u = e^x(x\cos y - y\sin y)$.

OR

Construct a function $f(z)$ which has a real function $u(x,y) = e^x(x\cos y - y\sin y)$ for its real part, satisfying Laplace's equation.

Solution: $\frac{\partial u}{\partial x} = e^x(x\cos y - y\sin y) + e^x\cos y$

$$\frac{\partial u}{\partial y} = e^x[-x\sin y - \sin y - y\cos y]$$

$$\left(\frac{\partial u}{\partial y}\right)_{y=0} = e^x x + e^x = e^x(x+1)$$

$$\left(\frac{\partial u}{\partial y}\right)_{y=0} = e^x.0 = 0$$

$$\phi_1(x,0) = \left(\frac{\partial u}{\partial x}\right)_{y=0} = e^x(x+1)$$

$$\phi_2(x,y) = \left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$$

By Milne's Thomson's method,

$$f(z) = \int\left[\phi_1(z,0) - i\phi_2(z,0)\right]dz + c$$

Put Your Own Notes

$f(z)=\int\left[e^{z}(z+1)-i.0\right]dz+c=\int\left(ze^{z}+e^{z}\right)dz+c$

$=(z-1)e^{z}+e^{z}+c=ze^{z}+c$

$\therefore\ f(z)=ze^{z}+c$

Method 2: Suppose $f(z)=u+iv$ is analytic and u is known. To determine $f(z)$.

Firstly we shall determine v.

$$dv=\frac{\partial v}{\partial x}dx+\frac{\partial v}{\partial y}dy=\left(-\frac{\partial u}{\partial y}\right)dx+\left(\frac{\partial u}{\partial x}\right)dy,$$

by Cauchy – Riemann equations.

Taking $M=-\left(\frac{\partial u}{\partial y}\right),\ N=\frac{\partial u}{\partial x}$, we get

$$dv=M\,dx+N\,dy \qquad \ldots(1)$$

$$\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}=-\frac{\partial^2 u}{\partial y^2}-\frac{\partial^2 u}{\partial x^2}=-\nabla^2 u=0.$$

(For u satisfies Laplace's equation)

or $\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}=0$ or $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$

Consequently (1) is exact differential equation.

So equation (1) can be integrated and v can be determined .Now u and v are known and hence $f(z)$ can be determined form the equation $f(z)=u+iv$. We illustrate this method by an example, given below:

Example 2: Find the analytic function of which the real part is $u=e^{x}(x\cos y-y\sin y)$

Solution: $dv=\frac{\partial v}{\partial x}dx+\frac{\partial v}{\partial y}dy=-\frac{\partial u}{\partial y}dx+\frac{\partial v}{\partial y}$

$=-e^{x}(-x\sin y-y\cos y-\sin y)dx+e^{x}(x\cos y-y\sin y+\cos y)dy$

Integrating, $v=\int e^{x}(x\sin y+y\cos y+\sin y)dx$ (treating y as constant).

$+\int$(those terms which do not contain x)$dy+c$

$=\sin y.\int xe^{x}dx+(y\cos y+\sin y)\int e^{x}dx+\int 0dy+c$

$=\left[(x-1)\sin y+y\cos y+\sin y\right]e^{x}+c$

$v=(x\sin y+y\cos y)e^{x}+c$

$f(z)=u+iv$

$=e^{x}(x\cos y-y\sin y)+i\left[e^{x}(x\sin y+y\cos y)+c\right]$

$=xe^{x}(\cos y+i\sin y)+iye^{x}(\cos y+i\sin y)+ic$

$=(x+iy)e^{x}.e^{iy}+c'=ze^{z}+c'.$

Put Your Own Notes

CHAPTER-5

COMPLEX INTEGRATION AND COUNTABILITY

5.1. Curve

Let $[a,b] \subseteq R$ & $x = x(t), y = y(t)$ are continuous function on $[a,b]$ then $\gamma(t) = x(t) + iy(t),\ t \in [a,b]$ is defined as a curve in complex plane

Note:

(i) This represents the same curve obtained by eliminating t between $x(t)$ and $y(t)$ to get $\phi(x,y) = 0$ in the Cartesian plane i.e. when Cartesian plane is visualized as argand plane $\phi(x,y) = 0$ represents $\gamma(t)$ and every order pair (a,b) on the curve gives a complex number on the curve γ as $a + ib$.

(ii) If $P = \{t_0, t_1 \ldots t_n = b\}$ be a partition of a closed interval $[a,b]$. Then $\gamma(t_r) = x(t_r) + iy(t_r) = z_r$ (say) is a complex number in $\mathbb{C}$.

(iii) $z_0 = \gamma(a),\ z_n = \gamma(b)$ are called initial and terminal point.

5.2. Definitions

1. **Closed Curve:** When $\gamma(a) = \gamma(b)$ then γ is defined as closed curve.

   Example: Let $\gamma(t) = e^{it}$ & $\gamma : [0, 2\pi] \to \mathbb{C}$ then $\gamma(0) = \gamma(2\pi)$ & hence γ is closed on $[0, 2\pi]$

2. **Simple Curve:** If γ is one-one on open interval (a,b) is called simple curve.
   i.e. it doesn't intersect itself.
   If $t_1 < t_2$ then $\gamma(t_1) \neq \gamma(t_2)\ \ \forall\ t_1, t_2 \in (a,b)$

3. **Jordan Curve:** Simple closed curve is Jordan curve.

4. **Reversal of the Curve:** If γ traces a curve from position P to Q i.e. P is initial point & Q is terminal point then the same curve when traversed from Q to P is called reversal of γ and denoted by $-\gamma$
   $(-\gamma)(t) = \gamma(a + b - t)\ \forall\, t \in [a,b]$ if $\gamma : [a,b] \to \mathbb{C}$.

   Example:

   Let $\gamma(t) = e^{it},\ \ t \in [0, \pi]$

   $= \cos t + i \sin t$ then reversal of γ i.e., $-\gamma(t)$ is

   $(-\gamma)(t) = \gamma(\pi + 0 - t) = e^{i(\pi - t)}$

   $= e^{i\pi} e^{-it}$

   $= -e^{-it}$

   $\Rightarrow (-\gamma)(t) = -(\cos t - i \sin t)$

Put Your Own Notes

5. **Smooth Curve:** If γ is differentiable with non vanishing derivative then called smooth curve.

Note: γ is smooth if $x(t)$ and $y(t)$ are differentiable and $x'(t), y'(t)$ does not vanish together.

Since $\gamma'(t) = x'(t) + iy'(t)$ and $\phi(x,y) = \phi(x(t), y(t))$

$$\Rightarrow \frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \tan\alpha \Rightarrow \alpha = \tan^{-1}\frac{y'(t)}{x'(t)}$$

$= \text{Arg } \gamma'(t)$

i.e., Arg $\gamma'(t)$ should defined

i.e., curve should not pass through origin.

5.3. Definitions

Suppose f is a complex-valued function that is continuous on an open set $D \subseteq \mathbb{C}$ and that $\gamma : [a,b] \to \mathbb{C}$ is a contour with $\gamma([a,b]) \subset D$. We define the complex line integral or contour integral of f along the contour γ, denoted by $\int_\gamma f(z)\,dz$, as follows

$$\int_\gamma f(z)\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt$$

1. Let γ be a smooth curve defined on $[a,b]$ and let f and g be continuous functions on an open set D containing $\gamma([a,b])$ and let α be a complex constant. Then

(i) $\int_\gamma f(z)\,dz = -\int_{-\gamma} f(z)\,dz$.

(ii) $\int_\gamma [\alpha f(z) + g(z)]\,dz = \alpha \int_\gamma f(z)\,dz + \int_\gamma g(z)\,dz$.

(iii) If $L = L(\gamma)$ is the length of the curve and $M = \max\limits_{t\in[a,b]} |f(\gamma(t))|$, then

$\left|\int_\gamma f(z)\,dz\right| \le ML$.

This property is called the standard estimate for integrals, or M - L inequality.

(iv) $\int_{\gamma_1+\gamma_2} f(z)\,dz = \int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz$ Whenever γ_1 and γ_2 are two smooth curves such that $\gamma_1(b) = \gamma_2(a)$ and $\gamma_j([a,b]) \subset D$ for $j = 1,2$.

2. If $f = u + iv$ is analytic in an open set D containing a contour γ with parametric interval $[a,b]$ i.e. $\gamma([a,b]) \subset D$, then

$\int_\gamma f'(z)\,dz = f(\gamma(b)) - f(\gamma(a))$.

Put Your Own Notes

3. **Cauchy Weak theorem:** If f is analytic with f' continuous inside and on a simple closed contour γ, then $\int_\gamma f(z)\,dz = 0$.

Winding number or index of a curve

Suppose that γ is a closed contour in $\mathbb{C}$. Let a be a given point in $\frac{\mathbb{C}}{\{\gamma\}}$.

Then, there is a useful formula that measures how often γ winds around.

For example if $\gamma : \gamma(t) = \{z : z - a = re^{it}, 0 \le 2\pi\}$ then γ encircles the point a k times (counterclockwise.) Further,

$$\int_\gamma \frac{dz}{z-a} = \int_0^{2k\pi} \frac{ire^{it}}{re^{it}} dt = 2k\pi i, \text{ i. e. } \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a} = k.$$

From this we also observe that if γ encircles the point a k-times in the clockwise direction, then

$$\frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a} = -k.$$

In either case $\frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a}$ is an integer. Here is the analytic deflection of the winding number of a, which captures the intuitive notion of "the number of times γ wraps around a in the counter clockwise direction"

Definition: Let γ be a closed contour in $\mathbb{C}$ that avoids a point $a \in \mathbb{C}$.. The index (or winding number) of γ about a, denoted by $n(\gamma; a)$ or Ind $(\gamma;a)$ is given by the integral

$$n(\gamma;a) = \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a}$$

Actually, from our later discussion, Cauchy's theorem will imply that $n(\gamma;a) = n(\gamma_0;a)$ for all closed curves γ_0 that are homotopic to γ as closed curves in $\frac{\mathbb{C}}{\{a\}}$.

In the following we will collect some properties of the index $n(\gamma;\alpha)$

4. For every closed contour γ in $\mathbb{C}$ and $a \in \mathbb{C} - \{\gamma\}, n(\gamma;a)$ is an integer.
5. If γ is a closed contour in $\mathbb{C}$, then the mapping $\zeta \mapsto n(\gamma;\zeta)$ is a continuous function of ζ at any point $\zeta \notin \gamma$.
6. The function $n(\gamma;\zeta), \zeta \in \mathbb{C} - \{\gamma\}$, is constant in the components of $\mathbb{C} - \{\gamma\}$.
7. We have $n(\gamma;\zeta) = 0$ in the unbounded component of the closed contour γ.
8. If γ consists of finitely many closed contours $\gamma_1, \gamma_2, ..., \gamma_k$ in $\mathbb{C}$, then for every $a \notin \gamma_k$ (i.e., not on any one of the γ_j),

   (i) $n(\gamma;a) = n(\gamma_1;a) + n(\gamma_2;a) + ... + n(\gamma_k;a)$

   $n(-\gamma;a) = -n(\gamma;a)$

Put Your Own Notes

5.4. Cauchy Goursat Theorem

Let f be analytic in an open set $D \subset \mathbb{C}$ and let z_1, z_2, z_3 be in D. Assume that the closed triangle T with vertices z_1, z_2, z_3 is contained in D. Then, $\int_{\partial T} f(z)\,dz = 0$.

5.5. Cauchy's Theorem

If a function $f(z)$ is analytic and continuous inside and on a simple closed contour C, then

$$\int_C f(z)\,dz = 0$$

5.6. Cauchy's Integral Formula

If $f(z)$ is analytic within and on a closed contour C and if a is any point within C, then

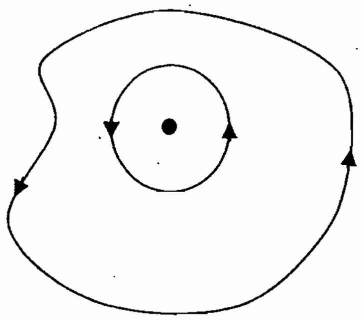
$$f(a) = \frac{1}{2\pi i}\int_C \frac{f(z)\,dz}{z-a}$$

Proof: Suppose $f(z)$ is analytic within and on a closed contour C and a is an interior point of C.

To prove that $f(a) = \frac{1}{2\pi i}\int \frac{f(z)\,dz}{z-a}$,,

Describe a circle γ about the centre $z = a$ of small radius r such that This circle $|z-a| = r$ does not intersect the curve C.

The function $\frac{f(z)}{z-a}$ is analytic in the annulus bounded by C



and γ. Hence by, corollary to Cauchy's theorem.

$$\int_C \frac{f(z)\,dz}{z-a} = \int_\gamma \frac{f(z)\,dz}{z-a} \qquad \ldots(1)$$

$$\text{Or} \int_C \frac{f(z)\,dz}{z-a} = \int_\gamma \frac{f(z)-f(a)}{z-a}\,dz + \int_\gamma \frac{f(a)\,dz}{z-a} \qquad \ldots(2)$$

Since $f(z)$ is analytic within C and so it is continuous at $z = a$ so that given $\varepsilon > 0$, there exists $\delta > 0$ such that. $|f(z) - f(a)| < \varepsilon$..........(3) for $|z-a| < \delta$..........(4). Since r is at our choice and so we can take $r < \delta$ so that (4). Since r is at our choice and so we can take $r < \delta$ so that (4) is satisfied $\forall z$ on the circle γ. For any point z on γ, $z - a = re^{i\theta}$.

Put Your Own Notes

$$\int_{\gamma}\frac{f(a)}{z-a}dz=\int_{0}^{2\pi}\frac{f(a)re^{i\theta}id\theta}{re^{i\theta}}=2\pi i\ f(a)$$

Hence, by (2), $\left|\int_C \frac{f(z)dz}{z-a}-2\pi i\ f(a)\right|=\left|\int_{\gamma}\frac{f(z)-f(a)}{z-a}\right|\le$

$$\int_{\gamma}\frac{|f(z)-f(a)|}{|z-a|}.|dz|<\frac{\varepsilon}{r}\int_{\gamma}|dz|=\frac{\varepsilon}{r}.2\pi r$$

Or $\int_{\gamma}\left|\frac{f(z)dz}{z-a}-2\pi i f(a)\right|<2\pi\in$

Since ε is arbitrary and so making $\varepsilon\to 0$, we get

$$\int_C\frac{f(z)dz}{z-a}-2\pi i f(a)\quad \text{Or}\quad f(a)=\frac{1}{2\pi i}\int_C\frac{f(z)dz}{z-a} \qquad ...(5)$$

5.7. Theorem (Extension of Cauchy's Integral Formula to Multiply Connected Regions)

If $f(z)$ is analytic in a ring shaped region bounded by two closed curves C_1 and C_2 and a is point in the region between C_1 and C_2.

$$f(a)=\frac{1}{2\pi i}\int_{C_2}\frac{f(z)}{z-a}dz-\frac{1}{2\pi i}\int_{C_1}\frac{f(z)}{z-a}dz$$

Where C_2 is outer curve.

5.8. Theorem (Cauchy Integral Formula for The Derivative of An Analytic Function)

If a function $f(z)$ is analytic within and on a closed contour C and a is any point lying in it, then

$$f'(a)=\frac{1}{2\pi i}\int_C\frac{f(z)}{(z-a)^2}dz$$

5.9. Theorem (Higher Order Derivatives)

If a function $f(z)$ is analytic within an on a closed contour C and a is any point within C then derivatives of all orders are analytic and are given by

$$f^{(n)}(a)=\frac{n!}{2\pi i}\int_C\frac{f(z)}{(z-a)^{n+1}}$$

Example 1: Based on Cauchy Integral Formula

(a) $\int_{|z|=1}\frac{\sin z}{z}dz=2\pi i(\sin z)\Big|_{z=0}=0$

(b) $\int_{|z|=1}\frac{\cos z}{z(z-4)}dz=2\pi i\frac{\cos}{z-4}\Big|_{z=0}=-\frac{\pi}{2}i$

(c) $\int_{|z-a|=1}\frac{e^{2\pi z}}{z-a}dz=2\pi i\,e^{2\pi a}$

Put Your Own Notes

(d) $\int_{|z|=2} \frac{e^z+z^2}{z-1}dz = 2\pi i[e+1]$

(e) If $f(z)=\sin(\pi z^2)+\cos(\pi z^2)$, then by Cauchy's deformation of contour

$$\int_{|z|=4} \frac{f(z)}{(z-2)(z-3)}dz \int |z-2| = \frac{1}{2}\frac{\frac{f(z)}{(z-3)}}{z-2}dz + \int_{|z-3|=\frac{1}{2}} \frac{\frac{f(z)}{(z-2)}}{z-3}dz$$

$=2\pi i(-f(2)+f(3))=0$

(f) By Cauchy's deformation of contour

$$\int_{|z|=4} \frac{e^z}{z(z-1)}dz = \int_{|z|=\frac{1}{2}} \frac{\frac{e^z}{(z-1)}}{z}dz + \int_{|z-1|=\frac{1}{2}} \frac{\frac{e^z}{z}}{z-1}dz = 2\pi i(-1+e).$$

(g) $\int_{|z|=1} \frac{\cos(e^z)}{z}dz = 2\pi i \cos 1 = \pi i(e^i + e^{-1}).$

Example 2: Evaluate $\int_C \frac{e^{2z}dz}{(z+1)^4}$ where c is $|z|=3$.

Solution: We know that $f^{(n)}(a) = \frac{n!}{2\pi i}\int_C \frac{f(z)dz}{(z-a)^{n+1}}$

Put $a=-1, n=3$

$$f^{(3)}(-1) = \frac{3!}{2\pi i}\int_C \frac{f(z)dz}{(z-1)^4} \quad \ldots(1)$$

Takes $f(z)=e^{2z}$, then $f^{(n)}(z)=2^n e^{2z}$

$\therefore f^{(3)}(-1) = 2^3 e^{-2} = \frac{8}{e^2}$ Now by (1)

$$\frac{8}{e^2} = \frac{3!}{2\pi i}\int_C \frac{e^{2z}dz}{(z+1)^4} \text{ or } \frac{8\pi i}{3e^2} = \int_C \frac{e^{2z}dz}{(z+1)^4}.$$

5.10. Theorem (Poisson's Integral Formula)

It $f(z)$ is analytic within and on a circle C defined by $|z|=R$ and if a is any point within C, then

$$f(a) = \frac{1}{2\pi i}\int_C \frac{(R^2 - a\bar{a})f(z)dz}{(z-a)(R^2 - z\bar{a})}$$

Hence deduce the Poisson's formula

$$f(re^{i\theta}) = \frac{1}{2\pi}\int_0^{\pi} \frac{(R^2-r^2)f(\text{Re}^{i\theta})d\phi}{R^2 - 2Rr\cos(\theta-\phi)+r^2}$$

where $a = re^{i\theta}$ is any point inside the circle $|z|=R$.

Put Your Own Notes

5.11. Morera's Theorem

If $f(z)$ is a continuous function in a domain D and if for every closed contour C in the domain D, $\int_C f(z)dz = 0$

Then $f(z)$ is analytic within D. (It is a sort of converse of Cauchy's theorem).

Proof: Let z_0 be a fixed point and z a variable point inside the domain D.

The value of the integral $\int_{z_0}^{z} f(t)dt$ is independent of the curve joining z_0 to z and depends on z only.

Write $F(z) = \int_{z_0}^{z} f(t)dt$

Let $z+h$ be a point in the neighbourhood of z.

$$f(z+h) - f(z) = \int_{z_0}^{z+h} f(t)dt - \int_{z_0}^{z} f(t)dt$$

$$= \int_{z_0}^{z+h} f(t)dt + \int_{z_0}^{z} f(t)dt = \int_{z}^{z+h} f(t)dt$$

$$\left|\frac{F(z+h)-F(z)}{h} - f(z)\right| = \left|\frac{1}{h}\int_{z}^{z+h} f(t)dt - f(z)\right| = \frac{1}{|h|}\left|\int_{z}^{z+h} \left[f(t)-f(z)\right]dt\right|$$

$$\leq \frac{1}{|h|}\int_{z}^{z+h} |f(t)-f(z)||dt| < \frac{\varepsilon}{|h|}|h|$$

$$\left[|f(t)-f(z)| < \varepsilon \; for \; |t-z| < \delta \; because \; of \; continuity \; of \; f(z)\right]$$

or $\left|\frac{F(z+h)-F(z)}{h} - f(z)\right| < \in$ which tends to 0 as $e \to 0$

Thus, $\lim_{h\to 0}\frac{F(z+h)-F(z)}{h} - f(z) = 0,$, or $F'(z) = f(z)$

Thus the derivative of $F(z)$ exists and so $F(z)$ is analytic in D. But we know that the derivative of analytic function is analytic.

Therefore $F'(z).i.e. f(z)$ is analytic in D.

Remark: Suppose that

$$f(z) = \begin{cases} \dfrac{1}{(z-1)^2} \; if \; z \in \Delta(1;r)\setminus\{1\} \\ 1 \; if \; z = 1 \end{cases}, \quad g(z) = \begin{cases} \dfrac{\cos}{z^2}.if \; z \in \Delta\setminus\{0\} \\ 0 \; if \; z = 0 \end{cases}$$

Then, for any simple closed contour in $\Delta(1;r)$ we have $\int_C f(z)dz = 0$.

Put Your Own Notes

However f is not analytic for $z \in \Delta(1;r)$ since f is not even continuous at $z=1$. Note that Morera's Theorem is not applicable since the continuity requirement is not satisfied.

Similarly for the $g(z)$ defined above we known that $\int_{|z|=1} g(z)dz = 0$. Again g is not analytic in Δ, since g is not continuous at $z=0$

5.12. Theorem

A necessary and sufficient condition for a function $f(z)$ to possess an indefinite integral in a simply connected domain D is that the function is analytic in D. Further, any two indefinite integrals differ by a constant.

5.13. Theorem (Cauchy's Inequality)

If $f(z)$ analytic within and on a circle C, given by $|z-a|=R$ and

If $|f(z)| \le M$ for every z on C, then $\left|f^{(n)}(a)\right| \le \frac{Mn!}{R^n}$

Proof: $|z-a| = R \Rightarrow z-a = \text{Re}^{i\theta} \Rightarrow dz = i\text{Re}^{i\theta}d\theta \Rightarrow |dz| = R\,d\theta$

We know that $f^{(n)}(a) = \frac{n!}{2\pi i}\int_C \frac{f(z)dz}{(z-a)^{n+1}}$

$$\left|f^{(n)}(a)\right| \le \frac{n!}{2\pi i}\int_C \frac{|f(z)|.|dz|}{|z-a|^{n+1}} \le \frac{Mn!}{2\pi R^{n+1}}\int_0^{2\pi} R\,d\theta = \frac{Mn!}{2\pi R^{n+1}}2\pi R$$

or $\left|f^{(n)}(a) \le \frac{Mn!}{R^n}\right|$

Remark: If take $a_n = \frac{f^{(n)}(a)}{n!}$ then $|a_n| \le \frac{M}{R^n}$

Put Your Own Notes

CHAPTER 6

SOME IMPORTANT THEOREMS AND ITS APPLICATIONS

6.1. Liouville's Theorem

A bounded & entire function is constant.

First Proof of Lowville's Theorem:

Since f is bounded on $\mathbb{C}$, there is a finite M such that $|f(z)| \leq M$ for $z \in \mathbb{C}$. Let a be an arbitrary complex number and let $\Gamma = \{z : z = \mathrm{Re}^{i\theta}, 0 \leq \theta < 2\pi\}$, where $|a| < R < \infty$. Then, according to the Cauchy integral formula, we have

$$f(a)-f(0)=\frac{1}{2\pi i}\int_{\Gamma}\left\{\frac{1}{z-a}-\frac{1}{z}\right\}f(z)dz=\frac{a}{2\pi i}\int_{\Gamma}\frac{f(z)}{z(z-a)}dz$$

so that for each fixed a, we have

$$|f(a)-f(0)| \leq \frac{|a|}{2\pi}\left\{\max_{|z|=R}\left|\frac{f(z)}{z(z-a)}\right|\right\}2\pi R \leq \frac{M|a|}{R-|a|}.$$

Which approaches zero as $R \to \infty$. Thus, $f(a) = f(0)$ for each $a \in \mathbb{C}$ and hence, f is constant.

Second Proof of Liouvill's Theorem

By hypothesis, there exists a finite $M > 0$ such that $|f(z)| \leq M$ for $|z| < R$ and for any $R > 0$. Equivalently, $|f(Rz)| \leq M$ for $|z| < 1$. In particular, $|f(0)| \leq M$. Set

$$g(z) = \frac{f(Rz) - f(0)}{2M}, \; |z| < 1.$$

Then g satisfies the hypotheses of Schwarz' lemma for each $R > 0$, since $g(0) = 0$ and

$$|g(z)| \leq \frac{|f(Rz)| + |f(0)|}{2M} \leq \frac{M + M}{2M} = 1.$$

Hence, we have $|g(z)| \leq |z|$ for $|z| < 1$. In other words,

$$|f(Rz) - f(0)| \leq 2M|z| \text{ for } |z| < 1,$$

or equivalently,

$$|f(z) - f(0)| \leq \frac{2M}{R}|z| \text{ for } |z| < R.$$

Put Your Own Notes

Again remember that M is a fixed constant, whereas R is at our disposal, and can be chosen as large as we please. Thus, restricting z in the unit disk $|z|<1$ and letting R tend to infinity in the inequality, we find that $f(z)=f(0)$ for $|z|<1$, which, by the Uniqueness theorem, gives $f(z)=f(0)$ for all $z\in\mathbb{C}$) and so f is constant.

Proof: Let $f(z)$ be an entire function such that $|f(z)|\le M \ \ \forall z\in\mathbb{C}$.

Then by Cauchy in equality we have

$$\left|f^n(z)\right|\le\frac{n!M}{R^n} \quad \forall z\in|z-z_0|\le R$$

taking $n=1$,

$$|f'(z)|\le\frac{M}{R}$$

Since $f(z)$ is entire, so we can take R as large as possible i.e., $R\to\infty$.

$$\Rightarrow |f'(z)|\le\frac{M}{R}\to 0 \text{ as } R\to\infty$$

i.e. $|f'(z)|=0 \quad \forall z\in\mathbb{C}$

$\Rightarrow f(z)$ is constant

Note: Non constant entire functions must be unbounded.

Example: If f is analytic & $f:\mathbb{C}\to\{z:|z|<1\}$ & $f(1+10i)=\dfrac{1}{10i}$ then find the value of $f(1-10i)$ is?

Solution: Since f is analytic on $\mathbb{C}\Rightarrow 1$ is an entire function also we have

$$\{f(z):z\in\mathbb{C}\}\subseteq\{z:|z|<1\}$$

$\Rightarrow f$ is a bounded function so by Liouvilles theorem f is a constant function

$$\Rightarrow f(z)=\frac{1}{10i} \quad \forall z\in\mathbb{C}$$

$$\Rightarrow f(1-10i)=\frac{1}{10i}$$

Results on Liouvilles Theorem:

1. If f is entire & if $\exists$ an $M>0$ with $|f(z)|>M \quad \forall z\in\mathbb{C}$, then f is constant.

   Solution: Since $M>0 \Rightarrow f(z)$ does not have any zero in $\mathbb{C}$.

   Let $\quad g(z)=\dfrac{1}{f(z)} \Rightarrow |g(z)|=\left|\dfrac{1}{f(z)}\right|\le\dfrac{1}{M}$

   $\Rightarrow g(z)$ is entire & bounded & hence $g(z)$ has to be constant $\Rightarrow$ $f(z)$ is also constant.

Put Your Own Notes

2. If $f(z)=u+iv$ is an entire function & either of the u & v are bounded then f is constant.

Solution:

Case (i) Let $f(z)=u+iv$ & $M>0$ such that $u \le M$. Let us construct a function $g(z)$, such that

$$g(z)=e^{f(z)}=e^{u+iv} \quad \forall\, z \in \mathbb{C}$$

$$\Rightarrow |g(z)|=\left|e^{f(z)}\right|=e^{u}<e^{M} \quad \forall\, z \in \mathbb{C}$$

Since $g(z)$ is entire & bounded $\Rightarrow$ $g(z)$ is a constant function & so $f(z)$ is also constant.

Case (ii) Let $v \le M$ again constant a function $\phi(z)$ such that

$$\phi(z)=e^{-if(z)} \quad \Rightarrow |\phi(z)|=\left|e^{-if(z)}\right|=e^{v}\le e^{M}$$

Now, since $\phi(z)$ is entire & bounded so $\phi(z)$ is constant & hence $f(z)$ is constant.

Corollary:

(i) If $f(z)=u+iv$ is entire function & $u(x,y)\ge M$ for some $M>0$ then f is constant.

(ii) If $f(z)=u+iv$ is entire function & $v(x,y)\ge M$ for some $M>0$ then f is constant.

3. Let $f(z)=u+iv$ is entire function such that, either $au+bv\le C$ or $au+bv\ge C$ then f is constant.

Solution: Let $f(z)=u+iv$ is an entire function.

Case (i): When $au+bv\le c$, construct a function $g(z)$ such that

$$g(z)=e^{(a-ib)(u+iv)}=e^{(a-ib)f(z)}$$

$\Rightarrow |g(z)|=e^{au+bv}\le e^{c}$ $\Rightarrow$ $g(z)$ is a bounded entire function so $g(z)$ is constant & hence $f(z)$ is constant.

Case (ii): When $au+bv\ge c$, construct a function $\phi(z)$ such that

$$\phi(z)=e^{-(a-ib)(u+iv)}=e^{-(au+bv+i(av-u))}$$

$$\Rightarrow |\phi(z)|=\left|e^{-(a-ib)(u+iv)}\right|=\left|e^{-(au+bv)}\right|\le \frac{1}{e^{c}}$$

$\Rightarrow$ $\phi(z)$ is constant & hence $f(z)$ is constant.

Put Your Own Notes

Lemma: Let f is an entire function & D is a bounded region of $\mathbb{C}$, then $f(D)$ is bounded.

Proof: Let if possible $f(D)$ is unbounded

Let $\langle z_n \rangle$ be a sequence in D such that for any $n \in \mathbb{N}$, $|f(z_n)| > n$.

Since, $\langle z_n \rangle$ is a bounded sequence so $\exists$ a subsequence $\langle z_{n_k} \rangle$ such that $\lim_{n_k \to \infty} z_{n_k} = z_0 \in D \cup \partial D$.

Also, since f is entire $\Rightarrow$ f is regular at $z_0 \Rightarrow \lim_{n_k \to \infty} f(z_{n_k}) = f(z_0)$

But $|f(z_{n_k})| > n_k$ for every n_k

$\Rightarrow f(z_{n_k})$ is a unbounded sequence which is a contradiction.

Hence, $f(D)$ has to be bounded.

4. Let $f(z) = u + iv$ is an entire function & D be a bounded region such that for any $z \in \mathbb{C}$, $\exists\, z' \in D$ such that $f(z) = f(z')$ then f is constant.

Solution: Since for any $z \in \mathbb{C}$, $\exists\, z' \in D$ such that

$f(z) = f(z') \Rightarrow f(\mathbb{C}) \subseteq f(D)$

But D is a bounded region $\Rightarrow f(D)$ is a bounded set. (using above lemma) & hence f is a bounded function.

$\Rightarrow f$ has to be constant.

5. Let $f(z) = u + iv$ is an analytic function in extended complex plane then $f(z)$ is constant.

OR

The only function which is analytic on the Riemann sphere is the constant function.

Proof: Since f is analytic at $z = \infty$, then $\lim_{|z| \to \infty} f(z)$ is finite. Let this limit be L. i.e., for given $\varepsilon > 0$, $\exists$ an $R > 0$ such that $|f(z)| - |L| \le |f(z) - L| < \varepsilon$ whenever $|z| > R$ & so, in particular, f is bounded for $|z| > R$ also by continuity of f on the compact set $\{z : |z| \le R\}$, f is bounded on the whole of $\mathbb{C}$. Hence by Liouville's theorem f is constant.

6. If $f : \mathbb{C} \to \mathbb{C}$ is analytic & $f(z) = f(z + z_1) = f(z + z_2)$ $\forall\, z \in \mathbb{C}$ where z_1 & z_2 are the two non zero complex numbers such that $\frac{z_1}{z_2} \notin R$, then f is constant.

OR

If f is an entire function & have two linearly independent periods over R then f is constant.

Put Your Own Notes

Proof: Since z_1/z_2 is not real, each $z \in \mathbb{C}$ can be written in the form $z = \lambda_1 z_1 + \lambda_2 z_2$

Where $\lambda_1, \lambda_2 \in \mathbb{R}$. But λ_1 and λ_2 may be written as $\lambda_1 = t_1 + n_1$ and $\lambda_2 = t_2 + n_2$, that is

$z = (t_1 + n_1) z_1 + (t_2 + n_2) z_2 = (t_1 z_1 + t_2 z_2) + (n_1 z_1 + n_2 z_2)$

for some integers n_1, n_2 and some $0 \le t_1, t_2 < 1$. Obviously if z_1 and z_2 are the periods of f, so is $m_1 z_1 + m_2 z_2$ for any integers m_1 and m_2 and hence, we must have $f(z) = f(t_1 z_1 + t_2 z_2)$. Thus, the behavior of f is now entirely characterized by its behavior on the parallelogram $\{t_1 z_1 + t_2 z_2 : 0 \le t_1, t_2 < 1\}$.

From the analyticity of f, it follows that $|f|$ is continuous on the closed parallelogram $D = \{t_1 z_1 + t_2 z_2 : 0 \le t_1, t_2 \le 1\}$ and so, f is bounded for all $z \in D$. Consequently, f is bounded on $\mathbb{C}$. We then conclude, by Liouville's theorem, that f is constant.

7. A function which is harmonic & bounded in $\mathbb{C}$ must be constant.

Proof: Let $u(x,y)$ is harmonic function & $|u| \le M$ for some $M \in R$

Since u is harmonic $\Rightarrow$, $\exists$ a v such that $f(z) = u + iv$ is analytic o $\mathbb{C} \Rightarrow f(z)$ is constant & hence $u(x,y)$ is constant.

Question: Let $f(z)$ be an analytic function on $\mathbb{C}$ and $f(z) = f(z+1) = f(z+i)\ \forall\, z \in \mathbb{C}$, then

(a) $\dfrac{1}{f}$ has no pole.

(b) f is analytic at $z = \infty$

(c) $f(z+\alpha) = f(z)\ \forall\, \alpha \in \mathbb{C}$

(d) $f(\mathbb{C}) = \mathbb{C} - \{0\}$

Ans: (a), (b) & (c)

Solution: since $\{1, i\}$ is a linearly independent set over $R \Rightarrow f(z)$ has to bounded and hence constant.

Question: Let V be a vector space of entire functions over $\mathbb{C}$. If f & $g \in V$ such that $|f| < |g|$ then f & g are linearly dependent.

Solution: Since $|f| < |g|$ $\Rightarrow$ g does not have any zero in $\mathbb{C}$.

Define, $h(z) = \dfrac{f(z)}{g(z)}\ \forall\, z \in \mathbb{C}$

$$\Rightarrow\ |h(z)| = \left|\frac{f(z)}{g(z)}\right| < 1$$

$\Rightarrow h(z)$ is entire & bounded so $h(z)$ has to be constant i.e., $h(z) = \lambda\ \forall\, z \in \mathbb{C}$ & $\lambda \in \mathbb{C}$.

$$\Rightarrow \frac{f}{g} = \lambda$$

$$\Rightarrow f = \lambda g.$$

Put Your Own Notes

8. If $f(z)$ is entire function satisfied by the condition $|f'(z).f(z)| \le 1$ then show that $f(z)$ is constant.

Proof: Let us constant a function $g(z)$ on $\mathbb{C}$ such that

$$g(z) = \frac{1}{2}[f(z)]^2$$

$\Rightarrow g'(z) = f(z).f'(z)$, since $f(z)$ & $f'(z)$ are entire function & hence so is $g'(z)$

Also $|g'(z)| = |f(z)f'(z)| \le 1$

Hence $g'(z)$ is bounded entire function so by Liouville's theorem we have, $g'(z)$ is constant

Let $g'(z) = A$

$\Rightarrow g(z) = Az + B$

$\Rightarrow f(z) = \sqrt{2(Az + B)}$

Since $f(z)$ is entire so it cannot have branch point at $z = -\dfrac{B}{Z}$

$\Rightarrow f(z) = \sqrt{2B}$ hence $f(z)$ is constant.

6.2. Fundamental Theorem of Algebra in $\mathbb{C}$

Every non-constant polynomial of degree $n \ge 1$ with complex coefficients has at least one complex zero & hence exactly n-zeros.

OR

Let $p(z) = \sum_{k=0}^{n} a_k z^k$ be a non constant polynomial of degree $n \ge 1$ with complex co-efficient. Then, $p(z)$ has n zeros in $\mathbb{C}$ i.e., $\exists$ n complex numbers $z_1, z_2, ..., z_n$, not necessarily distinct, such that

$$p(z) = a_n \prod_{k=1}^{n} (z - z_k)$$

Proof: If $a \in \mathbb{C}$, then by the 'division algorithm' there is a polynomial q of degree $n-1$ such that $p(z) = (z-a)q(z) + R$, where R is a constant. Clearly,

$R = 0 \Leftrightarrow p(a) = 0 \Leftrightarrow z - a$ is a factor of $p(z)$.

Since there exists a z_1 such that $p(z_1) = 0, z - z_1$ is a factor of $p(z)$ with no remainder term. By the 'division algorithm' there is a polynomial p_{n-1} of degree $n-1$ such that $p(z) = (z - z_1)p_{n-1}(z)$, because

$$p(z) - p(z_1) = a_1(z - z_1) + ... + a_{n-1}\left(z^{n-1} - z_1^{n-1}\right) + \left(z^n - z_1^n\right)$$

$$= (z - z_1)p_{n-1}(z).$$

Put Your Own Notes

This shows that p has a linear factor $z-z_1$. Thus, if $n>1$, then by applying the same principle we conclude that there is another complex number, say z_2, such that $p_{n-1}(z_2)=0$ and so p_{n-1} has a linear factor $z-z_2$. Proceeding in this manner, we can express p uniquely as a product of linear factors:

$$p(z)=a_n\prod_{k=1}^{n}(z-z_k)$$

Where $z_1, z_2, ..., z_n$ are (not necessarily distinct) the zeros of $p(z)$.

Corollary: Every polynomial $p(z)$ of positive degree omits no values, i.e., $p(\mathbb{C})=\mathbb{C}$.

Proof: If $p(z)$ is a polynomial of degree n, then $q(z)=p(z)-a$ is also a polynomial of degree n for each fixed $a\in\mathbb{C}$. Since, $p(z)$ has n-zeros. In other words, for each $a\in\mathbb{C}$ there are n points $z_1, z_2, ..., z_n$ such that $p(z_j)-a=0$ for $j=1,2,...,n$. Thus, $p(\mathbb{C})=\mathbb{C}$.

Theorem: If a polynomial $p(z)$ with real coefficients has a zero at α such that $\text{Im}(\alpha)\neq 0$, then the complex conjugate $\bar{\alpha}$ is also a zero of $p(z)$. Indeed, if α is a zero of order k then $\bar{\alpha}$ is also a zero with order k.

Proof: Set $p(z)=a_0+a_1z+a_2z^2+...+a_nz^n$, where $a_0, a_1, a_2, ..., a_n$ are all real and $a_n\neq 0$. Since α is a zero of $p(z)$, we have $p(\alpha)=0$. Taking conjugation on both sides, we have $p(\bar{\alpha})=0$. Hence, $\bar{\alpha}$ is also a zero of the polynomial $p(z)$. To prove the second assertion we note that if $p(z)$ has a zero at α of order k, then

$$p(\alpha)=p'(\alpha)=...=p^{(k-1)}(\alpha)=0 \text{ and } p^{(k)}(\alpha)\neq 0$$

Which would then imply that

$$p(\bar{\alpha})=p'(\bar{\alpha})=...=p^{(k-1)}(\bar{\alpha})=0 \text{ and } p^{(k)}(\bar{\alpha})\neq 0$$

Thus, complex zeros occur as conjugate pairs with the same multiplicity.

Convex Hull: The convex Hull of a set $D\subseteq\mathbb{C}$ is the intersection of all the convex sets containing D.

Note:

(i) Convex hull of D is the smallest convex set containing D.

(ii) The convex hull of empty set is the set $\{0\}$.

(iii)The convex hull of D is D itself provided D is convex.

(iv)The closed convex hull of D is the intersection of all the closed convex sets containing D.

(v) The convex hull of points $z_1, z_2, z_3, ..., z_n\in\mathbb{C}$ is the set of all linear combination $z=\sum_{j=1}^{n}\lambda_j z_j$ with $0\leq\lambda_j\leq 1$ & $\sum_{j=1}^{n}\lambda_j=1$.

Put Your Own Notes

6.3. Gauss's Theorem

Suppose that $p(z)$ is a polynomial of degree $n \geq 1$. Then every zero of $p'(z)$ lies in the convex hull of the set of zeros of $p(z)$.

Proof: Let $p(z)$ be a polynomial of the form

$$p(z)=\prod_{k=1}^{n}(z-z_k),$$

where $z_1, z_2, ..., z_n$ are (not necessarily distinct) the zeros of $p(z)$. Thus, by logarithmic differentiation, it follows from the above representation that for $z \neq z_k$

$$\frac{p'(z)}{p(z)}=\sum_{k=1}^{n}\frac{1}{z-z_k}=\sum_{k=1}^{n}\frac{\overline{(z-z_k)}}{|z-z_k|^2}$$

so that

$$\overline{\left(\frac{p'(z)}{p(z)}\right)}=\sum_{k=1}^{n}\left(\frac{1}{|z-z_k|^2}\right)(z-z_k).$$

If $c \in \mathbb{C}$ is such that $p(c) \neq 0$ and $p'(c)=0$, then the above equation becomes

$$c=\frac{\sum_{k=1}^{n} z_k|c-z_k|^{-2}}{\sum_{k=1}^{n}|c-z_k|^{-2}}.$$

Hence c is of the form $c=\sum_{k=1}^{n}\lambda_k z_k$, where

$$\lambda_j=\frac{|c-z_j|^{-2}}{\sum_{k=1}^{n}|c-z_k|^{-2}}, \quad j=1,2,...,n.$$

This shows that if $z_1, z_2, ..., z_n$ are the zeros of $p(z)$, then for every zero c of $p'(z)$ there are non-negative numbers $\lambda_1, \lambda_2, ..., \lambda_n$ such that

$$c=\sum_{k=1}^{n}\lambda_k z_k \text{, with } \sum_{k=1}^{n}\lambda_k=1.$$

The above construction uses the fact that $p(c) \neq 0$. If $p(c)=0=p'(c)$, then we simply take $\lambda_1=1$ and $c=1.c$.

Note: If all the zeros of a polynomial $p(z)$ have negative real parts, then all the zeros of $p'(z)$ have negative real part.

6.4. Luca's Theorem

If all the zero's of a polynomial $p_n(z)$ lie in a half plane, then the zeros of the derivative also lie in the same half plane.

Put Your Own Notes

Proof: If $z_1, z_2, ..., z_n$ are the zeros of the polynomial $p_n(z)$, then by Gauss's theorem all the zeros of $p'(z)$ are in the convex hull of the set $\{z_1, z_2, ..., z_n\}$. Given that $z_1, z_2, ..., z_n$ are in some half plane since every half plane is convex & hence contains the smallest convex set, i.e., the convex hull of $\{z_1, z_2, ..., z_n\}$.

Corollary: If all the zeros of a polynomial lie in the smallest convex polygon, then the zeros of its derivative also lie in the same polygon.

6.5. Generalized Version of Liouville's Theorem

An entire function f, which satisfies the inequality $|f(z)| \le M|z|^\alpha$ for some $\alpha \ge 0, \alpha \in R, M > 0$ & for all sufficiently large $|z|$, then

$$f(z) = a_0 + a_1 z + a_2 z^2 + ... + a_r z^{[\alpha]}$$

Where $[\alpha]$ is the greatest integer function of α.

Note:

(i) There can be no bianalytic together (i.e., a bijection which is analytic together with its inverse) mapping of the unit disk Δ onto the whole complex plane $\mathbb{C}$ or of the upper half-plane onto $\mathbb{C}$.

(ii) If f is entire & $|f(z)| \le A + B|z|^\alpha$ $\forall$ sufficiently large $|z|$ & for some fixed constants A, B & $0 \le \alpha < 1$, then f is constant.

Corollary*:* Let f is entire & $|f(z)| \le k|z|^\alpha$ $\forall z \in \mathbb{C}$ then $f(z) = \lambda z^{[\alpha]}$.

Proof: Since by extension of Liouville's theorem we have

$$f(z) = a_0 + a_1 z + ... + a_r z^r, \ r = [\alpha]$$

Since $|f(z)| \le k|z|^\alpha \Rightarrow f(o) = 0 \Rightarrow a_0 = 0$

Now consider $\left|\frac{f(z)}{z}\right| = \left|a_1 + a_2 z + ... + a_r z^{r-1}\right| \le k|z|^{r-1}$

$$\Rightarrow \lim_{z \to 0} \frac{f(z)}{z} = 0 \Rightarrow a_1 = 0$$

& similarity we have that $a_i = 0 \ \forall i = 1$ to $r - 1$

$$\Rightarrow f(z) = \lambda z^r.$$

Example: Let $f(z)$ be an entire function & $f(z) = f(kz)$, $|k| > 1$ $\forall z \in \mathbb{C}$ $\Rightarrow f(z)$ is constant.

Solution: Let z_1, z_2 be complex numbers such that $z_1 \ne z_2$

Since $f(z_1) = f(kz_1)$

$$\Rightarrow f\left(\frac{z_1}{k}\right) = f(z_1)$$

$$\Rightarrow f(z_1) = f\left(\frac{z_1}{k}\right) = f\left(\frac{z_1}{k^2}\right) = ... f\left(\frac{z}{k^n}\right) : n \in \mathbb{Z}$$

Put Your Own Notes

$\Rightarrow \exists$ a $\varepsilon > 0$ such that $|f(z_1) - f(0)| < \frac{\varepsilon}{2}$

Similarly we have $|f(z_2) - f(0)| < \frac{\varepsilon}{2}$

Consider $|f(z_2) - f(z_1)| = |f(z_1) - f(0) + f(0) - f(z_2)|$

$\leq |f(z_1) - f(0)| + |f(z_2) - f(0)|$

$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$

$\Rightarrow |f(z_2) - f(z_1)| < \varepsilon$

$\Rightarrow f(z_1) = f(z_2) \quad \forall z_1, z_2 \in \mathbb{C}$

$\Rightarrow f(z)$ is a constant function.

Put Your Own Notes

CHAPTER 7

TAYLOR AND LAURENT EXPANSION AND COUNTABILITY

7.1. Taylor Expansion

Suppose that $f(z)$ is analytic in a domain D. Then $\forall\, z \in \{z : |z - z_0| < R\}, f$ has the Taylor series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

$$a_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

Where $C = \{\zeta : |\zeta - z_0| = r\}$ & $0 < r < R$.

Results:

(i) The series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ converges to f uniformly in $|z - z_0| < \delta$, where δ can be taken as large as possible such that f remains analytic in $|z - z_0| < \delta$.

(ii) The Taylor series is nothing but the power series.

(iii)The radius of convergence of this Taylor series is the distance of nearest singularity from the point of expansion z_0 and is denoted by R.

Corollary: When $z_0 = 0$ then the Taylor series expansion becomes

$f(z) = \sum_{n=0}^{\infty} \frac{f^n(0)}{n!} z^n$ $(|z| < R)$ is called a Maclaurin series.

Note: If f is entire then for any $z_0 \in \mathbb{C}, f$ has infinite radius of convergence of its Taylor series expansions.

7.1.1. Zeros of Analytic Functions

Let f is analytic, $f(z) \not\equiv 0$ in an open set D then $z_0 \in D$ is said to be a zero of order m if $f(z_0) = f'(z_0) = \ldots = f^{m-1}(z_0) = 0$ & $f^m(z_0) \neq 0$.

Example: Let $f(z) = z \sin z$, then $f(z)$ is analytic on $\mathbb{C}$.

Now consider $f(0) = 0$

Also $f'(z) = z\cos z + \sin z \Rightarrow f'(0) = 0$

$f''(z) = -z \sin z + 2\cos z \Rightarrow f''(0) = 2 \neq 0$

Hence f has a zero of order 2 at $z = 0$.

Proposition: A function f analytic at z_0 has a zero of order m at z_0 if and only if $f(z) = (z - z_0)^m g(z)$, where g is analytic at z_0 & $g(z_0) \neq 0$.

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Proof: Since f is analytic at z_0 so f admits a Taylor series about z_0.

i.e. $f(z)=\sum_{n=0}^{\infty} a_n (z-z_0)^n$

Since z_0 is a zero of order m

$\Rightarrow a_0 = a_1 = ... = a_{m-1} = 0$ & $a_m = \dfrac{f^m(z_0)}{m!} \neq 0$

i.e. $f(z)=\sum_{n=0}^{\infty} a_{m+n} (z-z_0)^{m+n}$

or $f(z)=(z-z_0)^m \sum_{n=0}^{\infty} b_n (z-z_0)^n$ where $b_n = a_{n+m}$.

Let $\sum_{n=0}^{\infty} b_n (z-z_0)^n$ converges to $g(z)$ uniformly on $|z-z_0|<\delta$. we have

(i) $g(z)$ is analytic in $|z-z_0|<\delta$.

(ii) $g(z_0) \neq 0$ as $g(z_0)=b_0=a_m \neq 0$.

Example: Consider $f(z)=z-\sin z$

$\Rightarrow f'(z)=1-\cos z,\ f''(z)=\sin z$

$\Rightarrow f'''(z)=\cos z$

& $f(0)=f'(0)=f''(0)=0$ but $f'''(0)=1\neq 0$

Hence $f(z)$ has a zero of order 3.

Result: If f & g have zero of order m & n respectively at $z=z_0$. Then $h(z)=f(z).g(z)$ have zero of order $m+n$.

Proof: $\exists$ an analytic function f_1 such that $f_1(z_0)\neq 0$, $f(z)=(z-z_0)^m f_1(z)$

Let $g(z)$ has a zero of order n at $z=z_0$.

$\Rightarrow \exists$ an analytic function g_1 such that $g_1(z_0)\neq 0$

$g(z)=(z-z_0)^n g_1(z)$

Consider $h(z)=(z-z_0)^{n+m}.f_1(z)g_1(z)$

$\Rightarrow h$ has a zero of order $m+n$ since $f_1(z_0).g_1(z_0)\neq 0$.

Theorem: Every zero of an analytic function $f(\not\equiv 0)$ is isolated.

Proof: Suppose that f has a zero of order m at a. Then there exist a $R>0$ such that

$f(z)=(z-a)^m g(z),\ |z-a|<R,$

Where g is analytic at a and $g(a)\neq 0$. Let $|g(a)|=2\varepsilon>0$. Then for this ε, since g is continuous at a, there exists a $\delta>0$ such that

$|g(z)-g(a)|<\varepsilon$ whenever $|z-a|<\delta$.

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Therefore when $|z-a|<\delta$ we have

$|g(z)|=|g(a)-[g(a)-g(z)]|\geq|g(a)|-|g(z)-g(a)|>2\varepsilon-\varepsilon=\varepsilon$.

Thus, $g(z)\neq 0$ in $\Delta(a;\delta)$ (We remind the reader that we have already notice this point while discussing the limit of a function. But $|z-a|^m\neq 0$ in $|z-a|<\delta$. Hence, $f(z)=g(z)(z-a)^m\neq 0$ in this neighborhood except at a. This completes the proof.

Example: Find order of zero of the function $z^m \sin z$ at the origin.

Solution: Let $f(z)=z^m$ & $g(z)=\sin z$

$\Rightarrow$ $f(z)$ has a zero of order m at $z=0$ & $g(z)$ has a zero of order one at $z=0$

$\Rightarrow$ $h(z)=f(z).g(z)$ has a zero of order $(m+1)$

7.1.2. Identity/Uniqueness Theorem

Let f is analytic in D & $z_0\in D$, if there is a sequence $z_n\to z_0$, $z_n\neq z_0$ for any $n\in\mathbb{N}$ & $f(z_n)=0$ $\forall n\in\mathbb{N}$. Then $\exists\,\delta>0$ such that $f(z)\equiv 0$ $\forall z\in|z-z_0|<\delta$ & hence $f(z)\equiv 0$ in D.

OR

Let f be analytic in a domain D & let S be the set of all zeros of f which has a limit point in D then $f(z)\equiv 0$ in D i.e. $f(z)$ is identically zero in D.

Note: The f is defined on the connected & open set D.

Example: If $D=\mathbb{C}\setminus\{z:1\leq|z|\leq 3\}$ & if $f:D\to\mathbb{C}$ is defined by

$$f(z)=\begin{cases}0 & \text{for } |z|<1\\ 2 & \text{for } |z|>3\end{cases}$$

then f is an holomorphic function on D & its zero set $\{z:|z|<1\}$ has a limit point in D, yet f is not identically zero in D.

Corollary: Let f & g be analytic functions in a domain D & let them coincide on a set which has a limit point in D. Then $f(z)=g(z)$ for every z in D.

Proof: Define $h(z)=f(z)-g(z)$

Let z_0 be a limit point in D such that $f(z_0)=g(z_0)\Rightarrow h(z_0)=0$ in $D\Rightarrow h(z)\equiv 0$ by identity theorem.

Note: The Taylor series expansion in real & complex are same.

Example: Consider $f(z)=\sin^2 z+\cos^2 z$

& $g(z)=1$ then we have

$f(z)=g(z)$ $\forall z\in R\subseteq\mathbb{C}$

But every point of $z\in\mathbb{R}$ is a limit point in $\mathbb{C}\Rightarrow f(z)=g(z)$ in $\mathbb{C}$

i.e. $\sin^2 z+\cos^2 z=1$

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Note: From here we can conclude that all the trigonometric identity which holds in R also holds in $\mathbb{C}$.

Example: Let f is an entire function & $f\left(\frac{1}{2^n}\right)=0 \quad \forall n\in\mathbb{N}$

Now, consider $A=\{z:f(z)=0\}$

i.e., $A=\left\{\frac{1}{2^n}:n\in\mathbb{N}\right\}$

$\Rightarrow$ '0' is the limit point of A & $0\in\mathbb{C}\Rightarrow f(z)\equiv 0$ by identity theorem.

Example: Let f & g are entire functions & $f\left(\frac{1}{n}\right)=g\left(\frac{1}{n}\right) \quad \forall n\in\mathbb{N}$

Now consider $h(z)=f(z)-g(z)\Rightarrow h(z)$ is entire

& let $A=\{z:h(z)=0\}$

$$=\left\{\frac{1}{n}:n\in\mathbb{N}\right\}$$

$\Rightarrow$ '0' is the limit point of A & hence $h(z)\equiv 0$ on $\mathbb{C}$.

Example: Define $A=\left\{f \text{ is entire}: f\left(\frac{1}{n}\right)=f\left(\frac{-1}{n}\right)=\frac{1}{n^2}\right\}$

then find out the set A.

Solution: Define $g(z)=z^2$

Define $h(z)=f(z)-g(z)$

Then $h(z)=0$ on the set $A=\left\{\frac{1}{n}:n\in\mathbb{N}\right\}\cup\left\{-\frac{1}{n}:n\in\mathbb{N}\right\}$.

Since '0' is the limit point of A & $0\in\mathbb{C}\ \Rightarrow h(z)\equiv 0$ on $\mathbb{C}$

i.e. $f(z)=g(z)=z^2$

i.e. $A=\{f: f(z)=z^2\}$

Example: Define $A=\left\{f \text{ is entire}: f\left(\frac{1}{n}\right)=f\left(\frac{-1}{n}\right)=\frac{1}{n^3}\right\}$ then find the set A.

Solution: Since $f\left(\frac{1}{n}\right)=\frac{1}{n^3}$

Consider $g(z)=z^3$ & consider $h(z)=f(z)-g(z)$

$A_1=\{z:h(z)=0\}$ also zero is the limit point of A & $0\in\mathbb{C}\Rightarrow f(z)=z^3$

Also $f\left(\frac{-1}{n}\right)=\frac{1}{n^3}$

Consider $g_1(z)=-z^3$ & consider $h_1(z)=f(z)-g_1(z)$

Let $B=\{z:h_1(z)=0\}$ also zero is the limit point of B & $0\in\mathbb{C}\Rightarrow h_1(z)\equiv 0$

$\Rightarrow f(z)=g_1(z)=-z^3$

$\Rightarrow A$ is an empty set

Put Your Own Notes

Example: Let f be analytic function on $|z|<2$ & $f\left(\sqrt{2}+\frac{1}{n}\right)=\frac{1}{n^2}$ then find the function $f(z)$.

Solution: Consider $g(z)=\left(\frac{z-\sqrt{2}}{i}\right)^2$ then $f(z)=g(z)$ on the set A where $A=\{z\in\mathbb{C}:f(z)=g(z)\}$

i.e., $A=\left\{\sqrt{2}+\frac{1}{n}:n\in\mathbb{N}\right\}\Rightarrow\sqrt{2}$ is the limit point of A & $\sqrt{2}\in\{z:|z|<2\}$

Hence $f(z)=g(z)=\left(\frac{z-\sqrt{2}}{i}\right)^2$.

Results: If the zeros of an analytic function are not isolated then $f(z)\equiv 0$ throughout the domain of analyticity.

1. If $f(z)=\sum_{n=0}^{\infty}a_n z^n$ converges in $|z|<R, R>0$, such that $f^2(z)=f(z)$ for every z in the open interval $(0,R)$, then $f(z)$ is either 1 or 0 $\forall z\in\{z:|z|<R\}$.
2. If $f\in H(G)$ i.e. f is an analytic functions such that $f(x)=-f(-x)$ for every real number x in G, then $f(z)=-f(-z)$ $\forall z\in$ in G.
3. Suppose that $f\,\&\,g$ are analytic function on domain $\mathbb{C}$ & neither f nor g has a zero in $\mathbb{C}$. If $(f'/f)\left(\frac{1}{n}\right)=(g'/g)\left(\frac{1}{n}\right)$ $\forall n=2,3...$, then $f(z)=cg(z)$ in $\mathbb{C}$.

Proof: Define $h(z)=\left(\frac{f'}{f}\right)(z)-\left(\frac{g'}{g}\right)(z)$ in D.

and $A=\{z\in\mathbb{C}:h(z)=0\}$

$=\left\{\frac{1}{n}\in\mathbb{C}:n\in\mathbb{N}\right\}$.

since '0' is the limit point of the set A & $0\in\mathbb{C}\Rightarrow h(z)\equiv 0$ on $\mathbb{C}$.

i.e., $\frac{f'}{f}(z)=\frac{g'}{g}(z)$ on $\mathbb{C}$.

$\Rightarrow\ f'(z)g(z)-g'(z)f(z)=0$ on $\mathbb{C}$.

$\Rightarrow\ \left(\frac{f}{g}\right)'(z)=0$ on $\mathbb{C}$ $\Rightarrow\frac{f}{g}(z)$ has to be a constant function $\Rightarrow f(z)=cg(z)$.

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4. If f is an analytic function in the unit disk, $\{z_n\}_{n\geq 1}$ is a sequence of non zero complex numbers such that $z_n \to 0$ as $n \to \infty$ & $f(z_n) = f(-z_n)\ \forall\, n \in \mathbb{N}$, then f is even.

   Proof: Define $h(z) = f(z) - f(-z)$

   $A = \{z \in \mathbb{C} : h(z) = 0\}$ & $\Delta = \{z : |z| < 1\}$

   $= \{z_n \in \Delta : h(z_n) \to 0\}$

   Since 0 is the limit point of $A \,\&\, 0 \in \Delta$

   $\Rightarrow h(z) \equiv 0$ on Δ i.e. $f(z) = f(-z)\ \forall\, z \in \Delta$.

5. If $f \,\&\, g$ are entire functions which agree on some interval $[a,b] \subset R$, then $f(z) = g(z)$ in $\mathbb{C}$.

   Proof: Since the set of all limit points of $[a,b]$ is $[a,b]$ & $[a,b] \subseteq \mathbb{C}$.

   $\Rightarrow f(z) = g(z)$ in $\mathbb{C}$ by using identity theorem.

7.2. Laurent's Series/Expansion

If f is analytic in the annulus: $R_1 < |z| < R_2$ (Where $0 \leq R_1 < R_2 \leq \infty$), then f has a unique representation $f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$ for any z in the annulus, where

$$a_n = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta^{n+1}} d\zeta, \quad n \in \mathbb{Z}$$

with $C = \{\zeta : |\zeta| = r\}$ and $R_1 < z < R_2$.

Corollary*:* If f is analytic in the annulus: $R_1 < |z - a| < R_2$ (where $R_1 \geq 0$), then, for any z in the annulus, f has a unique representation

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n, \quad a_n = \frac{1}{2\pi i}\int_{|\zeta - a| = r} \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta \ (n \in \mathbb{Z}),$$

with $R_1 < r < R_2$.

OR

Let a function $f(z)$ be analytic inside the annulus $r < |z - z_0| < R$ & on the bounding circles $C_1 : |z - z_0| = r$ & $C_2 : |z - z_0| = R$. Then $f(z)$ can be represented by the Laurent series as

$$f(z) = \underbrace{\sum_{n=0}^{\infty} a_n (z - z_0)^n}_{\text{Regular Part}} + \underbrace{\sum_{n=+1}^{+\infty} a_{-n} (z-a)^{-n}}_{\text{Principal Part}}$$

Where $a_n = \dfrac{1}{2\pi i}\displaystyle\int_C \frac{f(z)}{(z - z_0)^{n+1}} dz$

& $a_{-n} = b_n = \dfrac{1}{2\pi i}\displaystyle\int_C \frac{f(z)}{(z - z_0)^{-n+1}} dz$

Note:

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1. $f(z)$ does not include any singularity in the deleted neighbourhood of z_0. i.e. Laurent series is expanded in a neighbourhood of isolated singularity.
2. The part containing negative powers of $(z-z_0)$ is called the principal part i.e. $\sum_{n=1}^{\infty} b_n (z-z_0)^{-n}$.
3. $\sum_{n=0}^{\infty} a_n (z-z_0)^n$ is called regular part.
4. In the Laurent series, let $r \to 0$. Then, $f(z)$ is analytic in $|z-z_0| < R$ except at the point $z = z_0$. If $f(z)$ is analytic at z_0 also, then the Laurent series is same as the Taylor's series.
5. If $|f(z)| \le M \quad \forall z$ in the annulus $r < |z-z_0| < R$ then

   $$|a_n| \le \frac{M}{2\pi P^{n+1}}.(2\pi P) = \frac{M}{P^n},\ n = 0, \pm 1, \pm 2, ...$$

   Valid for all P, $r < P < R$.
6. Since Laurent series is unique, it can be obtained by various methods like binomial expansions.
7. Every analytic function f in annulus $R_1 < |z| < R_2$ can be uniquely decomposed into a sum $f(z) = f_-(z) + f_+(z)$, where $f_+(z)$ is analytic for $|z| < r_2 (< R_1)$ & $f_-(z)$ is analytic for $|z| > r_1 (> R_1)$. Where

   $$f_+(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$$

   $$\& f_-(z) = \sum_{n=-1}^{-\infty} a_n (z-z_0)^n$$
8. Let $f(z)$ has Laurent expansion $\sum_{n=0}^{\infty} a_n (z-z_0)^n + \sum_{n=1}^{\infty} b_n (z-z_0)^{-n}$ then

   $$\frac{1}{2\pi i}\int_C f(z)dx = \frac{1}{2\pi i}\int_C \sum_{n=0}^{\infty} a_n (z-z_0)^n dz + \frac{1}{2\pi i}\int_C \sum_{n=1}^{\infty} b_n (z-z_0)^{-n} dz$$

   Where C is a closed curve.

   $$= \frac{1}{2\pi i}.0 + \sum_{n=1}^{\infty} \frac{1}{2\pi i}\int_C b_n (z-z_0)^{-n} dz$$

   $$\frac{1}{2\pi i} b_1 \int_C (z-z_0)^{-1} dz$$

   $$= \frac{1}{2\pi i}.b_1.2\pi i = b_1$$

   This is defined as residue of $f(z)$ at $z = z_0$.

Put Your Own Notes

9. Residue of $f(z)$ at $z=z_0$ is the co-efficient of $\frac{1}{(z-z_0)}$ in its Laurent expansion.

Example: Residue of $\sin z$ at $z=0$ is zero.

Note: Entire functions have residue zero $\forall\, z \in \mathbb{C}$.

Example: The function f defined by $f(z)=1/z$ is itself a Laurent series at $z=0$ in the annulus $0<|z|<\infty$. To determine the L

$$\frac{1}{z}=\frac{1}{z-a}\cdot\left[\frac{1}{1+a/(z-a)}\right]$$

$$=\frac{1}{z-a}\cdot\sum_{n\geq 0}(-1)^n\frac{a^n}{(z-a)^n},\ |z-a|>|a|,$$

valid for $|z-a|>|a|$. Similarly, for $|z-a|<|a|$, we have

$$\frac{1}{z}=\frac{1}{a}\left[\frac{1}{1+(z-a)/a}\right]=\sum_{n\geq 0}(-1)^n\frac{(z-a)^n}{a^{n+1}}$$

Which is the Taylor series expansion of f at a valid in $|z-a|<|a|$.

Example: Expand $z+\frac{1}{(1-z)}$ near zero.

Solution: Let $f(z)=z+\frac{1}{(1-z)}$

$$\Rightarrow\quad f(z)=z+(1-z)^{-1}=z+\left(1+z+\frac{z^2}{2}+\ldots\right)$$

Which is the Taylor series expansion of $f(z)$ near $z=0$.

7.3 Analysis of Singularities through Laurent Series

Let $f(z)$ be a function having isolated singularity at $z=z_0$ then we have

$$f(z)=\underbrace{\sum_{n=0}^{\infty}a_n(z-z_0)^n}_{\text{(Regular Part)}}+\underbrace{\sum_{n=1}^{\infty}b_n(z-z_0)^{-n}}_{\text{(Principal Part)}}$$ & the series on right hand side

converges uniformly in $0<|z-z_0|<R$ to $f(z)$ then

1. If there is no principal part then $z=z_0$ is either removable singularity or regular point of $f(z)$ i.e. if $b_n=0\ \forall\, n\in\mathbb{N}$ then $f(z)$ is either regular at $z=z_0$ or removable singularity.
2. If the principal part of the Laurent's expansion contain the finite number of terms then $z=z_0$ is the pole & the highest power of $\frac{1}{(z-z_0)}$ is defined as the order of the pole.
3. If the principal part of the Laurent's expansion contains infinite number of terms then $z=z_0$ is essential singularity.

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4. If $f(z)$ has pole of order m at $z=z_0$ then $\exists$ $\phi(z)$ analytic & non vanishing in $|z-z_0|<\delta$ for same $\delta>0$ such that

$$f(z)=\frac{\phi(z)}{(z-z_0)^m}$$

& conversely if $f(z)=\dfrac{\phi(z)}{(z-z_0)^m}$, $\phi(z)\neq 0$ & $\phi(z)$ is analytic in $|z-z_0|<\delta$.

$\Rightarrow$ $f(z)$ has pole of order m at $z=z_0$.

Example: Show that the function $\dfrac{\sin z}{z^r}$, $r\geq 2$ is a positive integer, has a pole of order $(r-1)$ at $z=0$.

Solution: We have that

$$f(z)=\frac{\sin z}{z^r}=\frac{1}{z^r}\left[z-\frac{z^3}{3!}+\ldots\right]=\frac{1}{z^{r-1}}g(z)$$

where $g(z)=1-\dfrac{z^2}{3!}+\ldots$ & $g(0)\neq 0$

hence, $f(z)$ has a pole of order $(r-1)$ at $z=0$.

Example: Show that the function $\operatorname{cosec} z$ has a simple pole at $z=0$.

Solution: Consider $f(z)=\operatorname{cosec} z=\dfrac{1}{\sin z}$

& also $$f(z)=\frac{1}{z-\left(\frac{z^3}{3!}\right)+\left(\frac{z^5}{5!}\right)\ldots}=\frac{1}{z}\left[1-\left[\frac{z^2}{3!}-\frac{z^4}{5!}+\ldots\right]\right]^{-1}$$

$$=\frac{1}{z}+\frac{z}{3!}+\text{higher powers of } z$$

since the principal part of the Laurent series is the single term $\dfrac{1}{z}$. Hence, $z=0$ is a simple pole.

7.4 Meromorphic Function

A function defined in $\mathbb{C}$ is said to be meromorphic at any point $z_0\in\mathbb{C}$, either $f(z)$ is regular or z_0 is non essential singularity.

Note:

1. Entire functions are meromorphic not conversely.
2. Infinity is the only possible point for a meromorphic function to have essential singularity.
3. The function f is said to be meromorphic at $z=\infty$ if $g(z)=f\left(\dfrac{1}{z}\right)$ is meromorphic in a neighbourhood of $z=0$.
4. Sums & Products of meromorphic functions are meromorphic.
5. The quotient of a meromorphic function is meromorphic, provided that the denominator term is not identically zero.

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6. Every meromorphic function in an open set D admits a representation as the quotient of two analytic functions in D.

Examples:

1. $f(z)=\dfrac{1}{\sin hz}$ is meromorphic in $\mathbb{C}$ since the singularities of f are poles at $z=k\pi i,\ k\in\mathbb{Z}$.

2. $f(z)=\dfrac{1}{\sin z}$ is a meromorphic function in $\mathbb{C}$ since $z=(2n+1)\dfrac{\pi}{2},0,n\in\mathbb{Z}$ are simple poles of $f(z)$.

3. Define $f(z)=\cot z$, since the singularity of $f(z)$ are $n\pi(n\in\mathbb{Z})$ & all the singularities are simple poles. But $f(z)$ is not meromorphic in $\mathbb{C}_\infty$ because ∞ is a limit point of poles of $f(z)$.

4. Consider $f(z)=\dfrac{1}{(e^z-1)}$ then the singularity of $f(z)$ are given by

   $e^z=1\Rightarrow z=2n\pi i,n\in\mathbb{Z}$ & since

   $\lim_{z\to 2n\pi i}(z-2n\pi i)f(z)=1\neq 0$

   $\Rightarrow$ $z=2n\pi i,\ n\in\mathbb{Z}$ are simple poles

5. The function $(z^2+4)^{-1}.e^{1/z}$ & $(z^2-4)\sin\left(\dfrac{1}{z}\right)$ are meromorphic in $\mathbb{C}-\{0\}$, however they are not meromorphic in $\mathbb{C}$. Because each of them has essential singularity at $z=0$.

7.5 Rational Function

A function is said to be rational if it is free from essential singularity in extended $\mathbb{C}$. More precisely $z_0\in\mathbb{C}\cup\{\infty\}$ then z_0 is either regular point of f or a non essential singularity.

Note: Every rational function is meromorphic.

Result: Every rational function can be expressed as a quotient of polynomials.

i.e. $f(z)=\dfrac{p(z)}{q(z)}$, $p(z)\,\&\,q(z)$ are polynomials.

Proof: Since $f(z)$ is a rational function $\forall\, z\in\mathbb{C}$, let $z=0\in\mathbb{C}$ be an isolated singularity then

$$f(z)=\sum_{n=0}^{\infty}a_n(z)^n+\sum_{n=1}^{m}b_n(z)^{-n}$$

Also $$g(z)=f\left(\frac{1}{z}\right)=\sum_{n=0}^{\infty}a_n\frac{1}{(z)^n}\sum_{n=1}^{m}b_n(z)^n$$

as $f(z)$ cannot have essential singularity at $z=\infty$. $\Rightarrow g(z)=f\left(\dfrac{1}{z}\right)$ cannot have essential singularity at $z=0$

$\Rightarrow\ \exists\ a\ k\in\mathbb{N}$ such that $a_n=0\quad\forall\, n>k$

Put Your Own Notes

$\Rightarrow \quad f(z)=\sum_{n=0}^{k} a_n z^n + \sum_{n=0}^{m} b_n z^{-n} = \dfrac{p(z)}{q(z)}$ where $f(z)$ is a meromorphic function in $\mathbb{C}$. But $f(z)$ is again not a meromorphic function in $\mathbb{C}_\infty$ as ∞ is the limit points of the poles of $f(z)$.

Result: A meromorphic function in $\mathbb{C}$ can have only a finite number of poles in a bounded subset D of $\mathbb{C}$.

Proof: Let if possible $f(z)$ be the meromorphic function & have infinite number of poles in bounded subset D of $\mathbb{C}$. Then by Bolzano Weierstrauss theorem $\exists$ a limit point of D & hence a non isolated singularity, contradicting the fact that f is meromorphic function $p(z)\,\&\,q(z)$ are polynomials.

Result: Let f is entire and at $z=\infty$ has a pole then f is polynomial of degree equal to the order of pole at $z=\infty$.

Proof: Since $f(z)$ is entire so its Taylor series expansion about $z=0$ is

$$f(z)=\sum_{n=0}^{\infty} a_n z^n.$$

then $f\left(\dfrac{1}{z}\right)=\sum_{n=0}^{\infty} a_n \dfrac{1}{z^n}$ but $f(z)$ have a pole at $z=\infty \Rightarrow \exists$ a $k\in\mathbb{N}$ such that

$$a_n = 0 \quad \forall n > k$$

$\Rightarrow \quad f(z)=\sum_{n=0}^{k} a_n z^n$ which is a polynomial.

7.5.1. Theorem: Let $f(z)$ be meromorphic in $\mathbb{C}$ and there exist a natural number n, $M>0$, and $R>0$ such that

$$|f(z)|\le M|z|^n \text{ for } |z|>R.$$

Then, f is a rational function.

7.5.2. Theorem: The range of a non-constant entire function is a dense subset of $\mathbb{C}$.

Proof: Let f be a non-constant entire function. Suppose on the contrary that $f(\mathbb{C})$ is not dense. Then, there would exist a point $w_0\in\mathbb{C}$ and a neighborhood $\Delta(w_0;\varepsilon)$ such that $\Delta(w_0;\varepsilon)\cap f(\mathbb{C})=\emptyset$. Then, for all $z\in\mathbb{C}$, we have $|f(z)-w_0|>\varepsilon$ so that $|g(z)|<1$ for $z\in\mathbb{C}$, where

$$g(z)=\frac{\varepsilon}{f(z)-w_0}$$

But then, g being a bounded entire function, would be constant, and hence f would be constant, which is a contradiction.

Remark: The range of a non-constant entire function is never contained in a half plane or in a bounded domain or in the complement of any simply connected domain in $\mathbb{C}$.

Put Your Own Notes

7.6 Picard's Little Theorem

Every non-constant entire function omits at most one complex number as its value. In other words, if an entire function omits two values, then it is constant.

7.7 Picard's Great Theorem

Suppose that f is analytic in $\Delta(z_0;r)\setminus\{z_0\}$ and $z=z_0$ is an essential singularity of f. Then $\mathbb{C}\setminus f(\Delta(z_0;r)\setminus\{z_0\})$ is a singleton set.

7.8 Casorati-Weierstrass Theorem

If f has an essential singularity at z_0 and if w_0 is a given finite complex number, then there exists a sequence $\{z_n\}$ with $z_n \to z_0$ such that $f(z_n)\to w_0$. In other words, f takes values arbitrarily close to every complex number in every neighborhood of an essential singularity.

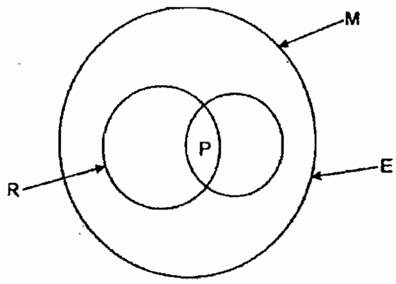
Results:

(i) If $g(z)$ is an entire transcendental function, then near ∞, the values assumed by $g(z)$ are dense in $\mathbb{C}$. i.e. $g(z)$ takes values arbitrary close to every complex number in every neighbourhood of ∞.

(ii) If f is an entire function such that $a \notin f(\mathbb{C})$ & $b \notin f(\mathbb{C})$, $a \neq b$ then f is constant.

(iii)Every meromorphic function Q in $\mathbb{C}$ that omits three distinct values $a,b,c \in \mathbb{C}$ is necessarily constant.

Geometrical Interpretation of Meromorphic Rational, Entire & Polynomial



M-Meromorphic

R-Rational

E-Entire

P-Polynomial

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CHAPTER 8

POWER SERIES AND COUNTABILITY

8.1. Power Series

Given a sequence $\langle a_n \rangle$ of complex numbers, the series $\sum_{n=0}^{\infty} a_n (z-z_0)^n$ is called the power series about the point z_0.

In particular, the series $\sum_{n=0}^{\infty} a_n z^n$ is a power series about the origin.

Example:

(i) $\sum_{n=0}^{\infty} (-1)^n . \frac{z^{2n}}{(2n)!}$

(ii) $\sum_{n=0}^{\infty} (-1)^n . \frac{z^{(2n+1)}}{(2n+1)!}$

(iii) $\sum_{n=0}^{\infty} \frac{z^n}{n!}$

Results:

1. This series converges always for $z = z_0$ & this may indeed be the only point for which it converges.
2. If the series $\sum_{n=0}^{\infty} a_n (z-z_0)^n$ converges at some point z, then the series converges (absolutely) at all points in the disk $|z-z_0| < |z_1 - z_0| = r$. If the series diverges at z_2, then it diverges $\forall \ z \in |z-z_0| > |z_2 - z_0|$
3. The largest R so that the power series converges uniformly at every compact subset of the disk $|z-z_0| < R$ is defined as radius of convergence (RoC).
4. If $\sum_{n=0}^{\infty} a_n (z-z_0)^n$ converges uniformly to $f(z)$ on $|z-z_0| < R$ then $f(z)$ is analytic & $\sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1}$ also converges on $|z-z_0| < R$

8.2. Radius of Convergence (R.O.C.)

The power series $\sum_{n=0}^{\infty} a_n (z-z_0)^n$ has radius of convergence R, if $\forall \, z \in |z-z_0| < R$, power series converges uniformly.

Note:

1. The power series may fail to converge on $|z-z_0| = R$.

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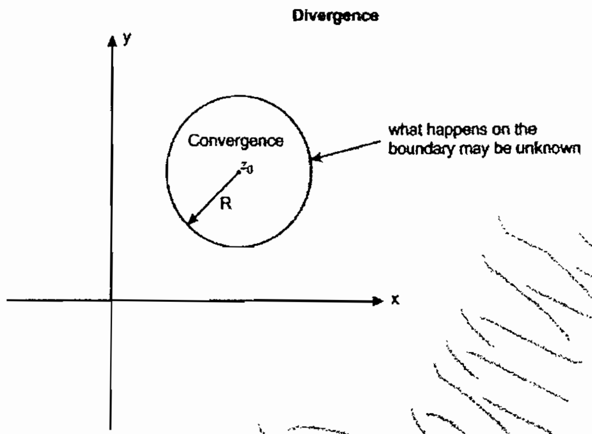
2. For any $S > 0 \ \exists \ z_1 \in |z - z_0| < R + S$ such that $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ fails to be convergent at z_1.

3. The region $|z - a| < R$ is called region of convergence of power series.



Result: For the power series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ we can find R, its radius of convergence, by any of the following methods:

(i) **Cauchy's root test:** $R = \dfrac{1}{\lim\limits_{n \to \infty} |a_n|^{1/n}}$ (provided the limit exists).

(ii) **Cauchy-Hadamard formula:** $R = \dfrac{1}{\limsup\limits_{n \to \infty} |a_n|^{1/n}}$ (this limits always exists).

(iii) **D' Alembert's ratio test:** $R = \dfrac{1}{\lim\limits_{n \to \infty} \left| \dfrac{a_{n+1}}{a_n} \right|}$ (provided the limit exists).

(iv) $R = \sup \left\{ P : \sum\limits_{n \geq 0} a_n (z_n - z_0)^n \text{ converges } \forall\, z \text{ satisfying } |z - z_0| \leq P \right\}$.

Note:

1. $R = 0$ if and only if $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ converges only for $z = z_0$.

2. If $R = \infty$, then the series $\sum_{n=1}^{\infty} a_n (z - z_0)^n$ converges $\forall\, z \in \mathbb{C}$.

8.3. Circle of Convergence

Let R be the radius of convergence of the series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ then the circle $|z - z_0| = R$ is called the circle of convergence & is the greatest circle about z_0 inside which the power series converges at each point.

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Examples:

1. Show that the series $\sum_{n=0}^{\infty}\left(\frac{n+2}{3n+1}\right)^n (z-4)^n$ has radius of convergence 3 .

Solution: Comparing the given series with the $\sum_{n=0}^{\infty} a_n (z-z_0)^n$ then we

ha $a_n = \left(\frac{n+2}{3n+1}\right)^n$.

By Cauchy's root test we have $\lim_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty}\left(\frac{n+2}{3n+1}\right) = \frac{1}{3}$

$\Rightarrow$ $\underset{n\to\infty}{\text{limit}} |a_n|^{1/n}$ exits. Hence radius of convergence of given power series is $R=3$.

2. Show that the series $\sum_{n=1}^{\infty} a_n z^n = 4z + 5^2 z^2 + 4^3 z^3 + 5^4 z^4 + 4^5 z^5 + \ldots$ has radius of convergence $\frac{1}{5}$.

Solution: Here $a_n = \begin{cases} 5^{2m} & \text{if } n = 2m \\ 4^{2m+1} & \text{if } n = 2m+1 \end{cases}$

i.e. $\left\{|a_n|^{1/n} : n \in \mathbb{N}\right\} = \{4,5,4,5,\ldots\}$

$\Rightarrow$ $\limsup_{n\to\infty} |a_n|^{1/n} = 5$.

3. Find the radius of convergence of the following power series.

(i) $\sum_{n=0}^{\infty} n z^n$

(ii) $\sum_{n=0}^{\infty} 2^n . z^n$

(iii) $\sum_{n=0}^{\infty} 2^n z^{2n}$

(iv) $\sum_{n=0}^{\infty} z^{n!}$

(v) $\sum_{n=0}^{\infty} 2^n z^{n!}$

(vi) $\sum_{n=0}^{\infty} z^n$

(vii) $\sum_{n=0}^{\infty} a_n . z^n$ Where

$$a_n = \begin{cases} (1+2i)^n; & \text{if } n \text{ is odd} \\ (2+3i)^n; & \text{if } n \text{ is even} \end{cases}$$

(viii) $\sum_{n=0}^{\infty} \frac{1}{2^n} z^n$

(ix) $\sum_{n=0}^{\infty} a_n z^n$

Where $a_n = \begin{cases} 5^n; & \text{where } n \text{ is even} \\ 3^n; & \text{where } n \text{ is odd.} \end{cases}$

Solutions:

(i) Consider $\sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} n z^n$

then we have $a_n = n$

$$\Rightarrow \frac{1}{R} = \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \limsup_{n\to\infty} \left|\frac{n+1}{n}\right|$$

$=1$

$\Rightarrow R = 1$. Where R is the radius of convergence.

(ii) Let R be the radius of convergence then $\frac{1}{R} = \limsup_{n\to\infty} \left|\frac{2^{n+1}}{2}\right| = \frac{1}{2}$

$\Rightarrow \quad R = 2$.

(iii) Put $z^2 = t \Rightarrow \sum_{n=0}^{\infty} 2^n z^{2n} = \sum_{n=0}^{\infty} 2^n . t^n$

$\Rightarrow R = \frac{1}{2}$

$\Rightarrow |t| < \frac{1}{2} \Rightarrow \left|z^2\right| < \frac{1}{2}$

Hence radius of convergence of $\sum 2^n . z^{2n}$ is $\frac{1}{\sqrt{2}}$.

(iv) $\sum_{n=0}^{\infty} z^{n!} = 1 + z + z^2 + z^6 + z^{24} + \ldots$

$$\Rightarrow \quad a_n = \begin{cases} 1; & \text{if } \exists k \in \mathbb{N} \cup \{0\} \text{ such that } n = k! \\ 0; & \text{otherwise} \end{cases}$$

$$\Rightarrow \quad \frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n} = 1$$

$\Rightarrow \quad R = 1$

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(v) Since $\sum 2^n z^{n!} = 1 + 2^1 z^1 + 2^2 z^2 + 2^3 z^3 + 2^4 z^{24} + \ldots$

$$\Rightarrow a_n = \begin{cases} 2^k; & n = k! \text{ for some } k \in \mathbb{N} \cup \{0\} \\ 0; & \text{otherwise} \end{cases}$$

$$\Rightarrow \frac{1}{R} = \lim_{n\to\infty} \sup |a_n|^{1/n} = 1$$

$$\Rightarrow R = 1$$

(vi) Radius of convergence $= 1$.

(vii) Consider $\sum_{n=0}^{\infty} a_n z^n$ where

$$a_n = \begin{cases} (1+2i)^n; & n \text{ is odd} \\ (2+3i)^n; & n \text{ is even} \end{cases}$$

Consider $|a_n|^{1/n} = \begin{cases} |1+2i|; & n \text{ is odd} \\ |2+3i|; & n \text{ is even} \end{cases}$

$$\Rightarrow \lim_{n\to\infty} \sup |a_n|^{1/n} = \sqrt{13}$$

$$\Rightarrow R = \frac{1}{\sqrt{13}}$$

(viii) Consider $\sum_{n=0}^{\infty} a_n z^n$ where $a_n = \frac{1}{2^n}$

$$\frac{1}{R} = \lim_{n\to\infty} \sup |a_n|^{1/n} = \frac{1}{2} \Rightarrow R = 2.$$

(ix) Consider $\sum_{n=0}^{\infty} a_n z^n$

where $a_n = \begin{cases} 5^n; & \text{when } n \text{ is even} \\ 3^n; & \text{when } n \text{ is odd} \end{cases}$

$$\frac{1}{R} = \lim_{n\to\infty} \sup |a_n|^{1/n} = \lim_{n\to\infty} \sup \{5, 3\} = 5$$

$$\Rightarrow R = \frac{1}{5}.$$

Results on the Radius of Convergence: Consider the power series $\sum_{n=0}^{\infty} a_n z^n$ which has radius of convergence R then

(i) $\sum_{n=0}^{\infty} a_n^2 z^n$ has radius of convergence R^2.

(ii) $\sum_{n=0}^{\infty} a_n z^{kn}$, $k \neq 0$ & $k \in R$ has radius of convergence is $R^{1/k}$.

(iii) $\sum_{n=0}^{\infty} \frac{a_n}{n!} z^n$ has radius of convergence ∞.

(iv) $\sum_{n=0}^{\infty} (1+b_0)^n a_n z^n$ has radius of convergence $\frac{R}{1+b_0}$.

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(v) $\sum_{n=0}^{\infty} k^n a_n z^n$ has radius of convergence $\frac{R}{k}$.

(vi) If the radii of convergence of the power series $\sum_{n=0}^{\infty} a_n z^n$ & $\sum_{n=0}^{\infty} b_n z^n$ are R_1 & R_2 respectively then

(a) RoC, i.e. R of $\sum_{n=0}^{\infty}(a_n+b_n)z^n$ is given by $R \geq \min\{R_1, R_2\}$.

(b) RoC, i.e. R of $\sum_{n=0}^{\infty} a_n b_n z^n$ is given by $R \geq R_1.R_2$.

(c) RoC, i.e. R of $\sum_{n=0}^{\infty} \frac{a_n}{b_n} z^n$, $b_n \neq 0$ is given by $R \geq \frac{R_1}{R_2}$.

Theorem: A power series $\sum_{n\geq 0} a_n z^n$ and the k-times derived series defined by $\sum_{n\geq k} n(n-1)...(n-k+1)a_n z^{n-k}$ have the same radius of convergence.

Theorem: If $\sum_{n\geq 0} a_n z^n$ has radius of convergence $R > 0$, then $f(z) = \sum_{n\geq 0} a_n z^n$ is analytic in $|z| < R$, $f^{(k)}(z)$ exists for every $k \in \mathbb{N}$ and

$$f^{(k)}(z) = k!a_k + \sum_{n\geq k+1} \frac{n!}{(n-k)!} a_n z^{n-k} \left(|z| < R\right)$$

Where $a_k = f^{(k)}(0)/k!$

For example, the geometric series $(1-z)^{-1} = \sum_{n\geq 0} z^n$ which converges for $|z| < 1$, after k-times differentiation yields

$$\frac{1}{(1-z)^{k+1}} = \sum_{n\geq k} \binom{n}{k} z^{n-k} = \sum_{m\geq 0} \frac{(m+k)!}{k!m!} z^m \text{ for } |z| < 1.$$

In particular, $z(1-z)^{-2} = \sum_{n\geq 1} nz^n$ for $|z| < 1$.

Corollary: If f is entire and if $z_0 \in \mathbb{C}$, then $f^{(n)}(z_0)$ exists for $n = 0, 1, ...$ and has the power series expansion

$$f(z) = \sum_{n\geq 0} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n \text{ for all } z.$$

Remark: Now it is clear that entire functions are just those functions which are defined by the sums of the series $\sum_{n\geq 0} a_n z^n$ where $\limsup_{n\to\infty} |a_n|^{1/n} = 0$.

Remark: Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ then

(a) $a_n = 0 \ \forall\, n = 0, 2, 4, ...$ if $f(z)$ is an odd function i.e. $f(-z) = -f(z)$.

(b) $a_n = 0 \ \forall\, n = 1, 3, ...$ if $f(z)$ is an even function i.e. $f(-z) = f(z)$.

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CHAPTER 9

ARGUMENT AND ROUCHE'S THEOREM AND COUNTABILITY

9.1. Argument Theorem

Let f is meromorphic function and C be a simple closed contour such that no zero or pole of f lie on C. Let $a_1, a_n, \ldots, a_k$ are zeros of order $n_1, n_2, n_3, \ldots, n_k$, respectively & $b_1, b_2, \ldots, b_r$ are poles of order $p_1, \ldots, p_r$ in C. Then

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}dz = \sum_{i=1}^{k} n_i - \sum_{i=1}^{r} p_i = N - P \text{ where}$$

N – Number of zero's counting multiplicity.

P – Number of poles counted upto order.

Corollary: If $g(z)$ is analytic within and on C then

$$\frac{1}{2\pi i}\int_C g(z)\frac{f'(z)}{f(z)}dz = \sum_{i=1}^{k} n_i g(a_i) - \sum_{i=1}^{r} p_i g(p_i)$$

Example: Consider $f(z) = \dfrac{(z^2-1)(z^2+1)}{(2z-3)^3(z-5)^{10}}$, then find $\displaystyle\int_C \frac{f'(z)}{f(z)}dz$ where $C : |z| = 6$

Solution: Since total number of zero's of $f(z)$ in $|z| = 6$ are 4 & that of poles are given by $2z-3=0$ and $z-5=0 \Rightarrow z = \dfrac{3}{2}$ & $z=5$, order of the poles at $z=\dfrac{3}{2}$ is 3 order of the poles at $z=5$ is 10.

$$\Rightarrow \int_C \frac{f'(z)}{f(z)}dz = 2\pi i(4-13) = -18\pi i$$

Example: Evaluate $\displaystyle\int_{|z|=10.5} z^2 \cot \pi z \, dz$.

Solution: $\displaystyle\int_{|z|=10.5} z^2 \cot \pi z \, dz = \int_{|z|=10.5} z^2 . \frac{\cos \pi z}{\sin \pi z} dz$

Let $f(z) = \sin \pi z$ then the zeros of $f(z)$ are $0, \pm 1, \pm 2, \pm 3, \ldots, \pm 10$ i.e. the total number of zero's are 21.

$$\Rightarrow \int_{|z|=10.5} z^2 \cot \pi z \, dz = \frac{1}{\pi}.2\pi i \sum_{i=1}^{21} \alpha_i^2 = 2i.2\sum_{r=1}^{10} r^2$$

$$= \frac{4i \times 10 \times 11 \times 21}{2} = 1240i$$

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Example: Let $f(z)=a_0+a_1z+a_2z^2+...+a_{n-1}z^{n-1}+a_nz^n$ & $C:a$ simple closed curve containing all the zeros of f then evaluate

(i) $\int\limits_C z\dfrac{f'(z)}{f(z)}dz$

(ii) $\int\limits_C z^2\dfrac{f'(z)}{f(z)}dz$

Solution:

(i) Let $f(z)$ have the zero's $\alpha_1,\alpha_2,...,\alpha_n$ then we have

$$\int\limits_C z\frac{f'(z)}{f(z)}dz=2\pi i(\alpha_1+\alpha_2+\alpha_3+...+\alpha_n)$$

$$=2\pi i.\left(-\frac{a_{n-1}}{a_n}\right)$$

(ii) $$\int\limits_C z^2\frac{f'(z)}{f(z)}=2\pi i\sum_{i=1}^{n}\alpha_i^2$$

$$=2\pi i\left[\left(\sum_{i=1}^{n}\alpha_i\right)^n-2\sum\alpha_i\alpha_j\right]$$

$$=2\pi i\left[\left(-\frac{a_{n-1}}{a_n}\right)^2-2\left(\frac{a_{n-2}}{a_n}\right)\right]$$

$$=2\pi i\left[\frac{a_{n-1}^2-2a_{n-2}a_n}{a_n^2}\right]$$

Example: Let α,β,γ be the roots of $f(z)=z^3-z-1$ then

Solution: $\dfrac{1}{2\pi i}\int z^9\dfrac{f'(z)}{f(z)}dz=\alpha^9+\beta^9+\gamma^9$

Since $\sum\alpha=0,\ \sum\alpha\beta=-1$

Consider $\sum\alpha^2=\left(\sum\alpha\right)^2-2\sum\alpha\beta=0-2(-1)=2$

$\alpha^3=\alpha+1=1\ \sum\alpha^3=\sum\alpha+\sum1$

$=0+3$

$\alpha^3=\alpha+1\Rightarrow\alpha^9=\alpha^3+1+3\alpha+3\alpha^2$

$\Rightarrow\sum\alpha^9=\sum\alpha^3+\sum3\alpha^2+\sum3\alpha+3$

$=3+6+3=12$

i.e., $\dfrac{1}{2\pi i}\int z^9\dfrac{f'(z)}{f(z)}dz=12$

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Example: Evaluate $\int\limits_{|Z|=1} \frac{dz}{\sin z}$

Solution: Consider $\int\limits_{|Z|=1} \frac{dz}{\sin z} = \int\limits_{|Z|=1} \frac{dz}{2\sin\frac{z}{2}\cos\frac{z}{2}}$

$= \int\limits_{|Z|=1} \frac{\sec^2\frac{z}{2}}{\tan\frac{z}{2}} dz = 2\pi i$ since in $|z|=1$, $f(z)=\tan\frac{z}{2}$ have only one zero.

Example: Evaluate $\int\limits_{|Z|=3\pi} \frac{e^z}{e^z-1} dz$

Solution: Let $f(z)=e^z-1 \ \ \forall z \in \mathbb{C}$ since $f(z)$ is entire so $f(z)$ have no singularity in $\mathbb{C}$. & the zero's of $f(z)$ are given by $e^z=1$ i.e. $z=2n\pi i, n\in\mathbb{Z}$, hence in $|z|=3\pi$ $f(z)$ have three zero's.

$\Rightarrow \int\limits_{|Z|=3\pi} \left(\frac{e^z}{e^z-1}\right) dz = 6\pi i$

9.2. Argument Principal

Let $f(z)$ is analytic with in & on a simple closed curve C & no zero of f lie on C then if N is the number of zero's of f with in C counting multiplicity. Then

$N = \frac{1}{2\pi i}\Delta_C \arg f(z)$

Where $\Delta_c \arg f(z)$ is referred to as the increase in the argument of $f(z)$ along C

i.e. $\frac{1}{2\pi i}\int\limits_C \frac{f'(z)}{f(z)} dz = \Delta_C \arg f(z)$

Example: Evaluate the integral $\int\limits_{|Z|=\pi} \tan \pi z \, dz = 12i$

Solution: Let $f(z)=\cos\pi z$, now

Consider $\int\limits_{|Z|=\pi} \frac{\sin\pi z}{\cos\pi z} dz = \int\limits_{|Z|=\pi} \frac{-f'(z)}{\pi f(z)} dz$

Since $\cos\pi z$ is entire so it does not have any singularity in $\mathbb{C}$ & its zero's are given by $z=\frac{(2n+1)}{n}, n\in\mathbb{Z}$. But in $|z|=\pi$, $\cos\pi z$ have six zero's in it

$\Rightarrow \int\limits_{|Z|=\pi} \tan\pi z = (2\pi i)\left(\frac{-6}{\pi}\right) = -12i$

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9.3. Rouche's Theorem

Let f and g be meromorphic in a domain $D \subseteq \mathbb{C}$ and have only finitely many zeros and poles in D. Suppose that C is a simple closed contour in D such that no zeros or poles of f or g lie on C, and, in addition, assume that

$|g(z)| < |f(z)|$ on C.

Then $\Delta_C f(z) = \Delta_C (f(z) + g(z))$; i.e., the difference between the number of zeros and number of poles is the same for f and $f + g$:

$N_f - P_f = N_{f+g} - P_{f+g}$.

Remarks:

1. If f & g have no poles in D, i.e. $f(z)$ & $g(z)$ are analytic with in & on simple closed contour C and $|f(z)| \geq |g(z)|$ then f & $f+g$ have the same number of zero's inside the contour C.
2. If $f(z)$ & $g(z)$ are analytic function with in & on a closed contour C & $|g(z) - f(z)| < |f(z)|$ on C then $f(z)$ & $g(z)$ has same number of zero's counting multiplicity with in C.

Example: Find the number of zero's of polynomial, $p(z) = z^4 - z^3 + 4z + 1$ inside $C : |z| = 1$.

Solution: Let $f(z) = 4z$

$g(z) = z^4 - z^3 + 1$

Inside $|z| = 1$

Consider $\left|\frac{g}{f}\right| = \left|\frac{z^4 - z^3 + 1}{4z}\right| \leq \frac{|z|^4 + |z|^3 + 1}{4|z|} \leq \frac{3}{4}$

$\Rightarrow |g| \leq |f|$

$\Rightarrow f + g$ & f have same number of zero's. Since $f(z) = 4z$ have one zero inside $|z| = 1$

$\Rightarrow \quad p(z) = f(z) + g(z)$ also have one zero inside C.

Example: Find the number of zero's of $p(z) = z^7 - 3z^3 + 1$ in the domain $D = \{z : 1 < |z| < 2\}$

Solution: Let $C_1 : |z| = 2$ be a closed

Curve, let $f(z) = z^7$ & $g(z) = -3z^3 + 1$

Inside $|z| = 2$ consider

$$\left|\frac{g}{f}\right| = \left|\frac{-3z^3 + 1}{z^7}\right| \leq \frac{3|z|^3 + 1}{|z|^7} < 1 \quad \forall\, z \in \{z : |z| < 2\}$$

$\Rightarrow$ Since $f(z) = z^7$ have 7 zero inside $|z| = 2 \Rightarrow f + g$ also have 7 zero.

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Now, let $C_2 : |z| = 1$ be a closed curve & $f_1(z) = -3z^3$

$g_1(z) = z^7 + 1$

Inside $|z| = 1$, we have

$$\left|\frac{g_1}{f_1}\right| \le \left|\frac{z^7+1}{3z^3}\right| \le \frac{2}{3} < 1$$

$\Rightarrow |g_1| < |f_1|$ Insides $|z| = 1$

So By Rouche's Theorem $p(z) = f_1(z) + g_1(z)$ have same number of zero's as that of $f_1(z)$ since $f_1(z)$ have three zeros in $|z| = 1 \Rightarrow p(z)$ also have three zeros in $|z| = 1$

Hence $p(z)$ have 4 zeros in the domain $D = \{z : 1 < |z| < 2\}$.

Example: Define $p(z) = a_0 + a_1 z + ... + a_n z^n$

$\& |a_r| > \sum_{\substack{i=1 \\ i \ne r}}^{n} |a_i|$ where $1 \le r \le n$ and r is fixed

Then $p(z)$ have r zero's inside $|z| = 1$

Solution: Consider $f(z) = a_r z^r$

& $g(z) = a_0 + a_1 z + ... + a_{r-1} z^{r-1} + a_{r+1} z^{r+1} + ... + a_n z^n$

On $|z| = 1$ we have

$$\left|\frac{g}{f}\right| = \frac{\left|\sum_{\substack{i=1 \\ i \ne r}}^{n} a_i z^i\right|}{|a_r||z|^r} < \frac{\sum_{i=1}^{n} |a_i|}{|a_r|} < 1$$

Then f & $f + g = p(z)$ have same number of zeros.

Since $f(z) = a_r z^r$ have r zeros in $|z| = 1$ hence $p(z) = (f + g)(z)$ have r zeros in $|z| = 1$

Example: Show that the equation $e^z = 2 + 3z$ has almost one solution in the unit disk $|z| < 1$

Solution: Let $f_1(z) = e^z - 2 - 3z$,

Consider $f(z) = e^z - 2$ & $g(z) = -3z$ for $|z| = 1$

Consider $\left|e^z - z\right| = \left|-1 + \sum_{n=1}^{\infty} \frac{z^n}{n!}\right| \le \sum_{n=1}^{\infty} \frac{1}{n!}$

$= e < 3 \le |-3z|$

Since $g(z)$ has only one zero in the unit disk, $f(z) + g(z)$ has only one zero in unit disk.

Put Your Own Notes

9.4. Riemann's Removable Singularity Theorem

If f has an isolated singularity at z_0, then $z = z_0$ is a removable singularity iff one of the following conditions holds:

(i) f is bounded in a deleted neighborhood of z_0,

(ii) $\lim\limits_{z \to z_0} f(z)$ exists,

(iii) $\lim\limits_{z \to z_0} (z - z_0) f(z) = 0$.

Note: If $f \,\&\, g$ are analytic & if both have a zero of order $n \,\&\, m$ at z_0 respectively, then

We have $f(z) = (z - z_0)^n f_0(z)$ & $g(z) = (z - z_0)^m g_0(z)$

Where f_0 & g_0 are both analytic & non-zero at z_0 & hence

(i) If $n = m$ then $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)} = \dfrac{f_o(z)}{g_0(z)}$ exists.

$\Rightarrow \dfrac{f(z)}{g(z)}$ has a removable singularity at z_0.

(ii) If $n > m$ then has a removable singularity.

(iii) If $n < m$ then $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)} = \infty$ & we have $\lim\limits_{z \to z_0} (z - z_0)^{m-n} \dfrac{f(z)}{g(z)} \neq 0$ & exists. Hence have a pole of order $(m - n)$ at $z = z_0$

(iv) If $n < m$ then $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)} = \infty$ & we have $\lim\limits_{z \to z_0} (z - z_0)^{m-n} \dfrac{f(z)}{g(z)} \neq 0$ & exists.

hence have a pole of order $(m - n)$ at $z = z_0$

Put Your Own Notes

CHAPTER 10

CALCULUS OF RESIDUES

10.1. Residue at A Finite Point

If f has an isolated singularity at z_0, then the residue of $f(z)$ at z_0 is

$$\operatorname{Res}[f(z);z_0]=\frac{1}{2\pi i}\int_C f(z)\,dz$$

Where C is any circle centered at z_0 & lying inside a disk about z_0.

Note: Let $f(z)=\sum_{n=0}^{\infty}a_n(z-z_0)^n+\sum_{n=1}^{\infty}b_n(z-z_0)^{-n}$ be the Laurent's expansion of $f(z)$ near $z=z_0$ then

$\operatorname{Res}[f(z);z_0]=b_1=\frac{1}{2\pi i}\int_C f(z)\,dz$ i.e. the coefficient of $\frac{1}{z-z_0}$ in Laurent's expansion.

Example: We know that the coefficient of z^k in $(1+z)^n$ is $\binom{n}{k}$.

So, we may write $\binom{n}{k}=$ coefficient of z^{-1} in $(1+z)^n/z^{k+1}$

$$=\operatorname{Res}[f(z);0],\quad f(z)=(1+z)^n/z^{k+1},$$

$$=\frac{1}{2\pi i}\int_C\frac{(1+z)^n}{z^{k+1}}dz,$$

Where C is any simple closed contour enclosing the origin (note that f is analytic for all $z\in\mathbb{C}\setminus\{0\}$).

Similarly, we also see that $\binom{n}{k}=$ coefficient of z^{-k} in $(1+1/z)^n$

$=$ constant term in $z^k(1+1/z)^n$

and therefore, we have

$$\sum_{k=0}^{n}\binom{n}{k}^2=\text{Coefficient of } z^{-1} \text{ in } \left(\frac{(1+z)^n}{z^{k+1}}\right)\left[z^k(1+1/z)^n\right]$$

$$=\frac{1}{2\pi i}\int_C F(z)\,dz,\quad F(z)=\frac{(1+z)^{2n}}{z^{n+1}},$$

Put Your Own Notes

$= \text{Res}\left[F(z);0\right]$

$=$ Coefficient of z^n in $(1+z)^{2n}$

$= \begin{pmatrix} 2n \\ n \end{pmatrix}.$

Example: Find the residue of $z^{-4}e^{iaz}$ at $z=0$

Solution: Taylor series expansion of e^{iaz} is given by

$$e^{iaz} = 1 + iaz + \frac{(iaz)^2}{2!} + \frac{(iaz)^3}{3!} + \dots$$

$$\Rightarrow \frac{e^{iaz}}{z^4} = \frac{1}{z^4} + \frac{ia}{z^3} + \frac{(ia)^2}{2!z^2} + \frac{(ia)^3}{z.3!} + \dots$$

Hence the $\text{Res}\left[z^{-4}e^{iaz};0\right]$ is the co-efficient of $\frac{1}{z}$ in the expansion of $\frac{e^{iaz}}{z^4}$

i.e. $\text{Res}\left[z^{-4}e^{iaz};0\right] = -\frac{ia^3}{3!}$

Results:

1. If f is analytic at z_0, then $\text{Res}\left[f(z);z_0\right]=0$

   Proof: Since $f(z)$ is analytic at z_0 so its Taylor series expansion is given by $f(z)=\sum_{n=0}^{\infty} a_n(z-z_0)^n$, Hence there does not exist any term of $\frac{1}{z-z_0}$ & hence $\text{Res}\left[f(z);z_0\right]=0$.

2. If f has a removable singularity at z_0, then we have $\text{Res}\left[f(z);z_0\right]=0$. In particular, if C is a simple closed contour containing only removable singularities at $z_k\,(k=1,2,\dots,n)$ inside C, then $\int_C f(z)dz=0$.

   Example: Consider $f(z)=\frac{(\cos z-1)^2}{z^2}$

   Now $\lim_{z\to 0}\frac{(\cos z-1)^2}{z^2} = \lim_{z\to 0}\frac{2(\cos z-1)}{2z}$

   $$= \lim_{z\to 0}\frac{(-\sin z)}{1} = 0$$

   i.e., $\lim_{z\to 0} f(z)$ exists finitely & hence $z=0$ is a removable singularity of $f(z)$.

   i.e., $\text{Res}\left[f(z),0\right]=0$.

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3. If $f(z)$ has an isolated singularity at z_0 & if f is even in $z-z_0$ i.e., $f(z-z_0)=f(-(z-z_0))$ then $\text{Res}[f(z),z_0]=0$.

Proof: Suppose that f is even in $z-z_0$. Then the Laurent series expansion around z_0 cannot have odd powers of $z-z_0$

Example: Consider the function f defined by

$$f(z)=\frac{1}{z^2(z^2-a^2)}(a\in R-\{0\}).$$

Since $f(z)=f(-z) \Rightarrow \text{Res}[f(z);0]=0$

10.2. Theorem

If f has a pole of order n at z_0, then

$$\text{Res}[f(z);z_0]=\lim_{z\to z_0}\frac{1}{(n-1)!}\frac{d^{n-1}}{dz^{n-1}}\left((z-z_0)^n f(z)\right)$$

The coefficient of $(z-z_0)^{-k}$ in the Laurent expansion is

$$a_{-k}=\frac{1}{(n-k)!}\lim_{z\to z_0}\frac{d^{n-k}}{dz^{n-k}}\left((z-z_0)^n f(z), \quad k=1,2,\ldots,n.\right.$$

(Case $k=n$ means that we have only to look at $(z-z_0)^n f(z)$.)

Example: Consider the function $f(z)=e^{2z}/\cosh\pi z$.

Since $\cos(iz)=\cosh z$, and $\cos z=0$ iff $z=(2k+1)\pi/2$, $k\in\mathbb{Z}$, we have

$$\cosh\pi z=0 \Leftrightarrow z=-\frac{(2k+1)i}{2}.$$

Thus, f has a simple pole at $z=-(2k+1)i/2\ (k\in\mathbb{Z})$. From this we see that if $C=\{z:|z-i/2|=1/2\}$, then

$$\int_C f(z)dz=2\pi i\,\text{Res}\left[\frac{e^{2z}}{\cosh\pi z};\frac{i}{2}\right]=2\pi i\left[\frac{e^i}{\pi\sinh(\pi i/2)}\right]=\frac{2\pi i e^i}{\pi i}=2e^i.$$

Inside the contour $\gamma:|z-i/2|=1$, f has singularity only at $z=i/2$. Therefore, if C is any circle around $i/2$ lying completely inside γ then we have $\int_C f(z)dz=2e^i$.

10.3. Theorem

If f has a simple pole at $z=z_0$ and if h is analytic at z_0 and with $h(z_0)\neq 0$, then $\text{Res}[f(z)h(z);z_0]=h(z_0)\text{Res}[f(z);z_0]$.

10.4. Theorem

Suppose ϕ is analytic at z_0 with $\phi(z_0)\neq 0$ and g has a simple zero at z_0. Then $\text{Res}[\phi(z)/g(z);z_0]=\phi(z_0)/g'(z_0)$. In particular, $\text{Res}[1/g(z);z_0]=1/g'(z_0)$.

10.5. Theorem

Put Your Own Notes

Suppose ϕ is analytic at z_0 with $\phi(z_0)\neq 0$, g has a pole of order two at z_0 and h has a zero of order two at z_0. Then we have

(i) $\operatorname{Res}[\phi(z)g(z);z_0]=\phi'(z_0)\operatorname{Res}[(z-z_0)g(z);z_0]$

$+\phi(z_0)\operatorname{Res}[g(z);z_0]$

(ii) $$\operatorname{Res}\left[\frac{\phi(z)}{h(z)};z_0\right]=\left.\frac{6\phi'(z)h''(z)-2\phi(z)h'''(z)}{3[h''(z)]^2}\right|_{z=z_0}$$

Example: Let $f(z)=\dfrac{\cos z}{\sin z}$, since $\sin z$ is analytic on $\mathbb{C}$ so the zero's of $\sin z$ are given by $z=k\pi : k\in\mathbb{Z}$ & they are the simple pole of $f(z)$.

Consider $\operatorname{Res}\left[\dfrac{\cos z}{\sin z};k\pi\right]=\cos(k\pi)\operatorname{Res}\left[\dfrac{1}{\sin z};k\pi\right]$

$$=\cos(k\pi)\lim_{z\to k\pi}\frac{(z-k\pi)}{\sin z}\ \forall z\in k$$

$$=\frac{\cos(k\pi)}{\cos(k\pi)}=1\ \forall k\in\mathbb{Z}$$

Example: Let $f(z)=\dfrac{\sinh z}{\cosh z}$, then $f(z)$ have the singularities at the points where $\cosh z=0$, i.e., where $z=\dfrac{(2k+1)\pi i}{2} : k\in\mathbb{Z}$ & these are the simple pole.

Also $\quad \sinh z\neq 0\ \ \forall z=\dfrac{(2k+1)\pi i}{2} : k\in\mathbb{Z}$

$$\operatorname{Res}\left[\frac{\sinh z}{\cosh z};\frac{(2k+1)\pi i}{2}\right]=\sinh\left(\frac{(2k+1)\pi i}{2}\right).\operatorname{Res}\left[\frac{1}{\cosh z};\frac{(2k+1)\pi i}{2}\right]$$

$$=\sinh\frac{[(2k+1)\pi i]}{2}.\lim_{z\to\frac{(2k+1)\pi i}{2}}\frac{\left(z-\frac{(2k+1)\pi i}{2}\right)}{\cosh z}$$

$$=\frac{\sinh\left[\frac{(2k+1)\pi i}{2}\right]}{\sinh\left[\frac{(2k+1)\pi i}{2}\right]}=1\ \ \forall k\in\mathbb{Z}.$$

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Example: Consider $f(z)=(z\sin z)^{-1}$ then f has a pole of order 2 at $z=0$ since $\lim\limits_{z\to 0}\frac{z^2}{z\sin z}=1\neq 0$ & a simple pole at $z=k\pi,\ k\in\mathbb{Z}$. But $f(-z)=f(z)$, i.e., is a even function so $\text{Res}[z\sin z;\, 0]=0$

$$\Rightarrow \int_C \frac{dz}{z\sin z}=2\pi i.\text{Res}[f(z);\,0]=0$$

Where $C:|z|=1$ is a unit closed circles.

10.6. Residue at the Point at Infinity

Let $z=\infty$ be an isolated singular point of $f(z)$, then Residue of $f(z)$ at $z=\infty$ is given by

$$\text{Res}[f(z);\,\infty]=-\text{Res}\left[\frac{1}{z^2}f\left(\frac{1}{z}\right);\,0\right]$$

Example:

Find the residue at $z=\infty$ for the function $f(z)$ where $f(z)$ is given by

(i) $f(z)=1+z^{-1},\ z\neq 0$

(ii) $f(z)=\dfrac{z}{e^{-z^2}+1}$

(iii) $f(z)=z^3.\cos\left(\dfrac{1}{z}\right)$

Solution:

(i) Let $f(z)=1+z^{-1}$

Define $g(z)=\dfrac{1}{z^2}.f\left(\dfrac{1}{z}\right)=\dfrac{1}{z^2}(1+z)=\dfrac{1}{z^2}+\dfrac{1}{z}$

Now $\text{Res}[f(z);\,\infty]=-\text{Res}\left[\left(\dfrac{1}{z}+\dfrac{1}{z^2}\right);\,0\right]$

(ii) Let $f(z)=\dfrac{z}{e^{-z^2}+1}$

Define $g(z)=\dfrac{1/z}{z^2\left(e^{-1/z^2}+1\right)}=\dfrac{1}{z^3\left[2-\dfrac{1}{z^2}+\dfrac{1}{2!z^2}....\right]}$

$$=\frac{1}{2z^3}\left[1-\left(\frac{1}{2z^2}....\right)\right]^{-1}$$

$$=\frac{1}{2z^3}\left[1+\frac{1}{2z^2}....\right]$$

$$\text{Res}[f(z);\,\infty]=-\text{Res}[g(z);\,0]=0$$

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(iii) Consider $f(z)=z^3\cos\left(\frac{1}{z}\right)$ & define

$$g(z)=\frac{1}{z^2}\left[\frac{1}{z^3}.\cos(z)\right]=\frac{1}{z^5}\cos z$$

$$=\frac{1}{z^5}\left[1-\frac{z^2}{2!}+\frac{z^4}{4!}-....\right]$$

$$=\frac{1}{z^5}-\frac{1}{2!z^3}+\frac{1}{4!z}-....$$

$$\Rightarrow \text{Res}[f(z);\infty]=-\text{Res}[g(z);0]=-\frac{1}{24}$$

10.7. Cauchy Residue Theorem

If f is analytic in a domain D except for isolated singularities at $a_1,a_2...a_n$, then, for any closed contour γ in D on which none of the points a_k lie, we have

$$\int_\gamma f(z)dz=2\pi i\sum_{k=1}^{n}n(\gamma;a_k)\text{Res}[f(z);a_k].$$

10.8. Residue Formula

If f is analytic in a domain D except for isolated singularities at $a_1,a_2,...,a_n$, then for any simple closed contour r in D on which none of the points a_k lie, we have

$$\int_r f(z)dz=2\pi i\sum\text{Res}[f(z);a_k]$$

Here the sum is taken over all $a_k'r$ inside r.

Note: $f(z)$ can have only a finite number of singularities, because otherwise singularities of $f(z)$ would have a limit point G possibly at the point at infinity & so G would not be an isolated singularity of $f(z)$, contrary to our assumption.

10.9. Extended Residue Formula

Let f be analytic in $\mathbb{C}$ except for isolated singularities at $a_1,a_2,...,a_n$. Then we have

(i) The sum of all residues (including the residue at infinity) of f is zero.

$$\text{Res}\left[\frac{1}{z^2}f\left(\frac{1}{z}\right);0\right]=\sum_{k=1}^{n}\text{Res}[f(z);a_k]$$

(ii) If γ is a simple closed contour in $\mathbb{C}$ such that all a_k's are interior to γ, then

$$\int_\gamma f(z)dz=2\pi i\,\text{Res}\left[\frac{1}{z^2}f\left(\frac{1}{z}\right);0\right].$$

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CHAPTER 11

CONFORMAL MAPPING

11.1. Definition

If z_1 & z_2 are two non-zero complex numbers, then $\theta(z_1,z_2)=\arg z_2-\arg z_1=\operatorname{Arg}\left(\frac{z_2}{z_1}\right)$ is defined as the oriented angle from z_1 to z_2.

Example:

(i) If $z_1=1+i$ & $z_2=1$, then $\frac{z_2}{z_1}=\frac{(1-i)}{2}$ so that

$$\theta(z_1,z_2)=\operatorname{Arg}\left(\frac{1-i}{2}\right)=-\frac{\pi}{4}=\arg(1)-\arg(1+i)$$

$$=-\operatorname{Arg}(1+i)$$

(ii) If $z_1=1+i$ & $z_2=-1$, then $\frac{z_2}{z_1}=\frac{(-1+i)}{2}$ so that

$$\theta(z_1,z_2)=\operatorname{Arg}\left(\frac{-1+i}{2}\right)=\frac{3\pi}{4}=\arg(-1)-\arg(1+i)$$

$$=\pi-\frac{\pi}{4}$$

i.e., the set of all analytic function on Ω.

11.2. Proposition

Let $f\in \boldsymbol{H}(\Omega)$, where Ω is a domain containing a smooth curve

$$\gamma:\gamma(t),\ t\in[0,1],$$

Passing through a point $z_0\in\Omega$ and $f'(z_0)\neq 0$. Then the tangent to the curve

$$\Gamma:\Gamma(t)=f(z)\big|_{z=\gamma(t)}=(f\circ\gamma)(t),\ t\in[0,1],$$

at $w_0=f(z_0)$ is

$$(f\circ\gamma)'(t_0)=f'(z_0)\gamma'(t_0)$$

(**Note** that curves are regarded as mappings so that the transformed curve Γ is simply the composite map $\Gamma=f\circ\gamma$).

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11.3. Conformal Mapping/Conformality

A function $f(z)$ is said to be conformal at $z_0 \in D$ (Domain) if, whenever γ_1 & γ_2 are two parameterized curves intersecting at $z_0 = \gamma_1(t_0) = \gamma_2(t_0)$ with non-zero tangents, then the following holds:

(i) The two transformed curves $\Gamma_1 = fo\gamma_1$ & $\Gamma_2 = fo\gamma_2$ have non-zero tangents at t_0.

(ii) The angle from $\Gamma_1'(t_0) = (fo\gamma_1)'(t_0)$ to $\Gamma_2'(t_0) = (fo\gamma_2)'(t_0)$ is the same as the angle from $\gamma_1'(t_0)$ to $\gamma_2'(t_0)$.

If it is conformal at each point of D then we say f is conformal in D

OR

A function $w = f(z)$ is said to be conformal at z_0 if curve in the z-plane passing through z_0 & image curve in w-plane passing $f(z_0)$ preserves the angle in magnitude & sense of rotation (orientation) of angle.

Isogonal: A function that preserves the magnitude (size) of the angle but not sense is said to be isogonal.

Example: Find the image of the region $y = x$ under the map $f(z) = z^2$.

Solution: Since $f(z) = z^2 \quad \forall z \in \mathbb{C}$

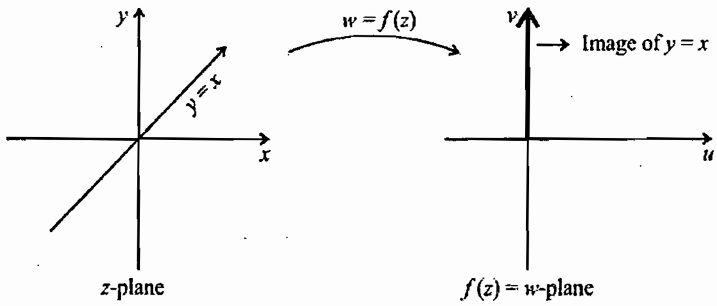
Let $z = x + iy \in \mathbb{C}$

$\Rightarrow f(z) = z^2 = x^2 - y^2 + i(2xy)$

Let $f(z) = u + iv$

But $f(z) = 0 + i(2x^2)$ on the curve $y = x$

$\Rightarrow u = 0$ & $v = 2x^2$



Example: Find the image of the region $y = x$ under the map $w = f(z) = \bar{z}$.

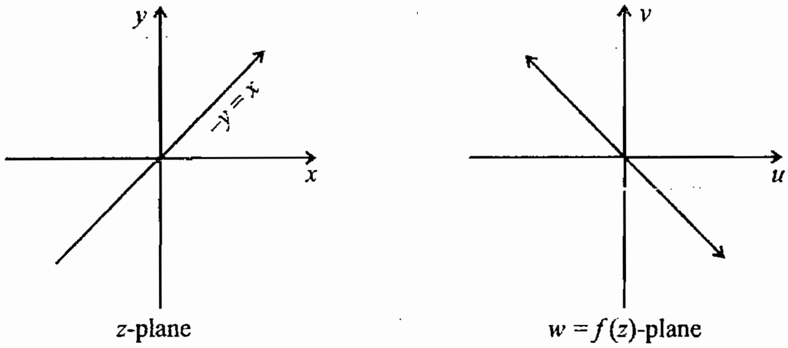
Solution: Let $w = f(z) = u + iv$ be a function

Such that $w = \bar{z} = x - iy$

$\Rightarrow u = x$ & $v = -y$

$\Rightarrow u + v = 0$ Hence under $w = f(z)$ the line $y = x$ in z-plane is mapped into a line $u + v = 0$ in w-plane i.e.

Put Your Own Notes



Example: Find the image of the region $y=x$ under the map $w=f(z)=\frac{1}{z}$

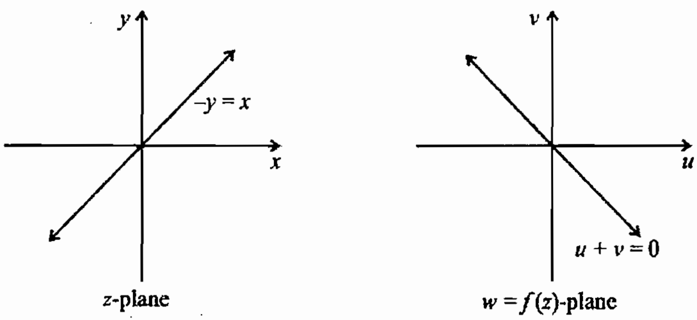
Solution: Let $f(z)=u+iv=\frac{1}{z}$

$\Rightarrow f(z)=\frac{\bar{z}}{|z|^2}=\frac{x-iy}{x^2+y^2}$ Where $z=x+iv\in\mathbb{C}$

$\Rightarrow u=\frac{x}{x^2+y^2}=\frac{x}{x^2+x^2}=\frac{1}{2x}$ along $y=x$

$\& \, v=-\frac{y}{x^2+y^2}=-\frac{x}{x^2+x^2}=-\frac{1}{2x}$ along $y=x$

$\Rightarrow u+v=0$ i.e., under $w=f(z)$ the line $y=x$ is mapped into the line $u+v=0$ in the w- plane i.e.,



Theorem: Let $f\in H(D)$ i.e, analytic on D & $z_0\in D$ such that $f'(z_0)\neq 0$. Then f is conformal at z_0.

Example: $f(z)=e^z$ is conformal everywhere since e^z is an entire function & $f'(z)=e^z\neq 0 \quad \forall z\in\mathbb{C}$.

11.4. Magnification Factor & Scale Factor

Let $w=f(z)$ is a conformal mapping at $z=z_0$ such that

$f(z)=u(x,y)+iv(x,y)$, then

Define $\lim\limits_{\Delta A_z\to 0}\frac{\Delta A_w}{\Delta A_z}=J$ at z_0

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i.e., $J = \begin{vmatrix} u_x(x,y) & u_y(x,y) \\ v_x(x,y) & v_y(x,y) \end{vmatrix}$ by using Cauchy Riemann's equation we have

$$J = \begin{vmatrix} u_x(x,y) & -v_x(x,y) \\ v_x(x,y) & u_x(x,y) \end{vmatrix}$$

$$= u_x^2(x,y) + v_x^2(x,y) = |f'(z)| \neq 0.$$

$J = |f'(z)|^2$ is defined as magnification factor at $z = z_0$

& $|f'(z)|$ is defined as the scale factor at $z = z_0$.

Note: When $|f'(z)| > 1$, then $w = f(z)$ is a magnification map & when $|f'(z)| < 1$ then called as contraction map since

$$|f'(z_0)| = \left| \lim_{z \to z_0} \frac{(f(z) - f(z_0))}{z - z_0} \right|$$

$$\Rightarrow |f(z) - f(z_0)| \approx |f'(z_0)||z - z_0|$$

which clearly represents an expansion if $|f'(z)| > 1$ & contraction $|f'(z)| < 1$.

Critical Point: If f is non constant analytic at z_0 & $f'(z_0) = 0$, then the conformal character fails at z_0, such a point z_0 is called a critical point of f.

Examples:

(i) Consider $w = f(z) = \sin z$ since $f(z)$ is entire, so for every $z_0 \in \mathbb{C}$ is a regular point $\Rightarrow$ $z = k\pi : k \in \mathbb{Z}$ are critical points of $f(z)$.

(ii) Consider the mapping $w = f(z) = e^{2z} - 2iz + 3$ then to find out the critical points of $w = f(z)$.

Put $f'(z) = 0$

i.e. $2e^{2z} - 2i = 0 \quad \Rightarrow e^{2z} = i$

$\Rightarrow z = n\pi + \dfrac{\pi}{4}$ Such that $n \in \mathbb{Z}$.

Remark: Constant map has no critical points & its scale factor is undefined.

Results on the Conformal Mapping

1. Let $f(z)$ be conformal at z_0. Then $f(z)$ has a local inverse at z_0. i.e., $\exists$ a unique transformation $z = g(w)$ defined & analytic in some neighborhood N of $w_0 = f(z_0)$ such that $g(w_0) = z_0$ & $f(z(w)) = w \ \forall w \in \mathbb{N}$.

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2. Suppose that f is analytic at z_0 & $f'(z)$ has a zero of order $(n-1)$ at z_0. If two smooth curves intersect at an angle α in the z- plane, then their images intersect at an angle $n\alpha$ in the w- plane.

   Example: Consider $f(z)=\sin z$. Then f is entire and $f'(z)=\cos z$ so that $f'(z_n)=0$ for $z_n=(2n+1)\pi/2,\ n\in\mathbb{Z}$. Thus, f is conformal on $\Omega=\mathbb{C}\setminus\{(2n+1)\pi/2 : n\in\mathbb{Z}\}$. Note that $f''(z)=-\sin z$ and $f''(z_n)\neq 0$ for each $n\in\mathbb{Z}$. The angle between any two smooth curves intersecting at $z_n(n\in\mathbb{Z})$ is increased by a factor of 2 by $w=f(z)$.

3. Let γ be a smooth arc in a domain D throughout which $w=f(z)$ is conformal & let Γ denote the image of γ under the transformation. Then Γ is also smooth.

4. Suppose that $f(z)$ is an entire function. If $f(z)$ is univalent in $\mathbb{C}$. Then it is conformal in the entire plane & will be of the form $a_0+a_1z,\ a_1\neq 0$.

   Example: Show that the map $w=\frac{1}{z}$ transforms circles & lines into circles & lines.

   Solution: Let a,b,c,d be real numbers such that $b^2+c^2>4ad$. Then

   $$a\left(x^2+y^2\right)+bx+cy+d=0 \qquad ...(1)$$

   Represents the equation of the circle in z- plane

   Since

   $$w=f(z)=u+iv=\frac{1}{z}=\frac{\bar{z}}{|z|^2}=\frac{x-iy}{x^2+y^2}\Rightarrow u=\frac{x}{x^2+y^2}, v=\frac{-y}{x^2+y^2}$$

   $$\Rightarrow \text{ i.e. } \left(x^2+y^2\right)=\frac{1}{u^2+v^2}$$

   $\Rightarrow$ Equation (1) under the mapping $w=\frac{1}{z}$ is mapped into $d\left(u^2+v^2\right)+bu-cv+a=0$ which is a circle or line in w- plane depending on the nature of real constants. We conclude the following.

11.5. Linear Fractional/Bilinear/Mobius Transformation

The map

$$w=\frac{az+b}{cz+d},\quad ad-bc\neq 0 \qquad (i)$$

Where a,b,c & d are complex constants, is called a linear fractional transformation, or Mobius transformation & equation (i) can also be written as

$$Azw+bz+cz+D=0,\ \ (AD-BC\neq 0) \qquad ...(ii)$$

Put Your Own Notes

and conversely any equation of the type (ii) can be put in the form **(i)** & since equation **(ii)** is linear in z & w, or bilinear in z & w so call it bilinear transformation.

Note: A mobius transformation is simply a composition of one, some or all of the following special types of transformation.

(i) **Translation:** It is a map of the form $z \mapsto z+\alpha,\ \alpha \in \mathbb{C}\setminus\{0\}$. If $\alpha = 0$ then it is an identity.

(ii) **Magnification or Contraction:** It is a map of the form $z \mapsto rz,\ r \in R-\{0\}$. For $r=1$, this is the identity map & for $r=0$ it is a constant map.

Case (i) When $r>1$, then this is a "magnification"

Case (ii) When $r<1$, then this is a contraction map.

Note: If $r<0$ then $w=rz$ gives the reflection through the origin followed by such a "magnification" or shrinking/contraction depending on $r<-1$ or $-1<r<0$.

(iii) **Rotation:** It is a map of the form $z \mapsto e^{i\theta}z,\ \theta \in \mathbb{R}$. This map produces a rotation through an angle about the origin with positive sense, $\theta > 0$.

Note: The rotation coupled with magnification is referred to as ***Dilation:*** $z \mapsto az\ (a \neq 0)$.

(iv) **Inversion:** It is a map of the form $z \mapsto \frac{1}{z}$ which produces a geometric inversion (or reciprocal map or the inversion map.)

Remark:

1. If we let $T(z)=T_{abcd}(z)$ & if $\alpha \in \mathbb{C}\setminus\{0\}$, then $\alpha a, \alpha b, \alpha c, \alpha d$ correspond to the same mobius transformation as

   $T_{abcd}(z)=T_{(a\alpha)(b\alpha)(c\alpha)(d\alpha)}(z)$

   i.e., behavior of T does not change when a,b,c,d are multiplied by a non-zero constant.

2. The mobius transformation $T(z)$ is analytic on $C\setminus\{d/c\}$.

3. If $c=0$ then $T(z)=\frac{az+b}{cz+d},\ ad-bc \neq 0$ reduces to

   $T(z)=\frac{a}{d}z+\frac{b}{d}=\alpha z+\beta\,(ad \neq 0, \alpha \neq 0)$ & called a linear map.

4. Every mobius transformation $T(z)$,

   $T(z)=\frac{az+b}{c+d},\ ad-bc \neq 0$, can be decompose as

   $$T(z)=\left[a\left(z+\frac{d}{c}\right)+b-\frac{ad}{c}\right]\frac{1}{c\left(z+\frac{d}{c}\right)}$$

   $$=\frac{a}{c}-\left(\frac{ad-bc}{c^2}\right)\frac{1}{\left(z+\frac{d}{c}\right)},\ c \neq 0$$

5. If $ad-bc=0$ then $T(z)$ is a constant map

11.6. Matrix Interpretation of a Mobius Transformation

Put Your Own Notes

Let $w = T(z) = \dfrac{az+b}{cz+d}$, $(a,b,c,d \in \mathbb{C}, (ad-bc) \neq 0)$ be a mobius transformation. Then every $T(z)$ can be associated with a 2×2 matrix via the map

$$z \mapsto A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \leftrightarrow T(z)$$

Note:

1. The mobius transformation T_0 given by $T_0(z) = z$ is the identity transformation which corresponds to the 2×2 identity matrix.
2. Composition of two mobius transformation is a mobius transformation.

   Result: Let $GL(2,\mathbb{C})$ denote the general linear group consisting of 2×2 invertible matrices A with complex entries

   i.e. $GL(2,\mathbb{C}) = \left\{ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a,b,c,d \in \mathbb{C}, |A| \neq 0 \right\}$

   then $GL(2,\mathbb{C})$ forms a subgroup of the group of all mobius transformations under operation of matrix multiplication, if defined by $T(z) = \dfrac{az+b}{cz+d}$, $ad-bc \neq 0$.
3. The inverse of a Mobius transformation is also a mobius transformation.

   Proof: Let $w = T(z) = \dfrac{az+b}{cz+d}$, $\left(z \neq -\dfrac{d}{c}\right)$ is a bilinear transformation solving above equation for z then we have $z = T^{-1}(w) = \dfrac{dw-b}{-cw+a}$, $\left(w \neq \dfrac{a}{c}\right)$ where $da-(-b)(-c) = ad-bc \neq 0$

   $\Rightarrow$ Inverse of mobius map is again a mobius map.
4. ***Define:*** $T : \mathbb{C} \setminus \left\{-\dfrac{d}{c}\right\} \to \mathbb{C} \setminus \left\{\dfrac{a}{c}\right\}$ such that $T(z) = \dfrac{az+b}{cz+d}$ then T is bianalytic (i.e., T & T^{-1} analytic) & T^{-1} is given by

   $$z = T^{-1}(w) = \frac{dw-b}{-cw+a}$$
5. Since $T'(z) = \dfrac{ad-bc}{(cz+d)^2}$ $\Rightarrow$ T is a conformal map.

11.7. Fixed Point

Let D be a subset of $\mathbb{C}_\infty$ and $f : D \to \mathbb{C}_\infty$. A point $z_0 \in D$ is said to be a fixed point of f if $f(z_0) = z_0$. The set of all fixed points of f is denoted by $\mathrm{Fix}(f)$.

Put Your Own Notes

Examples:

(a) The function $f(z)=z^2$ has exactly three fixed points, namely, 0, 1 and ∞ whereas the function $f(z)=z^{-1}$ has two fixed points namely 1 and -1.

(b) The function $f(z)=z-1$ has no fixed points in $\mathbb{C}$ whereas it has one fixed point in $\mathbb{C}_\infty$, namely the point at ∞.

(c) The reflection $z \mapsto \bar{z}$ is not a Mobius transformation but $f(z)=\bar{z}$ has all the points on $\mathbb{R}$ as its fixed points. What are the fixed points of $f(z)=i\bar{z}$?

(d) the function $f(z)=\dfrac{iz}{|z|}$, $z \neq 0$, has no fixed points in $\mathbb{C}\setminus\{0\}$.

(e) Every non-constant real-valued continuous function $f:(-1,1)\to(-1,1)$ has a fixed point in $(-1,1)$. However, a similar result does not hold for functions $f:\Delta\to\Delta$. For example $\phi_\alpha:\Delta\to\Delta, |\alpha|=1$, defined by

$$\phi_\alpha(z)=\frac{z-\alpha}{1-\bar{\alpha}z}$$

has no fixed points in Δ.

Proposition: Every Mobius transformation $T:\mathbb{C}_\infty\to\mathbb{C}_\infty$ has at most two fixed points in $\mathbb{C}_\infty$ unless $T(z)\equiv z$. Equivalently, if a Mobius transformation leaves three points in $\mathbb{C}_\infty$ fixed, then it is none other than the identity function.

Corollary: If S and T are two Mobius transformations which agree at three distinct points of $\mathbb{C}_\infty$, then $S=T$.

11.8. Normal Form or Canonical form of a Bilinear Transformation

If a bilinear transformation $w=T(z)$ has exactly two fixed points z_1 & z_2, then for some non-zero constant k it satisfies the equation

$$\left(\frac{w-z_1}{w-z_2}\right)=k\left(\frac{z-z_1}{z-z_2}\right) \qquad ...(i)$$

& if $T(z)$ has only one fixed point z_1 then it can be written as

$$\frac{1}{w-z_1}=k'+\frac{1}{z-z_1},\ k'\neq 0 \qquad ...(ii)$$

Equation **(i)** & **(ii)** are known as the normal form or canonical form of a bilinear transformation.

11.9. Lemma

Every Mobius transformation map circles & straight lines in the z-plane into circles or lines in w-plane.

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Results:

1. Every Mobius transformation maps circles in $\mathbb{C}_\infty$ onto circles in $\mathbb{C}_\infty$ (Here we regard straight line / as a limiting case of circle) in the extended complex plane $(\mathbb{C}_\infty)$ as a circle on the Riemann sphere using stereographic projection.
2. Under translation, magnification (scaling) & rotation, circles maps to circles & lines to lines.
3. Under the function $w=\frac{1}{z}$, we have

(i) The image of a line through the origin is a line through the origin.

(ii) The image of a line not through the origin is a circle through the origin.

(iii) The image of a circle through the origin is a line not through the origin.

(iv) The image of a circle not through the origin is a circle not through the origin.

11.10. Classification of Bilinear Transformation on the basis of Normal Form

Let $w=T(z)=\frac{az+b}{cz+d}$ be a bilinear transformation.

(i) **Parabolic:** The bilinear transformation with one fixed point is called parabolic i.e., $(a-d)^2+4bc=0$.

(ii) **Elliptic:** A bilinear transformation with two fixed points i.e., $(a-d)^2+4bc\neq 0$ such that $|k|=1,\ k\neq 1$ is said to be elliptic i.e., in the normal form k is of the form $k=e^{i\alpha}, \alpha\neq 0$.

(iii) **Hyperbolic:** A bilinear transformation with two fixed points i.e., $(a-d)^2+4bc\neq 0$ & $k>0$, $k\in\mathbb{R}$ is termed as hyperbolic.

(iv) **Loxodromic:** A bilinear transformation that is neither hyperbolic, elliptic, nor parabolic is called loxodromic, i.e., it has two fixed points & satisfies the condition $k=\alpha e^{i\alpha}$, $\alpha\neq 0,\ \alpha\neq 1$.

Example: Find all the bilinear transformations which have fixed points as -1 & 1.

Solution: Let $w=T(z)=\frac{az+b}{cz+d}$, $ad-bc\neq 0$ has two fixed point.

Consider $T(1)=1 \Rightarrow a+b=c+d$

$\Rightarrow\ a+b-c-d=0$

& $T(-1)=-1 \Rightarrow -a+b=-(-c+d) \Rightarrow -a+b-c+d=0$

on solving we set $a=d, b=c$

Hence, $T(z)=\frac{az+b}{cz+a}$, $a^2-b^2\neq 0$ is the required bilinear transformation.

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Example: Find the fixed points & what is the type of Bilinear transformation given that $w = T(z) = \frac{z}{z-2}$.

Solution: To find out the fixed points

Put $w = T(z) = \frac{z}{z-2} = z$

$\Rightarrow z^2 - 2z - z = 0 \quad \Rightarrow z = 0, 3$ are the fixed points

Now consider $\frac{w-0}{w-3} = \frac{(z/z-2)-0}{(z/z-2)-3} = \frac{z}{-2(z-3)}$

$$\Rightarrow \left(\frac{w-0}{w-3}\right) = -\frac{1}{2}\left(\frac{z-0}{z-3}\right)$$

$$\Rightarrow \qquad k = -\frac{1}{2}$$

Hence bilinear transformation is laxodromic.

11.11. Theorem

Given three distinct points z_1, z_2, z_3 in $\mathbb{C}_\infty$, $\exists$ a unique Mobius transformation $T(z)$ such that $T(z_1) = 0, T(z_2) = 1$ & $T(z_3) = \infty$.

Corollary: If $\{z_1, z_2, z_3\}$ & $\{w_1, w_2, w_3\}$ are two sets of triplets of distinct points in $\mathbb{C}_\infty$, then $\exists$ a unique Mobius transformation taking z_j to w_j $(j = 1,2,3)$ & that is given by

$$\frac{(w-w_1)(w_2-w_3)}{(w-w_3)(w_2-w_1)} = \frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}$$

11.12. Cross Ratio

For the set of three distinct points z_1, z_2, z_3 of $\mathbb{C}_\infty$, the expression

$\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)} = \frac{(z-z_1)/(z-z_3)}{(z_2-z_1)/(z_2-z_3)}$ is called the cross-ratio of the four points z, z_1, z_2, z_3 & is denoted by (z, z_1, z_2, z_3).

Notes:

(i) The cross ratio (z, z_1, z_2, z_3) is precisely the mobius transformation which sends z_1, z_2, z_3 into $0, 1, \infty$ respectively & is the unique mobius transformation.

(ii) For the set of three distinct points we have $(z, z_1, z_2, z_3) = (z_1, z, z_3, z_2) = (z_2, z_3, z, z_1) = (z_3, z_2, z_1, z)$.

(iii) There are $4! = 24$ cross ratios corresponding to the permutation performed on the four points z, z_1, z_2, z_3. But out of these only six are different.

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(iv) The cross ratio is invariant under Mobius transformation i.e., if z_2, z_3, z_4 are distinct points & T is any Mobius transformation then $(z, z_2, z_3, z_4) = (Tz, Tz_2, Tz_3, Tz_4)$ for any point $z \in \mathbb{C}_\infty$.

(v) Suppose that z, z_1, z_2, z_3 are for distinct points in $\mathbb{C}_\infty$ then

(z, z_1, z_2, z_3) is real $\Leftrightarrow \dfrac{1 - z_1 / z}{1 - z_3 / z}$ is real

$\Leftrightarrow$ either $z = \infty$ or $\dfrac{z - z_1}{z - z_3} = 1 - \dfrac{1}{s}$ for some $s \in \mathbb{R}$

$\Leftrightarrow$ either $z = \infty$ or $z = sz_1 + (1 - s) z_3$.

i.e., if z_1, z_2, z_3 are distinct & lie on a line L in $\mathbb{C}$ or $z = \infty$ or z on the line L.

(vi) The four distinct points z, z_1, z_2 in $\mathbb{C}_\infty$ all lie on a circle or on a line iff their cross ratio (z, z_1, z_2, z_3) is a real number.

11.13. Conformal Self-Maps of Disks And Half-Planes

Homeomorphism: A function $f : x \to y$ is called a homeomorphism if it has the following properties

(a) f is bijection

(b) f is continuous

(c) The inverse function f^{-1} is continuous

Note: An analytic map $f : \Omega \to \Omega'$ is said to be homeomorphic if it has an analytic inverse map $g : \Omega' \to \Omega$ i.e., $fog = I_\Omega$ & $gof = I_{\Omega'}$.

Conformal Self Map: If a function on a domain Ω is an analytic function from Ω into Ω i.e., one-one & onto.

Note: Every conformal self-map of a domain is called an automorphism of that domain & define $Aut(\Omega)$, the set of all automorphism.

Theorem 1: All conformal mappings which map H^+ onto the unit disk Δ such that $\beta \in H^+$ maps onto 0 are given by

$$f(z) = e^{i\theta}\left(\frac{z - \beta}{z - \bar{\beta}}\right) \text{ for some } \theta \in R$$

The inverse mapping $f^{-1} : D \to H^+$ is given by $f^{-1}(w) = \dfrac{\beta e^{i\theta} - \bar{\beta} w}{e^{i\theta} - w}$

Corollary: All conformal mappings which map the lower half-plane $H^- = \{z \in \mathbb{C} : \operatorname{Im} z < 0\}$ onto the unit disk Δ such that $b \in H^-$ maps onto 0 are given by

$$f(z) = e^{i\theta}\left(\frac{z - b}{z - \bar{b}}\right),\ \theta \in \mathbb{R}$$

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Theorem 2: All conformal mappings which map the right half-plane $\{z \in \mathbb{C} : \operatorname{Re} z > 0\}$ onto the unit disk Δ such that $\gamma(\operatorname{Re}\gamma > 0)$ maps onto 0 are given by

$$f(z) = e^{i\theta}\left(\frac{z-\gamma}{z+\bar{\gamma}}\right),\ \theta \in \mathbb{R}.$$

Where γ is in right half plane.

Example: Consider $w = f(z) = (i-z)/(i+z)$. Then we have

(i) f maps the unit disk Δ onto the right half-plane $\{w : \operatorname{Re} w > 0\}$

(ii) f maps the upper half-plane $\{z : \operatorname{Im} z > 0\}$ onto the unit disk Δ

(iii) f maps the open first quadrant $\{z : \operatorname{Im} z > 0, \operatorname{Re} z > 0\}$ onto the upper semi-disk $\{w : |w| < 1, \operatorname{Im} w > 0\}$.

Indeed, we see that

$$w = \frac{1+iz}{1-iz} \Leftrightarrow -iz = \frac{1-w}{1+w} = \frac{1-|w|^2 - 2i\operatorname{Im} w}{|1+w|^2}$$

So that

- $|z| < 1 \Leftrightarrow |w-1| < |w+1| \Leftrightarrow \operatorname{Re} w > 0$
- $\operatorname{Im} z > 0 \Leftrightarrow \operatorname{Re}(-iz) > 0 \Leftrightarrow |w| < 1$
- $\operatorname{Im} z > 0$ and $\operatorname{Re} z > 0 \Leftrightarrow |w| < 1$ and $\operatorname{Im} w > 0$.

11.14. Automorphism of the Unit Disk

Let $\theta \in \mathbb{R}$, $z_0 \in \Delta$ be fixed and ϕ be given by

$$\phi(z) = e^{i\theta}\left(\frac{z-z_0}{1-\bar{z}_0 z}\right).$$

Note that ϕ is one-to-one in $\mathbb{C}\setminus\{1/\bar{z}_0\}$. If $|z| = 1$, then $z^{-1} = \bar{z}$ so that $|\phi(z)| = 1$.

In addition, since every Mobius transformation maps a circle onto a circle or a straight line, ϕ must map the unit circle $|z| = 1$ onto itself. Also, $\phi(z_0) = 0$ and $\phi(\Delta) = \Delta$, because

$$|\phi(z)| < 1 \Leftrightarrow |z-z_0|^2 < |1-z\bar{z}_0|^2$$

$$\Leftrightarrow \left(1-|z_0|^2\right)\left(1-|z|^2\right) > 0$$

$$\Leftrightarrow |z| < 1 \ \left(\text{since } |\bar{z}_0| < 1\right)$$

So that Δ must be mapped onto itself by $\phi(z)$. Now ϕ has the inverse given by

$$\phi^{-1}(w) = e^{-i\theta}\left(\frac{w + e^{i\theta}z_0}{1+e^{-i\theta}\bar{z}_0 w}\right).$$

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Which has a similar form as ϕ and so, ϕ^{-1} shares similar properties as that of ϕ. Organizing these observations together, we can assert that for each z_0 with $|z_0|<1$ and $\theta\in\mathbb{R}$, ϕ is a bijective self mapping of the unit disk Δ. This result raises the following

Theorem 1: All conformal mappings which map the unit disk Δ onto itself and the point $z_0, |z_0|<1$, onto 0 must be of the form for some $\theta\in\mathbb{R}$. Equivalently,

$$\mathrm{Aut}(\Delta)=\left\{e^{i\theta}\frac{z-z_0}{1-\bar{z}_0 z}: z_0\in\Delta,\ 0\le\theta\le 2\pi\right\}.$$

Theorem 2: Every Mobius transformation of the form $T(z)=\dfrac{az+b}{cz+d}$ is a conformal self-map of the upper half-plane H^+ iff a,b,c,d are real numbers satisfying the condition $ad-bc>0$. Equivalently,

$$\mathrm{Aut}\left(H^+\right)=\left\{\frac{az+b}{cz+d}: a,b,c,d\in\mathbb{R},\ ad-bc>0\right\}.$$

11.15. Symmetric Point/Inverse Point

Definition: Let L be a line in $\mathbb{C}$. Two points a & a^* in $\mathbb{C}$ are said to be symmetric with respect to L if L is the perpendicular bisector of $[a, a^*]$ the line segment connecting a & a^*.

Examples:

(i) Two points z & z^* are symmetric w.r.t. the real axis when $z^*=\bar{z}$.

(ii) Two points z & z^* are symmetric w.r.t. the imaginary axis iff $z^*=-\bar{z}$

Definition: Suppose that k is a circle $|z-z_0|=r$ in $\mathbb{C}$. Two points a & a^* are said to be symmetric w.r.t. circle k (or inverse points w.r.t. the circle k) iff $|a-z_0||a^*-z_0|=r^2$ & $Arg(a-z_0)=Arg(a^*-z_0)$. i.e., a & a^* on the same ray emanating from the centre z_0 of k, & the product of their distances from the centre of the circle k is equal to the square of the radius of the circle.

If we let $(a-z_0)=R$, then $a=z_0+\mathrm{Re}^{i\alpha}$ for some $\alpha\in\mathbb{R}$ so we have

$$a=z_0+\mathrm{Re}^{i\alpha} \ \& \ a^*-z_0=\frac{r^2e^{i\alpha}}{R}$$

$$\Rightarrow(\bar{a}-\bar{z}_0)(a^*-z_0)-r^2 \text{ or } a^*=z_0+\frac{r^2}{\bar{a}-\bar{z}_0}$$

$$=z_0+\frac{r^2(a-z_0)}{(a-z_0)^2}$$

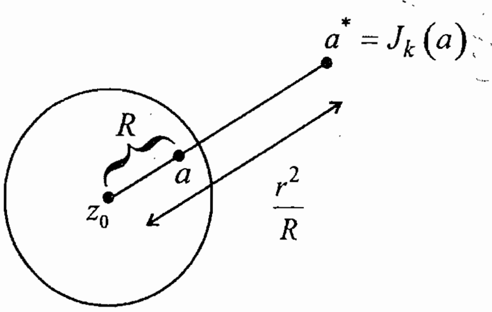
Note: z_0 & ∞ are symmetric w. r. t. the circle.

Put Your Own Notes

Reflection/Inversion Map: Suppose that k is a circle $|z-z_0|=r$ in $\mathbb{C}$ & a^* is the symmetric point of a w. r. t. the circle k in $\mathbb{C}$. Then define $J_k : C_\infty \to C_\infty$

$$a^* = J_k(a) = \begin{cases} z_0 + \dfrac{a^2}{\overline{a} - \overline{z}_0} & \text{if } a \neq z_0, \infty \\ \infty & \text{if } a = z_0 \\ z_0 & \text{if } a = \infty \end{cases}$$

Then the map J_k is called reflection or inversion map in the circle k.



The reflection/Inversion map J_k

Remark: If a is any point in $|z|<1$, then its inverse point with respect to the circle $|z|=1$ is $\dfrac{1}{\overline{a}}$.

Symmetry Principle for Mobius Maps: Let T be a Mobius transformation and K be a circle in $\mathbb{C}_\infty$ with $K' = T(K)$. Then

$T \circ J_K = J_{K'} \circ T$, i.e., $T(J_K(a)) = J_{K'}(T(a))$ for each $a \in \mathbb{C}_\infty$.

In particular, T maps any pair of symmetric points with respect to K onto a pair of symmetric points with respect to the image circle K' under T.

Remark:

(i) Two points are symmetric with respect to a circle in $\mathbb{C}_\infty$ if every circle containing the points intersect the given circle orthogonally.

(ii) Mobius transformation are conformal and preserve circles, and so preserve the orthogonality; hence they preserve the symmetry condition.

Theorem: Let K be a circle passing through three points in $\mathbb{C}_\infty$. Then the reflection J_K satisfies the relation

$$(J_K(a), z_1, z_2, z_3) = \overline{(a, z_1, z_2, z_3)}$$

for $a \neq z_1, z_2, z_3$. Conversely, if

$$(a^*, z_1, z_2, z_3) = \overline{(a, z_1, z_2, z_3)}$$

then a and a^* in $\mathbb{C}_\infty$ are symmetric with respect to the circle K.

Put Your Own Notes

CHAPTER 12

MAXIMUM & MINIMUM MODULUS PRINCIPLE & SCHWARZ LEMMA

12.1. Mean Value Property

A Complex valued function is said to have-Mean value property in a domain D if $\forall a \in D$, $f(a)=\frac{1}{2\pi}\int_0^{2\pi} f\left(a+re^{i\theta}\right)d\theta$ where $|z-a|=r\subset D$

(a) If f is analytic in D then it has mean value property.

(b) If f is continuous in $\bar{D}=DU\partial D$ and have the mean value property in D then $|f|$ attain-its maximum on ∂D (boundary of D).

12.2. Maximum Modulus Principle

If f is analytic in a domain D then $|f|$ does not attain its maximum value in D unless it is constant.

12.3. Maximum Modulus Theorem

If f is analytic in D and continuous on ∂D then maximum value of $|f|$ is always attains on ∂D but never inside unless it is constant.

12.4. Minimum Modulus Theorem

If f is analytic in D and continuous on ∂D and $f(z)\neq 0\ \forall z\in DU\partial D$ then minimum of $|f|$ is attained on the boundary of D

12.5. Minimum Modulus Principle

If f is analytic on D and does not vanish in D then minimum of $|f|$ is never attained inside D unless it is Constant.

12.6. Schwarz' Lemma and its Consequence

In this section we start with a simple but one of the classical theorems in complex analysis, namely, Schwarz' lemma which states that if f is analytic and satisfies $|f(z)|<1$ in Δ and $f(0)=0$, then $|f(z)|\leq|z|$ for each $z\in\Delta$ with the sign of equality if f has the form

$f(z)=e^{-i\alpha}z$

For some $\alpha\in\mathbb{R}$ Furthermore, $|f'(0)\leq 1|$ with the equality if f has the form.

Proposition: Let $f:\Delta\to\bar{\Delta}$ be analytic having a zero of order n at the origin. Then

(i) $|f(z)|\leq|z|^n$ for all $z\in\Delta$

Put Your Own Notes

(ii) $\left|f^{n}(0)\right| \le n!$

and the equality holds either in (i) some point $0 \ne z_0 \in \Delta$ or in (ii) occurs if $f(z) = \varepsilon z^n$ with $|\varepsilon| = 1$.

If f is analytic and satisfies $|f(z) \le M|$ in $\Delta(a;R)$ and $f(a) = 0$, then

(i) $|f(z)| \le M \dfrac{|z-a|}{R}$ for every $z \in \Delta(a;R)$

(ii)

(iii) $|f'(a)| \le \dfrac{M}{R}$

With the sign of equality if f has the form $f(z) = \dfrac{M\varepsilon(z-a)}{R}$ for some constant ε with $|\varepsilon| = 1$.

12.7. Schwarz Pick Lemma

Suppose that f is analytic on the unit disk Δ and satisfies the following two conditions

(i) $|f(z)| \le 1$ for all $z \in \Delta$

(ii) $f(a) = b$ for some $a, b \in \Delta$.

Then $|f'(a)| \le \dfrac{1-|f(a)|^2}{1-|a|^2}$.

Moreover, for a pair of elements a, a' in Δ, the following inequality holds:

$\rho(f(a), f(a')) \le \rho(a, a')$

Where $\rho(z,a) = \dfrac{|z-a|}{(1-\bar{a}z)}$ for $z, a \in \Delta$. Equality is obtain in at a point $a \in \Delta$, or equality is obtain in for a pair of points a, a' with $a \ne a'$ in Δ, if $f \in$ Aut (Δ); otherwise there is strict inequality in $|z| < 1$

Example: Suppose that $f : \Delta \to \overline{\Delta}$ is analytic such that $f(a) = 0$ and $f(a-1) = b$ for some $a \in (0,1)$ and $b \in \Delta$ then by

$|b| \le \dfrac{1}{1-a(a-1)} = \dfrac{1}{1+a-a^2}$.

In particular, this observation implies the following assertions:

(i) There exists no function f that is analytic in Δ such that $f\left(\dfrac{1}{4}\right) = 0$ and $f\left(\dfrac{-3}{4}\right) = \dfrac{17}{20}$.

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(ii) There exists no function f that is analytic in Δ such that

$$|f(z)|\le 1, f\left(\frac{1}{3}\right)=0 \text{ and } f\left(\frac{-2}{3}\right)=\frac{5}{6},$$

Example: Suppose we are given that $f:\Delta\to\bar{\Delta}$ is analytic and $f(a)=0$ for some point $a\in\Delta$. Suppose that we are asked to find an estimate for $|f(b)|$ for some $b\in\Delta$. To do this, by we see first that (since $f(a)=0$),

$$|f(z)|\le|\phi_a(z)|,\qquad \phi_a(z)=\frac{a-z}{1-\bar{a}z}\quad \text{for } z\in\Delta.$$

In particular, $|f(z)|\le|\phi_a(b)|$ and note that the maximum is achieved when $f(z)=\phi_a(z)$. for instance, if $f\left(\frac{1}{2}\right)=0$ then the estimate for $f\left(\frac{1}{4}\right)$ is given by

$$\left|f\left(\frac{1}{4}\right)\right|\le\left|\frac{\left(\frac{1}{2}\right)-\left(\frac{1}{4}\right)}{1-\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)}\right|=\frac{2}{7}$$

And the maximum is attained for $\phi_{\frac{1}{2}}(z)$.

Example: Suppose that $f:\Delta\to\bar{\Delta}$ is an analytic such that $f(0)=a$ for some $a\in\Delta$. We wish to verify whether there exists such an f with the property that $f'(0)=c$ for a complex constant c. if so, under what condition on c, such a function exist? To do this, according to we first observe that $|f'(0)|\le 1-|a|^2$ which means that c must satisfy the inequality $|c|\le 1-|a|^2$. for instance, consider the function $\phi_a(z)$ defined above. Then, $\phi_a(0)=a, \phi_a(0)=-\left(1-|a|^2\right)$ and so, the function f defined by $f(z)=k\phi_a(z)$ with $|k|\le 1$ does the job.

Example: Suppose that $f:\bar{\Delta}\to\bar{\Delta}$ is an analytic function, $|f(z)|\le 1$ on $|z|=1$, $f(a)=0$ and $f(-a)=b$ for some $a\in(0,1)$ and $b\in(0,1]$. Then, it follows that $|b|\le\frac{2a}{\left(1+a^2\right)}$. This observation is useful to conclude the following.

(i) There exists no analytic function $f:\bar{\Delta}\to\bar{\Delta}$ such that

$$|f(z)|\le 1 \text{ on } |z|=1,\ f\left(\frac{1}{2}\right)=0 \text{ and } f\left(-\frac{1}{2}\right)=\frac{19}{20}.$$

Note that the existence of such a function is guaranteed, for example, if we replace the condition $f\left(-\frac{1}{2}\right)=\frac{19}{20}$ by $f\left(-\frac{1}{2}\right)=\frac{4}{5}$.

There exists an analytic function $f:\bar{\Delta}\to\bar{\Delta}$ such that $f\left(\frac{1}{3}\right)=0$ and $f\left(-\frac{1}{3}\right)=\frac{3}{5}$. such an f is given by $\phi_{\frac{1}{3}}(z)$

Assignment Sheet -1

1. How many elements does the set $\{z \in \mathbb{C} \mid z^{60} = -1,\ z^k \neq -1 \text{ for } 0 < k < 60\}$ have?

(a.) 24.

(b.) 30.

(c.) 32.

(d.) 45.

2. Let f be a real valued harmonic function on $\mathbb{C}$, that is, f satisfies the equation $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$. Define the functions $g = \frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}$, $h = \frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}$ Then

(a.) g and h are both holomorphic functions.

(b.) g is holomorphic, but h need not be holomorphic.

(c.) h is holomorphic, but g need not be holomorphic.

(d.) both g and h are identically equal to the zero function.

3. Let $u(x,y) = x^3 - 3xy^2 + 2x$. for which of the following functions v, is $u + iv$ a holomorphic function on $\mathbb{C}$

(a.) $v(x,y) = y^3 - 3x^2y + 2y$

(b.) $v(x,y) = 3x^2y - y^3 + 2y$

(c.) $v(x,y) = x^3 - 3xy^2 + 2x$

(d.) $v(x,y) = 0$

4. Let $p(z), q(z)$ be two non-zero complex polynomials. Then $p(z)\overline{q(z)}$ is analytic if and only if

(a.) $p(z)$ is constant

(b.) $p(z)q(z)$ is constant

(c.) $q(z)$ is a constant

(d.) $\overline{p(z)}q(z)$ is a constant

5. If z_1 and z_2 are distinct complex numbers such that $|z_1| = |z_2| = 1$ and $z_1 + z_2 = 1$, then the triangle in the complex plane with z_1, z_2 and -1 as vertices

(a.) Must be equilateral

(b.) Must be right-angled

(c.) Must be isosceles, but not necessarily equilateral

(d.) Must be obtuse angled

6. Let $f : \mathbb{C} \to \mathbb{C}$ be a analytic function. For $z = x + iy$, let $u, v : \mathbb{R}^2 \to \mathbb{R}$ be such that $u(x,y) = \operatorname{Re} f(z)$ and $v(x,y) = \operatorname{Im} f(z)$. Which of the following are correct?

(a.) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

(b.) $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$

(c.) $\frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y \partial x} = 0$

(d.) $\frac{\partial^2 v}{\partial x \partial y} + \frac{\partial^2 v}{\partial y \partial x} = 0$

7. The number $\sqrt{2}e^{i\pi}$ is

(a.) A rational number

(b.) A transcendental number

(c.) An irrational number

(d.) An imaginary number

8. Let $f : \mathbb{C} \to \mathbb{C}$ be a complex valued function of the form $f(x,y) = u(x,y) + i\,v(x,y)$. Suppose that $u(x,y) = 3x^2y$. Then

(a.) f cannot be holomorphic on $\mathbb{C}$ for any choice of v

(b.) f is holomorphic on $\mathbb{C}$ for a suitable choice of v

(c.) f is holomorphic on $\mathbb{C}$ for all choices of v

(d.) u is not differentiable.

9. Let $f:\mathbb{C}\to\mathbb{C}$ be a complex valued function given by $f(z)=u(x,y)+iv(x,y)$. Suppose that $v(x,y)=3xy^2$. Then

(a.) f cannot be holomorphic on $\mathbb{C}$ for any choice of u

(b.) f is holomorphic on $\mathbb{C}$ for a suitable choice of u

(c.) f is holomorphic on $\mathbb{C}$ for all choices of u

(d.) v is not differentiable as a function of x and y.

10. The real part of square of a complex number is equal to

(a.) The square of its real part

(b.) The square of its imaginary part

(c.) The sum of squares of real and imaginary part.

(d.) The difference of squares of real and imaginary

11. If $f(z)=u(x,y)+ixy$ is analytic then

(a.) $u(x,y)=x^2-y^2$

(b.) $u(x,y)=\frac{1}{2}(x^2+y^2)$

(c.) $u(x,y)=\frac{1}{2}(x^2-y^2)$

(d.) $u(x,y)=x^2+y^2$

12. If $f(z)=e^{-x}\cos y+iv(x,y)$ is analytic then

(a.) $v(x,y)=-e^{-x}\sin y$

(b.) $v(x,y)=xy$

(c.) $v(x,y)=$ is constant

(d.) $v(x,y)=e^{x}\sin y$

13. Let $f(z)=\frac{z}{|z|}(z\neq 0)$ and $f(z)=0(z=0)$ then

(a.) f is discontinuous at 0

(b.) f is not analytic at 0 but it is continuous at 0.

(c.) f is differentiable at 0

(d.) f is not diff. at 0 but it is continuous at 0.

14. If $z=x+iy=3e^{i\pi/4}$ then

(a.) $x=y$

(b.) $x=3$

(c.) $y=3$

(d.) $x^2+y^2=3$

15. If $f=u+iv$ is analytic and if $u(x,y)=x^3-3xy^2$ then

(a.) $v(x,y)=0$

(b.) $v(x,y)=3x^2y-y^3$

(c.) $v(x,y)=y^3-3x^2y$

(d.) $v(x,y)=u(x,y)$

16. The function $f:\mathbb{C}\to\mathbb{C}$ defined by $f(z)=|z|$ is

(a.) analytic when $z\neq 0$

(b.) analytic every where

(c.) analytic no where

(d.) analytic in region $\{z:|z|>1\}$.

17. $\alpha\neq 1$ is a complex numbers s.t. $\alpha^5=1$ then which of the following is true ?

(a.) $1+\alpha+\alpha^2+\alpha^3+\frac{1}{\alpha}=0$

(b.) $\alpha=\alpha^2$

(c.) α^4 is also equal to 1

(d.) $|\alpha|=5$

18. If $f=u+iv$ is an analytic function it satisfies C.R equation

(a.) $u_x=v_y\ \&\ u_y=v_x$

(b.) $u_x=v_x\ \&\ u_y=v_y$

(c.) $u_x=v_x\ \&\ u_y=-v_y$

(d.) $u_x=v_y\ \&\ u_y=-v_x$

19. Which of following function is bounded as z varies over the complex plane

(a.) $\exp(z)$

(b.) $\sin z$

(c.) $\frac{\sin z}{e^z}$

(d.) None of the above.

20. If z_1, z_2 are two complex numbers such that $|z_1+z_2| = |z_1|+|z_2|$ then it is necessary that

(a.) $z_1 = z_1$

(b.) $z_2 = 0$

(c.) $z_1 = \lambda z_2$ for some real numbers λ.

(d.) $z_1 . z_2 = 0$ or $z_1 = \lambda z_2$ for some real numbers λ.

21. By stereographic projection with the South Pole at the origin (0,0,0) the point (1,0,0) goes to complex numbers.

(a.) $z = i$

(b.) $z = 1+i$

(c.) $z = 1$

(d.) $z = -1$

22. The values of i where i is square roots of -1 is

(a.) $e^{\pi/2}$

(b.) $e^{-\pi/2}$

(c.) $e^{i\pi/2}$

(d.) $e^{-i\pi/2}$

23. The function $f(z) = |z|^2 + i\bar{z} + 1$ is differentiable at

(a.) i

(b.) 1

(c.) $-i$

(d.) No point in $\mathbb{C}$

24. Let $u(x, y) = 2x(1-y)$ for all real x and y. Then a function $v(x, y)$, so that $f(z) = u(x, y) + iv(x, y)$ is analytic, is

(a.) $x^2 - (y-1)^2$

(b.) $(x-1)^2 - y^2$

(c.) $(x-1)^2 + y^2$

(d.) $x^2 + (y-1)^2$

25. Let $u(x,y)$ be the real part of an entire function $f(z) = u(x,y) + iv(x,y)$ for $z = x+iy \in \mathbb{C}$. If C is +ve oriented boundary of a rectangular region R in $\mathbb{R}^2$, then $\oint_C \left[\frac{\partial u}{\partial y}dx - \frac{\partial u}{\partial x}dy\right] =$

(a.) 1

(b.) 0

(c.) 2π

(d.) π

26. Define $f : \mathbb{C} \to \mathbb{C}$ by $f(z) = \begin{cases} 0 & \text{if } Re(z) = 0 \text{ or } \operatorname{Im}(z) = 0 \\ z & \text{otherwise} \end{cases}$ then the set of point where f is analytic is

(a.) $\{z : \operatorname{Re}(z) \neq 0 \text{ and } \operatorname{Im}(z) \neq 0\}$

(b.) $\{z : \operatorname{Re}(z) \neq 0\}$

(c.) $\{z : \operatorname{Re}(z) \neq 0 \text{ or } \operatorname{Im}(z) \neq 0\}$

(d.) $\{z : \operatorname{Im}(z) \neq 0\}$

27. Which of following is not the real part of an analytic function?

(a.) $x^2 - y^2$

(b.) $\frac{1}{1+x^2+y^2}$

(c.) $\cos x \cosh y$

(d.) $x + \frac{x}{x^2+y^2}$

28. The principal value of $\log(i^{1/4})$ is

(a.) $i\pi$

(b.) $\frac{i\pi}{2}$

(c.) $\frac{i\pi}{4}$

(d.) $\frac{i\pi}{8}$

29. Consider the functions $f(z) = x^2 + iy^2$ and $g(z) = x^2 + y^2 + ixy$ at $z = 0$

(a.) f is analytic but not g

(b.) g is analytic but not f

(c.) Both f and g are analytic

(d.) Neither f nor g is analytic

30. Consider a function $f(z)=u+iv$ defined on $|z-i|<1$ where u, v are real valued functions of x, y. Then $f(z)$ is analytic for u equals to

(a.) x^2+y^2

(b.) $\ln(x^2+y^2)$

(c.) e^{xy}

(d.) $e^{x^2-y^2}$

31. At $z=0$, the function $f(z)=z^2\overline{z}$

(a.) Does not satisfy Cauchy-Riemann equations

(b.) Satisfy Cauchy-Riemann equations but not differentiable

(c.) Is differentiable

(d.) If analytic.

32. The function $\sin z$ is analytic in

(a.) $\mathbb{C}\cup\{\infty\}$ Where $\mathbb{C}$ denotes complex plane.

(b.) $\mathbb{C}$ except on negative real axes.

(c.) $\mathbb{C}-\{0\}$

(d.) $\mathbb{C}$

33. The function $f(z)=|z|^2$ is

(a.) Differentiable every where

(b.) Differentiable only at origin

(c.) Not differentiable any where

(d.) Differentiable only on real axes

34. An analytic function $f(z)$ is such that $\operatorname{Re}\{f'(z)\}=2y$ and $f(1+i)=2$, then the imaginary part of $f(z)$ is

(a.) $-2xy$

(b.) x^2-y^2

(c.) $2xy$

(d.) y^2-x^2

35. The function $f(z)=\begin{cases}\dfrac{(\overline{z})^2}{z^2} & \text{if } z\neq 0\\ 0 & \text{if } z=0\end{cases}$

(a.) Satisfy C-R equation at $z=0$

(b.) Is not continuous at $z=0$

(c.) Is differentiable at $z=0$

(d.) Is analytic at $z=1$

36. The harmonic conjugate of $u(x,y)=x^2-y^2+xy$ is

(a.) x^2-y^2-xy

(b.) x^2+y^2-xy

(c.) $2xy+\frac{1}{2}(y^2-x^2)$

(d.) $\frac{1}{2}xy+2(y^2-x^2)$

37. Let $f:\mathbb{C}\to\mathbb{C}$ be given by

$$f(z)=\begin{cases}\dfrac{(\overline{z})^2}{z} & \text{when } z\neq 0\\ 0 & \text{when } z=0\end{cases}$$

(a.) Is not continuous at $z=0$

(b.) It differentiable but not analytic at $z=0$

(c.) Is analytic at $z=0$

(d.) Satisfy the C-R equations at $z=0$.

38. The complex analytic function $f(z)$ with the imaginary part $e^x(y\cos y+x\sin y)$ is

(a.) ze^{z+c}

(b.) $(z+c)e^z$

(c.) ze^z

(d.) $(z^2+z)e^{z^2+z}$

39. For $z\in\mathbb{C}$, define $f(z)=\dfrac{e^z}{e^z-1}$. Then

(a.) f is entire

(b.) The only singularities of f are poles

(c.) f has infinitely many poles on the imaginary axis

(d.) Each pole of f is simple

40. Let $f(z)=\dfrac{z-1}{\exp\left(\dfrac{2\pi i}{z}\right)-1}$. Then,

(a.) f has an isolated singularity at $z=0$.

(b.) f has a removable singularity at $z=1$.

(c.) f has infinitely many poles.

(d.) each pole of f is of order 1.

41. For the function $f(z)=\sin\left(\frac{1}{\cos(1/z)}\right)$, the point $z=0$ is

(a.) a removable singularity

(b.) a pole

(c.) an essential singularity

(d.) a non-isolated singularity

42. An example of a function with a non-isolated essential singularity at $z=2$ is

(a.) $\tan\left(\frac{1}{z-2}\right)$

(b.) $\sin\left(\frac{1}{z-2}\right)$

(c.) $e^{-(z-2)}$

(d.) $\tan\left(\frac{z-2}{z}\right)$

43. For the function $f(z)=\sin\left(\frac{1}{z}\right), z=0$ is a

(a.) Removable singularity

(b.) simple pole

(c.) branch point

(d.) Essential singularity

44. If $f(z)=z^3$ then it

(a.) has an essential singularity at $z=\infty$

(b.) has a pole of order 3 at $z=\infty$

(c.) has a pole of order 3 at $z=0$

(d.) is analytic at $z=\infty$.

45. Let f be an entire function on $\mathbb{C}$. Let $g(z)=\overline{f(\bar{z})}$. Which of the following statements is/are correct?

(a.) If $f(z)\in\mathbb{R}$ for all $z\in\mathbb{R}$ then $f=g$

(b.) If $f(z)\in\mathbb{R}$ for all $z\in\{z \mid \operatorname{Im} z=0\}\cup\{z \mid \operatorname{Im} z=a\}$ for some $a>0$, then $f(z+ia)=f(z-ia)$ for all $z\in\mathbb{C}$

(c.) If $f(z)\in\mathbb{R}$ for all $z\in\{z \mid \operatorname{Im} z=0\}\cup\{z \mid \operatorname{Im} z=a\}$, for some $a>0$, then $f(z+ia)=f(z)$ for all $z\in\mathbb{C}$

(d.) If $f(z)\in\mathbb{R}$ for all $z\in\{z \mid \operatorname{Im} z=0\}\cup\{z \mid \operatorname{Im} z=a\}$, for some $a>0$, then $f(z+ia)=f(z)$ for all $z\in\mathbb{C}$

Assignment Sheet - 2

1. $\int_{|z+1|=2} \frac{z^2}{4-z^2}dz =$

(a.) 0.

(b.) $-2\pi i$.

(c.) $2\pi i$.

(d.) 1.

2. Let $\gamma_k = \{ke^{ik\theta} : 0 \le \theta \le 2\pi\}$ for $k = 1, 2, 3$. Which of the following are necessarily correct?

(a.) $\frac{1}{2\pi i}\int_{\gamma_k} \frac{1}{z}dz = 0$ for $k = 1, 2, 3$

(b.) $\frac{1}{2\pi i}\int_{\gamma_1} \frac{1}{z}dz = 1$

(c.) $\frac{1}{2\pi i}\int_{\gamma_2} \frac{1}{z}dz = 4$

(d.) $\frac{1}{2\pi i}\int_{\gamma_3} \frac{1}{z}dz = 3$

3. Let $I_r = \int_{C_r} \frac{dz}{z(z-1)(z-2)}$, where $C_r = \{z \in \mathbb{C} : |z| = r\}, r > 0$. Then

(a.) $I_r = 2\pi i$ if $r \in (2, 3)$

(b.) $I_r = \frac{1}{2}$ if $r \in (0, 1)$

(c.) $I_r = -2\pi i$ if $r \in (1, 2)$

(d.) $I_r = 0$ if $r > 3$

4. Let $f(z) = \frac{\cos(2\pi z)}{z^2+1}$ and $c = \left\{z : \left|z - \frac{1}{2}i\right| = \frac{1}{4}\right\}$ then $\int_c f(z)dz$ is equal to

(a.) zero

(b.) πi

(c.) $-\pi i$

(d.) $2\pi i$

5. Let $\gamma(t) = e^{it}, 0 \le t \le 2\pi$ then $\frac{1}{2\pi i}\int_\gamma \frac{z^2}{\left(z - \frac{1}{2}\right)}dz$ is equal to

(a.) 1/4

(b.) 2

(c.) 4

(d.) 0

6. Let $f(z) = \frac{z^2+3z+2}{z^2-1}$ and $c = \left\{z : \left|z - \frac{1}{2}\right| = 1\right\}$ is then $\int_c f(z)dz$ is equal to

(a.) zero

(b.) $2\pi i$

(c.) $4\pi i$

(d.) $6\pi i$

7. $f : \mathbb{C} \to \mathbb{C}$ is a non-constant analytic function which of the following possible?

(a.) $\operatorname{Re} f(z) = 1$ for all z.

(b.) $f(z)$ takes only integer values.

(c.) $f(z)$ is an integer for every integer n.

(d.) $f(z)$ is bounded.

8. The contour integral $\int_C \frac{\cos z}{z}dz$ where C is circle $|z| = 1$ is

(a.) $2\pi i$

(b.) zero

(c.) 1

(d.) does not exists

9. $\int_C z^2 dz$, where C is circle with centre 0 and radius 2 equal.

(a.) $4\pi^2$

(b.) $2\pi i$

(c.) 4

(d.) 0

10. $\Upsilon(t)=2e^{it}$ where $t\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ then $\int_{\Upsilon} z^2dz=?$

(a.) $-i$

(b.) 0

(c.) i

(d.) $1/2$

11. If f and g are analytic on the same region D and if $f(z)g(z)=0$ all z in the region D then

(a.) at least one of f and g vanishes identically on D.

(b.) f and g both vanishes everywhere on D.

(c.) f vanishes only at finitely. Many points of D and g never vanishes

(d.) f and g vanish only at finitely many points of D.

12. Let $C=\{z\in\mathbb{C}:|z-i|=2\}$. Then $\frac{1}{2\pi}\oint_C \frac{z^2-4}{z^2+4}dz$ is equal to ___________

13. Let $\Omega=\{z\in\mathbb{C}:\text{Im}(z)>0\}$ and let C be a smooth curve lying in Ω with initial point $-1+2i$ and final point $1+2i$. The value of $\int_c \frac{1+2z}{1+z}dz$ is

(a.) $4-\frac{1}{2}\ln 2+i\frac{\pi}{4}$

(b.) $-4+\frac{1}{2}\ln 2+i\frac{\pi}{4}$

(c.) $4+\frac{1}{2}\ln 2-i\frac{\pi}{4}$

(d.) $4-\frac{1}{2}\ln 2+i\frac{\pi}{2}$

14. Let $\int_c\left[\frac{1}{(z-2)^4}-\frac{(a-2)^2}{z}+4\right]dz=4\pi$, where the close curve C is the triangle having vertices at i, $\left(\frac{-1-i}{\sqrt{2}}\right)$ and $\left(\frac{1-i}{\sqrt{2}}\right)$,

(a.) $1+i$

(b.) $2+i$

(c.) $3+i$

(d.) $4+i$

15. Let $I=\int_c \frac{f(z)}{(z-1)(z-2)}dz$, where $f(z)=\sin\frac{\pi z}{2}+\cos\frac{\pi z}{2}$ and C is the curve $|z|=3$ oriented anti-clockwise. Then the value of I is

(a.) $4\pi i$

(b.) 0

(c.) $-2\pi i$

(d.) $-4\pi i$

16. Let S be open unit disk and $f:S\to\mathbb{C}$ be a real valued analytic function with $f(0)=1$, then the set $\{z\in S: f(z)\neq 1\}$ is

(a.) empty

(b.) non empty finite

(c.) countable infinite

(d.) uncountable

17. Which one of the following does not hold for all continuous functions $f:[-\pi,\pi]\to\mathbb{C}$?

(a.) If $f(-t)=f(t)$ for each $t\in[-\pi,\pi]$ then $\int_{-\pi}^{\pi} f(t)dt=2\int_0^{\pi} f(t)dt$

(b.) If $f(-t)=-f(t)$ for each $t\in[-\pi,\pi]$ then $\int_{-\pi}^{\pi} f(t)dt=0$

(c.) $\int_{-\pi}^{\pi} f(-t)dt=-\int_{-\pi}^{\pi} f(t)dt$

(d.) there is an α with $-\pi<\alpha<\pi$ such that $\int_{-\pi}^{\pi} f(t)dt=2\pi f(\alpha)$

18. Let S be the positive oriented circle given by $|z-2i|=2$, then the value of $\int_S \frac{dz}{z^2+4}$ is

(a.) $-\frac{\pi}{2}$

(b.) $\frac{\pi}{2}$

(c.) $\frac{-i\pi}{2}$

(d.) $\frac{i\pi}{2}$

19. Let $f(z)$ be an analytic function, then the value of $\int_0^{2\pi} f(e^{it})\cos t\,dt$ equals

(a.) 0

(b.) $2\pi f(0)$

(c.) $2\pi f'(0)$

(d.) $\pi f'(0)$

20. Define $f:\mathbb{C}\to\mathbb{C}$ by $f(z)=\begin{cases}0 & \text{if } Re(z)=0 \text{ or } \mathrm{Im}(z)=0\\ z & \text{otherwise}\end{cases}$ then the set of point where f is analytic is

(a.) $\{z:\mathrm{Re}(z)\neq 0 \text{ and } \mathrm{Im}(z)\neq 0\}$

(b.) $\{z:\mathrm{Re}(z)\neq 0\}$

(c.) $\{z:\mathrm{Re}(z)\neq 0 \text{ or } \mathrm{Im}(z)\neq 0\}$

(d.) $\{z:\mathrm{Im}(z)\neq 0\}$

21. Let r be a simple closed curve in the complex plane then the set of all possible values of $\oint_r \frac{dz}{z(1-z^2)}$ is

(a.) $\{0,\pm\pi i\}$

(b.) $\{0,\pm\pi i,2\pi i\}$

(c.) $\{0,\pm\pi i,\pm 2\pi i\}$

(d.) $\{0\}$.

22. For the positively oriented unit circle, $\int_{|z|=1}\frac{2\mathrm{Re}(z)}{z+2}dz$ equals -

(a.) 0

(b.) πi

(c.) $2\pi i$

(d.) $4\pi i$

23. Let r be curve: $r=2+4\cos\theta$. $(0\le\theta\le 2\pi)$.

If $I_1=\int_r\frac{dz}{(z-1)}$ and $I_2=\int_r\frac{dz}{(z-3)}$ then

(a.) $I_1=2I_2$

(b.) $I_1=I_2$

(c.) $2I_1=I_2$

(d.) $I_1=0, I_2\neq 0$

24. Let $I=\int_C\frac{\cot(\pi z)}{(z-i)^2}dz$, where C is contour $4x^2+y^2=2$ (counter clockwise) then I is equal to

(a.) 0

(b.) $-2\pi i$

(c.) $2\pi i\left(\frac{\pi}{\sinh^2\pi}-\frac{1}{\pi}\right)$

(d.) $\frac{-2\pi^2 i}{\sinh^2\pi}$

25. Let r be any circle enclosing the origin and oriented counter-clockwise. then the value of the integral $\int_r\frac{\cos z}{z^2}dz$ is

(a.) $2\pi i$

(b.) 0

(c.) $-2\pi i$

(d.) Undefined.

26. The value of the integral $\int_c\frac{dz}{z^2-1}$ $c:|z|=4$ is equal to

(a.) πi

(b.) 0

(c.) $-\pi i$

(d.) $2\pi i$

27. The value of integral $\int_C\frac{\sin\pi z^2+\cos\pi z^2}{(z-4)(z-2)}dz$ where C is the circle $|z|=3$ traced anti-clockwise is

(a.) $-2i\pi$

(b.) $i\pi$

(c.) $-i\pi$

(d.) $2i\pi$

28. The value of $\int_{|z|=2}\frac{e^z}{(z+1)^2}dz$ is

(a.) $2\pi i e^{-1}$

(b.) $\frac{8\pi i}{3}e^{-2}$

(c.) $\frac{2\pi i}{3}e^{-2}$

(d.) 0

29. Let, for each $n \geq 1, C_n$ be the open disc in $\mathbb{R}^2$, with centre at the point $(n,0)$ and radius equal to n. Then $C = \bigcup_{n\geq 1} C_n$

(a.) $\{(x,y)\in\mathbb{R}^2 : x>0 \text{ and } |y|<x\}$.

(b.) $\{(x,y)\in\mathbb{R}^2 : x>0 \text{ and } |y|<2x\}$.

(c.) $\{(x,y)\in\mathbb{R}^2 : x>0 \text{ and } |y|<3x\}$.

(d.) $\{(x,y)\in\mathbb{R}^2 : x>0\}$.

30. Let f be an entire function. Which of the following statements are correct?

(a.) f is constant if the range of f is contained in a straight line.

(b.) f is constant if f has uncountable many zeros.

(c.) f is constant if f is bounded on $\{z\in\mathbb{C} : \text{Re}(z)\leq 0\}$.

(d.) f is constant if the real part of f is bounded.

31. Let $f(z)=\sum_{n=0}^{\infty} a_n z^n$ be an entire function and let r be a positive real number. then

(a.) $\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq \sup_{|z|=r} |f(z)|^2$

(b.) $\sup_{|z|=r} |f(z)|^2 \leq \sum_{n=0}^{\infty} |a_n|^2 r^{2n}$

(c.) $\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq \frac{1}{2\pi} |f(re^{i\theta})|\, d\theta$

(d.) $\sup_{|z|=r} |f(z)|^2 \leq \frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^2\, d\theta$

32. Let f be a non constant entire function. Which of the following properties is possible for f for each $z\in\mathbb{C}$?

(a.) $\text{Re} f(z) = \text{Im} f(z)$.

(b.) $|f(z)|<1$.

(c.) $\text{Im} f(z)<0$.

(d.) $f(z)\neq 0$.

33. Consider the function $f,g:\mathbb{C}\to\mathbb{C}$ defined by $f(z)=e^z$, $g(z)=e^{iz}$. Let $S=\{z\in\mathbb{C} : \text{Re}\, z\in[-\pi,\ \pi]\}$. Then

(a.) f is an onto entire function

(b.) g is a bounded function on $\mathbb{C}$

(c.) f is bounded on S

(d.) g is bounded on S.

34. Let f be an entire function. If $\text{Im} f \geq 0$, then

(a.) Re f is constant

(b.) f is constant

(c.) $f=0$

(d.) $f(z)=z, z\in D$

35. Let f be an entire function. If $\text{Re} f$ is bounded then

(a.) $Im\, f$ is constant

(b.) f is constant

(c.) $f\equiv 0$

(d.) f' is a non zero constant

36. f is an entire function. If f satisfies the following two equations $f(z+1)=f(z)$ and $f(z+i)=f(z)$ for every z in $\mathbb{C}$ then

(a.) $f'(z)=f(z)$

(b.) $f(z)\in\mathbb{R}\ \forall z$

(c.) $f\equiv$ constant

(d.) f is a non constant polynomial.

37. Let p be a polynomial in one complex variable. Suppose all zeroes of p are in the upper half plane $H=\{z\in\mathbb{C} | \text{Im}(z)>0\}$. Then

(a.) $\text{Im}\frac{p'(z)}{p(z)}>0$ for $z\in\mathbb{R}$.

(b.) $\text{Re}\, i\frac{p'(z)}{p(z)}<0$ for $z\in\mathbb{R}$.

(c.) $\text{Im}\frac{p'(z)}{p(z)}>0$ for $z\in\mathbb{C}$, with $\text{Im}\, z<0$.

(d.) $\text{Im}\frac{p'(z)}{p(z)}>0$ for $z\in\mathbb{C}$, with $\text{Im}\, z>0$.

38. Let a, b, c be non-collinear points in the complex plane and let Δ denote the closed triangular region of the plane with vertices a, b, c. For $z \in \Delta$, let $h(z) = |z-a| \cdot |z-b| \cdot |z-c|$. The maximum value of the function h:

(a.) is not attained at any point of Δ.

(b.) is attained at an interior point of Δ.

(c.) is attained at the centre of gravity of Δ.

(d.) is attained at a boundary point of Δ.

39. The maximum modulus of e^{z^2} on the set $S = \{z \in \mathbb{C} : 0 \le \mathrm{Re}(z) \le 1, 0 \le Im(z) \le 1\}$ is

(a.) 2/e

(b.) e

(c.) e+1

(d.) e^2

40. Let T be the closed unit disk and ∂T be the unit circle. Then which one of following holds for every analytic function $f : T \to C$.

(a.) $|f|$ attains its minimum and maximum on ∂T.

(b.) $|f|$ attains its minimum on ∂T but need not attain its maximum on ∂T.

(c.) $|f|$ attains maximum on ∂T but need not attain its minimum on ∂T.

(d.) $|f|$ need not attain its maximum on ∂T and also it need not attain its minimum on ∂T.

41. Let $f(z) = 2z^2 - 1$ then the maximum value of $|f(z)|$ on the unit disc $D = \{z \in \mathbb{C} : |z| \le 1\}$ equals.

(a.) 1

(b.) 2

(c.) 3

(d.) 4

Assignment Sheet -3

1. If f and g are analytic on the same region D and if $f(z)g(z)=0$ all z in the region D then
(a.) at least one of f and g vanishes identically on D.
(b.) f and g both vanishes everywhere on D.
(c.) f vanishes only at finitely many points of D and g never vanishes
(d.) f and g vanish only at finitely many points of D.

2. The value of $\frac{i}{4-\pi}\int_{|z|=4}\frac{dz}{z\cos(z)}$ is equal to ______

3. Let S be the disk $|z|<2$ in complex plane and let $f:S\to\mathbb{C}$ be an analytic function such that $f\left(1+\frac{\sqrt{2}}{n}i\right)=\frac{-2}{n^2}$ for each natural number n, then $f(\sqrt{2})$ is equal to
(a.) $3-2\sqrt{2}$
(b.) $3+2\sqrt{2}$
(c.) $2-3\sqrt{2}$
(d.) $2+3\sqrt{2}$

4. Let f be an analytic function defined on the open unit disc in $\mathbb{C}$. Then f is constant if
(a.) $f\left(\frac{1}{n}\right)=0$ for all $n\geq 1$.
(b.) $f(z)=0$ for all $|z|=\frac{1}{2}$.
(c.) $f\left(\frac{1}{n^2}\right)=0$ for all $n\geq 1$.
(d.) $f(z)=0$ for all $z\in(-1,1)$.

5. Let $\sum_0^\infty a_n z^n$ be a convergent power series such that $\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=R>0$. Let p be a polynomial of degree d. then the radius of convergence of the power series $\sum_{n=0}^\infty p(n)a_n z^n$ equals
(a.) R
(b.) D
(c.) $R.d$
(d.) $R+d$

6. Let $p(x)$ be a polynomial of the real variable x of degree $k\geq 1$. Consider the power series $f(z)=\sum_{n=0}^\infty p(n)z^n$ where z is a complex variable. Then the radius of convergence of $f(z)$ is
(a.) 0
(b.) 1
(c.) k
(d.) ∞

7. For $z\in\mathbb{C}$, define $f(z)=\frac{e^z}{e^z-1}$. Then
(a.) f is entire
(b.) The only singularities of f are poles
(c.) f has infinitely many poles on the imaginary axis
(d.) Each pole of f is simple

8. Let f be an entire function. Suppose, for each $a\in\mathbb{R}$, there exists at least one coefficient c_n in $f(z)=\sum_{n=0}^\infty c_n(z-a)^n$, which is zero. Then
(a.) $f^{(n)}(0)=0$ for infinitely many $n\geq 0$
(b.) $f^{(2n)}(0)=0$ for every $n\geq 0$
(c.) $f^{(2n+1)}(0)=0$ for every $n\geq 0$
(d.) There exists $k\geq 0$ $f^{(n)}(0)=0$ for all $n\geq k$

9. Let f be a holomorphic function on the unit disc $\{|z|<1\}$ in the complex plane. Which of the following is/are necessarily true?
(a.) If for each positive integer n we have $f\left(\frac{1}{n}\right)=\frac{1}{n^2}$ then $f(z)=z^2$ on the unit disc.
(b.) If for each positive integer n we have $f\left(1-\frac{1}{n}\right)=\left(1-\frac{1}{n}\right)^2$ then $f(z)=z^2$ on the unit disc.
(c.) f cannot satisfy $f\left(\frac{1}{n}\right)=\frac{(-1)^n}{n}$ for each positive integer n.
(d.) f cannot satisfy $f\left(\frac{1}{n}\right)=\frac{1}{n+1}$ for each positive integer n.

10. Let $f(z)=\dfrac{z-1}{\exp\left(\dfrac{2\pi i}{z}\right)-1}$. Then,

(a.) f has an isolated singularity at $z=0$.

(b.) f has a removable singularity at $z=1$.

(c.) f has infinitely many poles.

(d.) each pole of f is of order 1.

11. Consider the function $f(z)=z^2(1-\cos z), z\in\mathbb{C}$. Which of the following are correct?

(a.) The function f has zeros of order 2 at 0

(b.) The function f has zeros of order 1 at $2\pi n,\ n=\pm1,\ \pm2,...$

(c.) The function f has zeros of order 4 at 0

(d.) The f has zeros of order 2 at $2\pi n, n=\pm1,\pm2,...$

12. Consider the power series $\sum_{n=1}^{\infty} z^{n!}$. The radius of convergence of this series is

(a.) 0

(b.) ∞

(c.) 1

(d.) A real number greater than 1

13. Which of the following functions f are entire functions and have simple zeros at $z=ik$ for all $k\in\mathbb{Z}$.

(a.) $f(z)=a_nz^n+a_{n-1}z^{n-1}+....+a_0$ for some $n\geq1$ and some $a_0,a_1,...,a_n\in\mathbb{C}$

(b.) $f(z)=a\sin 2\pi i z$, for some $a\in\mathbb{C}$

(c.) $f(z)=b\cos 2\pi\left(iz-\dfrac{1}{4}\right)$, for some $b\in\mathbb{C}$

(d.) $f(z)=e^{cz}$, for some $c\in\mathbb{C}$.

14. The power series $\sum_{n=0}^{\infty}3^{-n}(z-1)^{2n}$ converges if

(a.) $|z|\leq3$

(b.) $|z|<\sqrt{3}$

(c.) $|z-1|<\sqrt{3}$

(d.) $|z-1|\leq\sqrt{3}$

15. At $z=0$, the function $f(z)=\exp\left(\dfrac{z}{1-\cos z}\right)$ has

(a.) a removable singularity

(b.) a pole

(c.) an essential singularity

(d.) The Laurent expansion of $f(z)$ around $z=0$ has infinitely many positive and negative powers of z.

16. Consider the power series $\sum_{n\geq1}a_nz^n$ where $a_n=$ number of divisors of n^{50}. Then the radius of convergence of $\sum_{n\geq1}a_nz^n$ is

(a.) 1

(b.) 50

(c.) $\dfrac{1}{50}$

(d.) 0

17. Let $D=\{z\in\mathbb{C}:|z|<1\}$ and let $f_n:D\to\mathbb{C}$ be defined by $f_n(z)=\dfrac{z^n}{n}$ for $n=1,2,...$

Then

(a.) The sequence $\{f_n(z)\}$ and $\{f_n'(z)\}$ converge uniformly on D

(b.) The series $\sum_{n=1}^{\infty}f_n(z)$ converges uniformly on D

(c.) The series $\sum_{n=1}^{\infty}f_n'(z)$ converges for each $z\in D$

(d.) The sequence $\{f_n''(z)\}$ does not converge unless $z=0$

18. Let f, g be holomorphic functions defined as $A \cup D$, where

$A = \left\{ z \in \mathbb{C} : \frac{1}{2} < |z| < 1 \right\}$ and

$D = \{ z \in \mathbb{C} : |z-2| < 1 \}$

Which of the following statements are correct?

(a.) If $f(z)g(z) = 0$ for all $z \in A \cup D$, then either $f(z) = 0$ for all $z \in A$ or $g(z) = 0$ for all $z \in A$

(b.) If $f(z)g(z) = 0$ for all $z \in D$, then either $f(z) = 0$ for all $z \in D$ or $g(z) = 0$ for all $z \in D$

(c.) If $f(z)g(z) = 0$ for all $z \in A$, then either $f(z) = 0$ for all $z \in A$ or $g(z) = 0$ for all $z \in A$

(d.) If $f(z)g(z) = 0$ for all $z \in A \cup D$, then either $f(z) = 0$ for all $z \in A \cup D$, or $g(z) = 0$ for all $z \in A \cup D$.

19. Let $f : \mathbb{C} \to \mathbb{C}$ be an entire function and let $g : \mathbb{C} \to \mathbb{C}$ be defined by $g(z) = f(z) - f(z+1)$ for $z \in \mathbb{C}$. Which of the following statements are true?

(a.) If $f\left(\frac{1}{n}\right) = 0$ for all positive integers n, then f is a constant function

(b.) If $f(n) = 0$ for all positive integers n, then f is a constant function

(c.) If $f\left(\frac{1}{n}\right) = f\left(\frac{1}{n} + 1\right)$ for all positive integers n, then g is a constant function

(d.) If $f(n) = f(n+1)$ for all positive integers n, then g is a constant function.

20. The power series $\sum_{0}^{\infty} 2^{-n} z^{2n}$ converges if

(a.) $|z| \le 2$

(b.) $|z| < 2$

(c.) $|z| \ge \sqrt{2}$

(d.) $|z| < \sqrt{2}$

21. Let $\mathbf{D} = \{ z \in \mathbb{C} : |z| < 1 \}$ be the unit disc. Let $f : \mathbf{D} \to \mathbb{C}$ be an analytic function satisfying $f\left(\frac{1}{n}\right) = \frac{2n}{3n+1}$ for $n \ge 1$. Then

(a.) $f(0) = \frac{2}{3}$

(b.) f has a simple pole at $z = -3$

(c.) $f(3) = \frac{1}{3}$

(d.) No such f exists.

22. At $z = 0$ the function $f(z) = \frac{e^z + 1}{e^z - 1}$ has

(a.) A removable singularity

(b.) A pole

(c.) An essential singularity

(d.) The residue of $f(z)$ at $z = 0$ is 2.

23. The poles of function $\frac{\cos z}{z^2 + 1}$ are

(a.) Integral multiple of 2π

(b.) only

(c.) i and $-i$

(d.) ± 1

24. Let r be radius of convergence of $\sum_{n=0}^{\infty} a_n z^n$ let $b_n = a_{n+2}$ for all n then the radius of convergence of $\sum b_n z^n$ is

(a.) $r + 2$

(b.) r

(c.) r^2

(d.) $r - 2$

25. In the Laurrent series of $\frac{\sin z}{z^2}$ at $z = 0$ the coefficient of $\frac{1}{z}$ is

(a.) zero

(b.) π

(c.) -1

(d.) 1

26. The function $\frac{\sin z}{z^2}$ has

(a.) pole of order 2 at origin with residue 1.

(b.) pole of order 1 at origin with residue 1.

(c.) pole of order 1 at origin with residue 2.

(d.) an essential singularity at origin with residue 1.

27. $f(z)=\dfrac{\sin z}{z^2}$ has

(a.) no singularities

(b.) an essential singularity at $z=0$

(c.) a removable singularity at $z=0$

(d.) a pole of order 1 at $z=0$

28. The coeff. of $\frac{1}{z}$ in the Laurent series of $\dfrac{\sin 2z}{z^2}$ is

(a.) 0

(b.) 2

(c.) -1

(d.) 1

29. An example of a power series whose radius of convergence is 0, is

(a.) $\sum_{n=0}^{\infty} z^n$

(b.) $\sum_{n=0}^{\infty} nz^n$

(c.) $\sum_{n=0}^{\infty} \frac{1}{z^n}$

(d.) $\sum_{n=0}^{\infty} n^n z^n$.

30. Consider the series $f(z)=\sum_{n=-\infty}^{\infty}\frac{z^n}{n!}$ in the region $\{z\in\mathbb{C}:1<|z|<2\}$ which of the following is true?

(a.) There is a z in this region for which the series does convergence.

(b.) f is not analytic

(c.) f takes all values in the complex plane.

(d.) f is bounded.

31. Let f and g be analytic functions on the unit disk such that $f\left(\frac{1}{n}\right)=g\left(\frac{1}{n}\right)$ for all non-zero integers.

(a.) f and g are equal

(b.) f and g are constant

(c.) $f(z)-g(z)=\sin(\pi/2)$

(d.) $f(z)\neq g(z)$ if z is purely imaginary.

32. The function $f(z)=e^z-2i$

(a.) has no zeros

(b.) has only one zero

(c.) has a countable numbers of zero

(d.) has an uncountable numbers of zeros

33. The function $f(z)=\tan z$

(a.) has poles at $z=(2n+1)\frac{\pi}{2}$ $\quad n\in z$

(b.) is an entire function

(c.) has no zeros in C

(d.) has removable singularity at $z=0$

34. Consider the power series $\sum_{n=0}^{\infty} a_n z^n$, where $a_n=\begin{cases}\frac{1}{3^n} & \text{if } n \text{ is even}\\ \frac{1}{5^n} & \text{if } n \text{ is odd.}\end{cases}$ The radius of convergence of the series is equal to ________

35. Let $\sum_{n=-\infty}^{\infty} a_n z^n$ be the Laurent series expansion of $f(z)=\dfrac{1}{2z^2-13z+15}$ in the annulus $\frac{3}{2}<|z|<5$. Then $\dfrac{a_1}{a_2}$ is equal to ________

36. The radius of convergence of the power series $\sum_{n=0}^{\infty} 4^{(-1)^n n} z^{2n}$ is ____________

37. The coefficient of $(z-\pi)^2$ in the Taylor series expansion of $f(z)=\begin{cases}\dfrac{\sin z}{z-\pi} & \text{if } z\neq\pi\\ -1 & \text{if } z=\pi\end{cases}$ around π is

(a.) $\frac{1}{2}$

(b.) $-\frac{1}{2}$

(c.) $\frac{1}{6}$

(d.) $-\frac{1}{6}$

38. Let $\sum_{n=\infty}^{\infty} b_n z^n$ be the Laurent series expansion of the function $\frac{1}{z\sinh z}$, $|z|<\pi$. Then which one of the following is correct?

(a.) $b_{-2}=1, b_0=-\frac{1}{6}, b_2=\frac{7}{360}$

(b.) $b_{-3}=1, b_{-1}=-\frac{1}{6}, b_1=\frac{7}{360}$

(c.) $b_{-2}=0, b_0=-\frac{1}{6}, b_2=\frac{7}{360}$

(d.) $b_0=1, b_2=-\frac{1}{6}, b_4=\frac{7}{360}$

39. For the function $f(z)=\sin\left(\frac{1}{\cos(1/z)}\right)$, the point $z=0$ is

(a.) a removable singularity

(b.) a pole

(c.) an essential singularity

(d.) a non-isolated singularity

40. Let $\sum_{n=-\infty}^{\infty} a_n(z+1)^n$ be the Laurent series expansion of $f(z)=\sin\left(\frac{z}{z+1}\right)$. Then $a_{-2}=$

(a.) 1

(b.) 0

(c.) cos (1)

(d.) $\frac{-1}{2}\sin(1)$

41. Consider the function $f(x)=\frac{e^{iz}}{z(z^2+1)}$ the residue of $f(z)$ at the isolated singular point in the upper half plane $\{z=x+iy\in\mathbb{C}:y>0\}$ is

(a.) $-\frac{1}{2e}$

(b.) $-\frac{1}{e}$

(c.) $\frac{e}{2}$

(d.) 1

42. Let $f(z)=\cos z-\frac{\sin z}{z}$ for non-zero $z\in c$ and $f(0)=0$. Also let $g(z)=\sinh z$ for $z\in c$, then $\frac{g(z)}{zf(z)}$ has a pole at $z=0$ of order

(a.) 1

(b.) 2

(c.) 3

(d.) greater than 3

43. Let $S=\{0\}\cup\left\{\frac{1}{4n+7}:n=1,2,.....\right\}$ then number of analytic functions which vanish only on S.

(a.) Infinite

(b.) 0

(c.) 1

(d.) 2

44. It is given that $\sum_{n=0}^{\infty} a_n z^n$ converges at $z=3+i4$, then the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n z^n$ is

(a.) ≤ 5

(b.) ≥ 5

(c.) <5

(d.) >5.

45. Let $f(z)=\frac{1}{z^2-3z+2}$ then the coefficient of $\frac{1}{z^3}$ in the Laurrent series expansion of $f(z)$ for $|z|>2$ is

(a.) 0

(b.) 1

(c.) 3

(d.) 5

46. The coefficient of $\frac{1}{z}$ in expansion of $Log\left(\frac{z}{z-1}\right)$ valid in $|z|>1$ is

(a.) -1

(b.) 1

(c.) -1/2

(d.) 1/2

47. If $\sin z = \sum_{n=0}^{\infty} a_n \left(z-\frac{\pi}{4}\right)^n$, then a_6 equal

(a.) 0

(b.) $\frac{1}{720}$

(c.) $\frac{1}{720\sqrt{2}}$

(d.) $-\frac{1}{720\sqrt{2}}$

48. In the Laurrent series expansion of $f(z)=\frac{1}{z-1}-\frac{1}{z-2}$ valid in region $|z|>2$ then the coeff. of $\frac{1}{z^2}$ is

(a.) -1

(b.) 0

(c.) 1

(d.) 2

49. Let $f(z)$ be an analytic function with a simple pole at $z=1$ and a double pole at $z=2$ with residues 1 and -2 respectively. Further if $f(0)=0$, $f(3)=-\frac{3}{4}$ and f is bounded as $z\to\infty$, then $f(z)$ must be

(a.) $z(z-3)-\frac{1}{4}+\frac{1}{(z-1)}-\frac{2}{(z-1)}+\frac{1}{(z-2)^2}$

(b.) $-\frac{1}{4}+\frac{1}{(z-1)}-\frac{2}{(z-2)}+\frac{1}{(z-2)^2}$

(c.) $\frac{1}{(z-1)}-\frac{2}{(z-2)}+\frac{5}{(z-2)^2}$

(d.) $\frac{15}{4}+\frac{1}{z-1}-\frac{2}{z-2}+\frac{7}{(z-2)^2}$

50. An example of a function with a non-isolated essential singularity at $z=2$ is

(a.) $\tan\left(\frac{1}{z-2}\right)$

(b.) $\sin\left(\frac{1}{z-2}\right)$

(c.) $e^{-(z-2)}$

(d.) $\tan\left(\frac{z-2}{z}\right)$

51. Let $f(z)=u(x,y)+iv(x,y)$ be an entire function having taylor series expansion as $\sum_{n=0}^{\infty} a_n z^n$. If $f(x)=u(x,0)$ and $f(iy)=iv(0,y)$ then

(a.) $a_{2n}=0\;\forall n$

(b.) $a_0=a_1=a_2=a_3=0.\ a_4\neq 0$

(c.) $a_{2n+1}=0$ for all n

(d.) $a_0\neq 0$ but $a_2=0$

52. The radius of convergence of the power series of the function $f(z)=\frac{1}{1-z}$ about $z=\frac{1}{4}$ is

(a.) 1

(b.) 1/4

(c.) 3/4

(d.) 0

53. For the function $f(z)=\sin\left(\frac{1}{z}\right), z=0$ is a

(a.) Removable singularity

(b.) simple pole

(c.) branch point

(d.) Essential singularity

54. The series $\sum_{n=1}^{\infty}\frac{z^n}{n\sqrt{n+1}}$ $,|z\leq 1|$ is

(a.) uniformly but not absolutely convergent

(b.) uniformly and absolutely convergent

(c.) absolutely convergent but not uniformly convergent

(d.) convergent but not uniformly convergent

55. For the function $f(z)=\frac{1-e^{-z}}{z}$, the point $z=0$ is

(a.) An essential singularity

(b.) A pole of order zero

(c.) A pole of order one

(d.) A removable singularity.

56. Expansion of function $f(z)=\frac{1}{(3-2z)}$ in power of $(z-3)$ and the radius of convergence of the series so obtained are

(a.) $-\frac{1}{3}\left[1-\frac{2}{3}(z-3)+\frac{4}{9}(z-3)^2-......\right]|z-3|<3$

(b.) $-\frac{1}{3}\left[1-\frac{2}{3}(z-3)+\frac{4}{9}(z-3)^2-......\right]$
$|z-3|<\frac{3}{2}$

(c.) $\frac{1}{3}\left[1+\frac{2}{3}(z-3)+\frac{4}{9}(z-3)^2+......\right]|z-3|<\frac{3}{2}$

(d.) $\frac{1}{3}\left[1-\frac{2}{3}(z-3)-\frac{4}{9}(z-3)^2-......\right]|z-3|<\frac{3}{2}$

57. For the function $f(z)=\frac{z-\sin z}{z^3}$, the point $z=0$ is

(a.) A pole of order 3

(b.) A pole of order 2

(c.) An essential singularity

(d.) A removable singularity

58. The singularity of $e^{\sin z}$ at $z=\infty$ is

(a.) a pole

(b.) a removable singularity

(c.) non isolated singularity

(d.) isolated essential singularity

59. Let $p(z)=a_0+a_1z+...+a_nz^n$ and $q(z)=b_1z+b_2z^2+...+b_nz^n$ be complex polynomials. If a_0, b_1 are non-zero complex numbers then the residue of $\frac{p(z)}{q(z)}$ at 0 is equal to

(a.) $\frac{a_0}{b_1}$

(b.) $\frac{a_1}{a_0}$

(c.) $\frac{a_1}{b_1}$

(d.) $\frac{a_0}{a_1}$

60. The function $\frac{1}{z(z-1)^3}$ has a pole of order p and residues r where

(a.) $p=1 \quad r=1$

(b.) $p=3 \quad r=\frac{1}{3}$

(c.) $p=3 \quad r=1$

(d.) $p=1 \quad r=0$

61. Residue of $\frac{\sin z}{z^4}$ at its singularity is

(a.) 0

(b.) 1

(c.) -1/6

(d.) 2

62. The residue of $e^{1/z}$ at $z=0$ is

(a.) 2

(b.) ∞

(c.) 1

(d.) -2

63. Let $f:\mathbb{C}\to\mathbb{C}$ be analytic for a simple pole at $z=0$ and let $g:\mathbb{C}\to\mathbb{C}$ be analytic. Then, the value of $\frac{\operatorname{Res}_{z=0}\{f(z)g(z)\}}{\operatorname{Res}_{z=0} f(z)}$ is

(a.) $g(0)$

(b.) $g'(0)$

(c.) $\lim_{z\to 0} z f(z)$

(d.) $\lim_{z\to 0} z f(z)g(z)$

64. Let $f(z)=\frac{z}{8-z^3}, z=x+iy$ then $\operatorname{Res}_{z=2} f(z)$ is

(a.) $-\frac{1}{8}$

(b.) $\frac{1}{8}$

(c.) $-\frac{1}{6}$

(d.) $\frac{1}{6}$

65. The sum of the residues at all the poles of $f(z)=\dfrac{\cot \pi z}{(z+a)^2}$, where a is constant. $(a \neq 0, \pm 1, \pm 2, ...)$ is

(a.) $\dfrac{1}{\pi}\sum_{n=-\infty}^{\infty}\dfrac{1}{(n+a)^2}-\pi \operatorname{cosec}^2 \pi a$

(b.) $-\dfrac{1}{\pi}\sum_{n=-\infty}^{\infty}\dfrac{1}{(n+a)^2}+\pi \operatorname{cosec}^2 \pi a$

(c.) $-\dfrac{1}{\pi}\sum_{n=-\infty}^{\infty}\dfrac{1}{(n+a)^2}-\pi \operatorname{cosec}^2 \pi a$

(d.) $\dfrac{1}{\pi}\sum_{n=-\infty}^{\infty}\dfrac{1}{(n+a)^2}+\pi \operatorname{cosec}^2 \pi a$

66. $\displaystyle\int_0^{2\pi}\frac{d\theta}{13-5\sin\theta}$ is equals

(a.) $-\dfrac{\pi}{6}$

(b.) $-\dfrac{\pi}{12}$

(c.) $\dfrac{\pi}{12}$

(d.) $\dfrac{\pi}{6}$

67. The residue of $\dfrac{\sin z}{z^8}$ at $z=0$ is

(a.) 0

(b.) $\dfrac{1}{7!}$

(c.) $-\dfrac{1}{7!}$

(d.) None of these

Assignment Sheet -4

1. Let $p(z,\ w)=\alpha_0(z)+\alpha_1(z)w+....+\alpha_k(z)w^k$, where $k\geq 1$ and $\alpha_0,....,\alpha_k$ are non-constant polynomials in the complex variable z. Then $\{(z,\ w)\in\mathbb{C}\times\mathbb{C}: p(z,\ w)=0\}$ is

(a.) Bounded with empty interior

(b.) Unbounded with empty interior

(c.) Bounded with nonempty interior

(d.) Unbounded with nonempty interior

2. $f(z)$ is single valued complex function then

(a.) It is rational implies it is meromorphic

(b.) It is rational if it is meromorphic

(c.) It is meromorphic if and only if it is rational

(d.) Let z is rational function as it has simple poles as its singularities

3. If $\phi(z)=\mathrm{Re}(z)+f(z)$ where $f(z)$ is meromorphic. Then

(a.) $\phi(z)$ is meromorphic

(b.) $\phi(z)$ and $f(z)$ has same number of singularities

(c.) $\phi(z)$ is analytic in every closed and bounded region provided it has no poles

(d.) $\phi(z)$ has no singularities

4. Select the incorrect statement

(a.) There doesn't exists an analytic function f in neighbourhood of 0 whose square is z.

(b.) There doesn't exists non constant entire function $f(z)$ s.t. $e^{f(z)}$ is bounded

(c.) Every rational function is meromorphic

(d.) None of these

5. If M = set of all meromorphic function

R = set of rational function

I = set of functions having essential singularity at infinity then

(a.) $M\cap I$ is empty

(b.) $R\cap I$ is empty

(c.) $M\cap R$ is empty

(d.) Intersection of any pair of $M\cap R$ or I is non empty

6. If real part of a function is constant in a domain D, then f is

(a.) Meromorphic

(b.) Number of poles of function are finite

(c.) No relation of with poles

(d.) Can't say any thing

7. If f is an meromorphic function on C then

(a.) Poles of f are isolated but not zeroes

(b.) Zeroes of f are isolated but poles of f need not be isolated

(c.) Zeroes and poles of f are isolated

(d.) Neither poles nor zeroes of f are isolated

8. Let f,g be meromorphic functions on $\mathbb{C}$. If f has a zero of order k at $z=a$ and g has a pole of order m at $z=a$ then $g(f(z))$ has

(a.) zero of order km at $z=a$

(b.) pole of order km at $z=a$

(c.) zero of order $|k-m|$ at $z=a$

(d.) pole of order $|k-m|$ at $z=a$

9. Let f be a meromorphic function on $\mathbb{C}$ such that $|f(z)|\geq|z|$ at each z where f is holomorphic. Then which of the following is/are true?

(a.) The hypotheses are contradictory, so no such f exists.

(b.) Such an f exists.

(c.) There is a unique f satisfying the given conditions.

(d.) There is an $A\in\mathbb{C}$ with $|A|\geq 1$ such that $f(z)=Az$ for each $z\in\mathbb{C}$.

10. Let f be an analytic function defined on $\mathbb{D}=\{z\in\mathbb{C}:|z|<1\}$ such that the range of f is confined in the set $\mathbb{C}\setminus(-\infty,\ 0]$. Then

(a.) f is necessarily a constant function.

(b.) There exists an analytic function g on $\mathbb{D}$ such that $g(z)$ is a square root of $f(z)$ for each $z\in\mathbb{D}$.

(c.) There exists an analytic function g on $\mathbb{D}$ such that $\mathrm{Re}\,g(z)\geq 0$ and $g(z)$ is a square root of $f(z)$ for each $z\in\mathbb{D}$

(d.) There exists an analytic function g on $\mathbb{D}$ such that $\mathrm{Re}\,g(z)\leq 0$ and $g(z)$ is a square root of $f(x)$ for each $z\in\mathbb{D}$.

11. Let f be an entire function such that $\lim_{|z|\to\infty}|f(z)|=\infty$. Then

(a.) $f\left(\frac{1}{z}\right)$ has an essential singularity at 0

(b.) f cannot be a polynomial

(c.) f has finitely many zeros

(d.) $f\left(\frac{1}{z}\right)$ has a pole at 0

12. If f is entire $|f(z)|<A|z|^n$ for some positive integers n and then

(a.) $f(z)=z^n$

(b.) $f(z)=e^{nz}$

(c.) $f(z)$ is polynomial of deg less or equal to n.

(d.) $f(z)=ne^z$

13. Let f be a entire function on $\mathbb{C}$ such that $|f(z)|\le 100\log|z|$ for each z with $|z|\ge 2$. If $f(i)=2i$, then $f(1)$

(a.) Must be 2

(b.) Must be $2i$

(c.) Must be i

(d.) Cannot be determined from the given data

14. Let $f(z)$ be an entire function such that $|f(z)|\le K|z|, \forall z\in\mathbb{C}$, for some $K>0$. If $f(1)=i$, the value of $f(i)$ is

(a.) 1

(b.) -1

(c.) i

(d.) $-i$

15. Let $f:\mathbb{C}\to\mathbb{C}$ be an arbitrary analytic function satisfying $f(0)=0$ and $f(1)=2$ then

(a.) there exists a unique $\{z_n\}$ st $|z_n|>n$ and $|f(z_n)|>n$

(b.) there exists a sequence $\{z_n\}$ st $|z_n|>n$ and $|f(z_n)|<n$

(c.) there exists a bounded sequence $\{z_n\}$ such that $|f(z_n)|>n$.

(d.) there exists a sequence $\{z_n\}$ s.t. $z_n\to 0$ and $f(z_n)\to 2$.

16. Let $f(z)$ be an entire function st for some constant α. $|f(z)|\le\alpha|z|^3$ for $|z|\ge 1$ and $f(z)=f(iz)$ for all $z\in c$ then

(a.) $f(z)=\alpha z^3$ for all $z\in c$

(b.) $f(z)$ is constant.

(c.) $f(z)$ is quadratic polynomials

(d.) No such $f(z)$ exists.

17. Let B be an open subset of $\mathbb{C}$ and ∂B denote the boundary of B. Which of the following statements are correct?

(a.) For every entire function f, we have $\partial(f(B))\subseteq f(\partial B)$

(b.) For every entire function f and a bounded open set B, we have $\partial(f(B))\subseteq f(\partial B)$

(c.) For every entire function f, we have $\partial(f(B))=f(\partial B)$

(d.) There exist an unbounded open subset B of $\mathbb{C}$ and an entire function f such that $\partial(f(B))\subseteq f(\partial B)$

18. Let $f:\mathbb{C}\to\mathbb{C}$ be a meromorphic function analytic at 0 satisfying $f\left(\frac{1}{n}\right)=\frac{n}{2n+1}$ for $n\ge 1$

(a.) $f(0)=\frac{1}{2}$

(b.) f has a simple pole at $z=-2$

(c.) $f(2)=\frac{1}{4}$

(d.) No such meromorphic function exists

19. Let f be a polynomial function on the entire complex plane s.t. $f(z)\ne 0$ for z s.t $|z|=1$ then $\frac{1}{2\pi i}\int_{|z|=1}\frac{f'(z)}{f(z)}dz$

(a.) can take only integer values

(b.) can take only value

(c.) is zero

(d.) is equal to degree of f

20. If $f:\mathbb{C}\to\mathbb{C}$ is analytical and real valued then

(a.) f is non-constant

(b.) f maps open sets to open

(c.) f is constant

(d.) $f(z)=z^n$ for some $n>0$

21. Suppose $f(z)=(z-1)(z-2)(z-3)^3$ and $\Upsilon(t)=4e^{it}$, $0\le t\le 4$ then the argument principle implies that $\int\frac{f'(z)}{f(z)}dz$ is

(a.) $2\pi i$

(b.) $-2\pi i$

(c.) 0

(d.) 1

22. The number of roots of the equation $z^5-12z^2+14=0$ that lie in the region $\left\{z\in\mathbb{C}:2\le|z|<\frac{5}{2}\right\}$ is

(a.) 2

(b.) 3

(c.) 4

(d.) 5

23. The nos of zeros, counting multiplicities of the polynomial $z^5+3z^3+z^2+1$ inside the circle $|z|=2$ is

(a.) 0

(b.) 2

(c.) 3

(d.) 5

24. Let $f(z)=1+2z+3z^2+...nz^n$ and let γ be a closed curve which contains all the zeros of $f(z)$ then $\frac{1}{2\pi i}\int_\gamma z^2\frac{f'(z)}{f(z)}dz$ equal to –

(a.) 0

(b.) n

(c.) $\frac{2n-n^2+1}{n^2}$

(d.) $\frac{2n-n^2-1}{n^2}$

25. The value of $\int_C\frac{f'(z)}{f(z)}dz$ where $f(z)=\frac{(z^2+1)^2}{(z^2+3z+2)^3}$ and C is the circle $|z|=3$ with positive sense

(a.) $2\pi i$

(b.) $-2\pi i$

(c.) $4\pi i$

(d.) $-4\pi i$

26. If $f(z)=z^2+2$ then the minimum value of $|f(z)|$ over the closed region $|z|\le 1$ is ...... Fill the space from the following choices

(a.) 0

(b.) 1

(c.) 3

(d.) 2

27. Consider the polynomial $p(z)=z^5+z^3+5z^2+2$ then the number of zeros in $1<|z|<2$?

(a.) 2

(b.) 3

(c.) 5

(d.) none

28. The number of zeros of $p(z)=4z^2-3iz^2+iz-9=0$ in the a disc $|z|<1$ is

(a.) 0

(b.) 1

(c.) 2

(d.) 4

29. If $g(z)=z^3$ and $f(z)=z^3-z+1$ then the value of $\frac{1}{2\pi i}\int_C g(z)\frac{f'(z)}{f(z)}dz$ where C contains all the zeros of $f(z)$ is

(a.) –1

(b.) 0

(c.) –3

(d.) –9

30. If $f(z)$ is analytic on D where $D=\{(x,y):|x|\le a,|y|\le b, a\ge b\}$ if $f(z)$ satisfies the inequality $|f(z)|\le M$ on the boundary of D then which can taken as an upper bound of $|f'(0)|$

(a.) $\frac{2M(a+b)}{\pi ab}$

(b.) $\frac{2\pi ab}{M^2}$

(c.) $\frac{2M^2}{2\pi(ab)}$

(d.) $\frac{2M(a+b)}{\pi b^2}$

31. If C be the unit circle $z=e^{i\theta},(0<\theta\le 2\pi)$ the value of $\frac{1}{2\pi i}\Delta\log\left(\frac{1}{z^2}\right)$ is

(a.) 0

(b.) 1

(c.) –1

(d.) –2

Assignment Sheet -5

1. Let $D=\{z\in\mathbb{C}:|z|<1\}$. Then there exists a holomorphic function $f:D\to\bar{D}$ with $f(0)=0$ with the property

(a.) $f'(0)=\frac{1}{2}$

(b.) $\left|f\left(\frac{1}{3}\right)\right|=\frac{1}{4}$

(c.) $f\left(\frac{1}{3}\right)=\frac{1}{2}$

(d.) $|f'(0)|=\sec\left(\frac{\pi}{6}\right)$

2. Let $f(z)=\frac{1+z}{1-z}$. Which of the following is/are true?

(a.) f maps $\{|z|<1\}$ onto $\{\text{Re}(z)>0\}$.

(b.) f maps $\{|z|<1, \text{Im}(z)>0\}$ onto $\{\text{Re}(z)>0, \text{Im}(z)>0\}$.

(c.) f maps $\{|z|<1, \text{Im}(z)<0\}$ onto $\{\text{Re}(z)<0, \text{Im}(z)<0\}$.

(d.) f maps $\{|z|>1\}$ onto $\{\text{Im}(z)>0\}$.

3. Let $\mathbb{D}=\{z\in\mathbb{C}:|z|<1\}$. Which of the following are correct?

(a.) There exists a holomorphic function $f:\mathbb{D}\to\mathbb{D}$ with $f(0)=0$ and $f'(0)=2$

(b.) There exists a holomorphic function $f:\mathbb{D}\to\mathbb{D}$ with $f\left(\frac{3}{4}\right)=\frac{3}{4}$ and $f'\left(\frac{2}{3}\right)=\frac{3}{4}$

(c.) There exist a holomorphic function $f:\mathbb{D}\to\mathbb{D}$ with $f\left(\frac{3}{4}\right)=\frac{-3}{4}$ and $f'\left(\frac{3}{4}\right)=\frac{-3}{4}$

(d.) There exists a holomorphic function $f:\mathbb{D}\to\mathbb{D}$ with $f\left(\frac{1}{2}\right)=\frac{-1}{2}$ and $f'\left(\frac{1}{4}\right)=1$

4. The minimum possible value of $|z|^2+|z-3|^2+|z-6i|^2$, where z is a complex number and $i=\sqrt{-1}$, is

(a.) 15

(b.) 45

(c.) 30

(d.) 20

5. Let $f:\mathbb{D}\to\mathbb{D}$ be a holomorphic function with $f(0)=0$, where $\mathbb{D}$ is the open unit disc $\{z\in\mathbb{C}:|z|<1\}$. Then

(a.) $|f'(0)|=1$

(b.) $\left|f\left(\frac{1}{2}\right)\right|\le\frac{1}{2}$

(c.) $\left|f\left(\frac{1}{2}\right)\right|\le\frac{1}{4}$

(d.) $|f'(0)|\le\frac{1}{2}$

6. Let $f(z)=z+\frac{1}{z}$ for $z\in\mathbb{C}$ with $z\neq0$. Which of the following are always true?

(a.) f is an analytic function on $\mathbb{C}\setminus\{0\}$

(b.) f is a conformal map on $\mathbb{C}\setminus\{0\}$

(c.) f maps the unit circle to a subset of the real axis

(d.) The image of any circle in $\mathbb{C}\setminus\{0\}$ is again a circle.

7. Let $f:D\to D$ be holomorphic with $f(0)=0$ and $f\left(\frac{1}{2}\right)=0$ where $D=\{z:|2|<1\}$ which of the following statements are correct?

(a.) $\left|f'\left(\frac{1}{2}\right)\right|\le\frac{4}{3}$

(b.) $|f'(0)|\le1$

(c.) $\left|f'\left(\frac{1}{2}\right)\right|\le\frac{4}{3}$ and $|f'(0)|\le1$

(d.) $f(z)=z, z\in D$

8. For $z \in D$ of the form $z = x + iy$, define

$H^+ = \{z \in C : y > 0\}$

$H^- = \{z \in C : y < 0\}$

$L^+ = \{z \in C : x > 0\}$

$L^- = \{z \in C : x < 0\}$

The function $f(z) = \dfrac{2z+1}{5z+3}$

(a.) Maps H^+ onto H^+ and H^- onto H^-

(b.) Maps H^+ onto H^- and H^- onto H^+

(c.) Maps H^+ onto L^+ and H^- onto L^-

(d.) Maps H^+ onto L^- and H^- onto L^+

9. Let U be an open subset of $\mathbb{C}$ containing $D = \{z \in \mathbb{C} : |z| \le 1\}$ and let $f : U \to \mathbb{C}$ be the map defined by

$f(z) = e^{i\psi} \dfrac{z-a}{1-\bar{a}z}$ for $a \in D$, and $\psi \in [0,\ 2\pi]$.

Which of the following statements are true?

(a.) $\left|f\left(e^{i\theta}\right)\right| = 1$ for $0 \le \theta \le 2\pi$

(b.) f maps $\{z \in \mathbb{C} : |z| < 1\}$ onto itself

(c.) f maps $\{z \in \mathbb{C} : |z| < 1\}$ into itself

(d.) f is one-one

10. Let $f : \mathbf{D} \to \mathbf{D}$ be holomorphic with $f(z) = \dfrac{1}{2}$ and $f\left(\dfrac{1}{2}\right) = 0$, where $\mathbf{D} = \{z \in \mathbb{C} : |z| \le 1\}$.

Which of the following is correct?

(a.) $|f'(0)| \le \dfrac{3}{4}$

(b.) $\left|f'\left(\dfrac{1}{2}\right)\right| \le \dfrac{4}{3}$

(c.) $|f'(0)| \le \dfrac{3}{4}$ and $\left|f'\left(\dfrac{1}{2}\right)\right| \le \dfrac{4}{3}$

(d.) $f(z) = z, z \in \mathbf{D}$

11. Define

$H^+ = \{z \in \mathbb{C} : y > 0\}$

$H^- = \{z \in \mathbb{C} : y < 0\}$

$L^+ = \{z \in \mathbb{C} : x > 0\}$

$L^- = \{z \in \mathbb{C} : x < 0\}$

The function $f(z) = \dfrac{z}{3z+1}$

(a.) Maps H^+ onto H^+ and H^- onto H^-

(b.) Maps H^+ onto H^- and H^- onto H^+

(c.) Maps H^+ onto L^+ and H^- onto L^-

(d.) Maps H^+ onto L^- and H^- onto L^+

12. The region $\{z \in \mathbb{C} : -1 < \operatorname{Re}(z) < 1\}$ can be mapped conformally onto

(a.) the upper half plane.

(b.) The annulus

(c.) the plane with two pts removed.

(d.) the punctured disk.

13. The transformation $w = \dfrac{z-i}{z+i}$ maps

(a.) the upper half plane to itself

(b.) the upper half plane to lower half plane.

(c.) the upper half plane to unit disk

(d.) the unit disk to itself

14. An example of map which is conformal in the whole of its domain of definition is

(a.) $f(z) = z^2$

(b.) $f(z) = e^{z+2}$

(c.) $f(z) = \cos z$

(d.) $f(z) = e^{z^2}$

15. Let $f : \mathbb{C} \setminus \{3i\} \to \mathbb{C}$ be defined by $f(z) = \dfrac{z-i}{iz+3}$. Which of the following statements about f is **FALSE**?

(a.) f is conformal on $\mathbb{C} \setminus \{3i\}$

(b.) f maps circles in $\mathbb{C} \setminus \{3i\}$ onto circles in $\mathbb{C}$

(c.) All the fixed points of f are in the region $\{z \in \mathbb{C} : \operatorname{Im}(z) > 0\}$

(d.) There is no straight line $\mathbb{C} \setminus \{3i\}$ which is mapped onto a straight line in $\mathbb{C}$ by f

16. The image of the region $\{z \in \mathbb{C} : \mathrm{Re}(z) > \mathrm{Im}(z) > 0\}$ under the mapping $z \mapsto e^{z^2}$ is

(a.) $\{w \in \mathbb{C} : \mathrm{Re}(w) > 0, \mathrm{Im}(w) > 0\}$

(b.) $\{w \in \mathbb{C} : \mathrm{Re}(w) > 0, \mathrm{Im}(w) > 0, |w| > 1\}$

(c.) $\{w \in \mathbb{C} : |w| > 1\}$

(d.) $\{w \in \mathbb{C} : \mathrm{Im}(w) > 0, |w| > 1\}$

17. Consider the function $f(z) = \dfrac{z^2 + \alpha z}{(z+1)^2}$ and $g(z) = \sinh\left(z - \dfrac{\pi}{2\alpha}\right), \alpha \neq 0$.

(i) The residue of $f(z)$ at its pole is equal to 1. Then the value of α is

(a.) -1

(b.) 1

(c.) 2

(d.) 3

(ii) For the value of α obtained in the function $g(z)$ is not conformal at a point

(a.) $\dfrac{\pi(1+3i)}{6}$

(b.) $\dfrac{\pi(3+i)}{6}$

(c.) $\dfrac{2\pi}{3}$

(d.) $\dfrac{i\pi}{2}$

18. Under the transformation $w = \sqrt{\dfrac{1-iz}{z-i}}$, the region $D = \{z \in \mathbb{C} : |z| < 1\}$ is transformed to

(a.) $\{z \in \mathbb{C} : 0 < \arg z < \pi\}$

(b.) $\{z \in \mathbb{C} : -\pi < \arg z < 0$

(c.) $\left\{z \in \mathbb{C} : 0 < \arg z < \dfrac{\pi}{2} \text{ or } \pi < \arg z < \dfrac{3\pi}{2}\right\}$

(d.) $\left\{z \in \mathbb{C} : \dfrac{\pi}{2} < \arg z < \pi \text{ or } \dfrac{3\pi}{2} < \arg z < 2\pi\right\}$

19. Let f be a bilinear transformation that maps -1 to 1, i to ∞ and $-i$ to 0 then $f(1)$ is equal to

(a.) -2

(b.) -1

(c.) i

(d.) $-i$

20. Let G_1 and G_2 be the images of the disc $\{z \in \mathbb{C} : |z+1| < 1\}$ under the transformations.

$w = \dfrac{(1-i)z+2}{(1+i)z+2}$ and $w = \dfrac{(1+i)z+2}{(1-i)z+2}$ resp. then.

(a.) $G_1 = \{w \in \mathbb{C} : \mathrm{Im}(w) < 0\}$ and $G_2 = \{w \in \mathbb{C} : \mathrm{Im}(w) > 0\}$

(b.) $G_1 = \{w \in \mathbb{C} : \mathrm{Im}(w) > 0\}$ and $G_2 = \{w \in \mathbb{C} : \mathrm{Im}(w) < 0\}$

(c.) $G_1 = \{w \in \mathbb{C} : |w| > 2\}$ and $G_2 = \{w \in \mathbb{C} : |w| < 2\}$

(d.) $G_1 = \{w \in c : |w| < 2\}$ and $G_2 = \{w \in c : |w| > 2\}$.

21. Let $w = f(z)$ be the bilinear transformation that maps -1, 0 and 1 to $-i$, 1 and i respectively. Then $f(1-i)$ equals

(a.) $-1+2i$

(b.) $2i$

(c.) $-2+i$

(d.) $-1+i$

22. The bilinear transformation as which maps the points $0, 1, \infty$ in the z-plane onto the points $-i, \infty, 1$ in w-plane is

(a.) $\dfrac{z-1}{z+i}$

(b.) $\dfrac{z-i}{z+1}$

(c.) $\dfrac{z+i}{z-1}$

(d.) $\dfrac{z+1}{z-i}$

23. The function $f(z)=z^2$ maps the I[st] quadrant onto

(a.) Itself

(b.) Upper half plane

(c.) Third Quadrant

(d.) Right half plane.

24. Let $w=\dfrac{az+b}{cz+d}$ and $f=\dfrac{\alpha z+\beta}{rz+s}$ be bilinear (Mobius) transformation, then the following is also a bilinear transformation

(a.) $f(z)w(z)$

(b.) $f\{w(z)\}$

(c.) $f(z)+w(z)$

(d.) $f(z)+\dfrac{1}{w(z)}$

25. The fixed points of $f(z)=\dfrac{2iz+5}{z-2i}$ are

(a.) $1\pm i$

(b.) $1\pm 2i$

(c.) $2i\pm 1$

(d.) $i\pm 1$.

26. The function $w(z)=-\left(\dfrac{1}{z}+bz\right),-1<b<1$, maps $|z|<1$ onto

(a.) A half plane

(b.) Exterior of the circle

(c.) Exterior of an ellipse

(d.) Interior of an ellipse

27. The transformation $w=e^{i\theta}\left(\dfrac{3-p}{\bar{p}z-1}\right)$, where p is a constant, maps $|z|<1$ onto

(a.) $|w|<1$ if $|P|<1$

(b.) $|w|>1$ if $|P|>1$

(c.) $|w|=1$ if $|P|=1$

(d.) $|w|=3$ if $P=0$

28. The conjugate (also called symmetric) point of $1+i$ with respect to circle $|z-1|=2$ is

(a.) $1-i$

(b.) $1+4i$

(c.) $1+2i$

(d.) $-1-i$

29. The bilinear transformation $w=\dfrac{2z}{z-2}$ maps $\{z:|z-1|<1\}$ onto.

(a.) $\{w:\operatorname{Re}w<0\}$

(b.) $\{w:\operatorname{Im}w>0\}$

(c.) $\{w:\operatorname{Re}w>0\}$

(d.) $\{w:|w+2|<1\}$

30. The transformation $w=\dfrac{1}{2}\left[z+\alpha^2 z^{-1}\right]$, $\alpha\in\mathbb{R}$ maps the circle $|z|=r\ (r\neq\alpha)$ into

(a.) Circle

(b.) Ellipse

(c.) Hyperbola

(d.) Line

31. The magnification factor of the conformal mapping $W=\sqrt{2}e^{\pi i/4}z(1-2i)$ is

(a.) 1

(b.) 2

(c.) 3

(d.) 10

32. The circle $a(x^2+y^2)+bx+cy+d=0$ is transformed by $f(z)=\dfrac{1}{z}$ into a circle the centre of the circle is given by

(a.) $\left(\dfrac{2ab}{b^2+c^2},\dfrac{2ac}{b^2+c^2}\right)$

(b.) $\left(\dfrac{-2ab}{b^2+c^2},\dfrac{2ac}{b^2+c^2}\right)$

(c.) $\left(\dfrac{2ab}{b^2+c^2},\dfrac{-2ac}{b^2+c^2}\right)$

(d.) $\left(\dfrac{-2ab}{b^2+c^2},\dfrac{-2ac}{b^2+c^2}\right)$

33. If $f(z)=e^{2z}-2iz+3$ then which of the following is incorrect

(a.) $f(z)$ has infinite critical point

(b.) scale factor of $f(z)$ at every real number is defined

(c.) scale factor at every point on imaginary axis is defined

(d.) $f(z)$ is not conformal

34. The transformation $z=a\cos\omega+ib\sin\omega$ maps the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ in z plane to

(a.) The real axis of w-plane

(b.) The imaginary axis of w-plane

(c.) An ellipse in w-plane

(d.) The complete w-plane

35. The rectangular region R bounded by $x=0,\ y=0;\ x=2,\ y=3$ in z-plane is mapped into the rectangular region R of w-plane through the transformation $w=\sqrt{2}e^{i\pi/4}z$ this transformation performs

(a.) A rotation

(b.) A magnification

(c.) A translation and magnification

(d.) A rotation and magnification

36. Choose the correct statements

(a.) Let $z_1,z_2,z_3,z_4,$ are fixed complex number then we can define at most 6 different cross ratio.

(b.) The cross ratio is invariant under B.T. and this property can be used in obtaining specific B. T. mapping three points into three other points

(c.) A B.T. which maps $z=0,i,-i$ into $w=i,1,0$ as $w=-\frac{i(1+z)}{(z+1)}$

(d.) $(z_1,z_2,z_3,z_4)=\frac{(z_1-z_2)(z_3,-z_4)}{(z_1-z_4)(z_3-z_2)}$